

Renewal Geometry and the Emergence of Lorentzian Spacetime

Aurélien Pélissier^{1,†}

¹École normale supérieure Paris-Saclay, 91190 Gif-sur-Yvette, France

[†]Correspondence: aurelien.pelissier.38@gmail.com

Abstract

We construct a finite predictive framework in which Lorentzian spacetime and vacuum Einstein dynamics emerge from operational renewal data rather than being assumed at the outset. Histories indistinguishable by all possible futures are identified, yielding a canonical predictive structure from which relational geometry is reconstructed. Two independent finite criteria select three spatial dimensions, while one additional protected signed direction fixes a $(1, 3)$ Lorentzian signature. Calibrated renewal statistics then reconstruct the spatial metric, lapse, and shift. A common variational structure is reconstructed directly from the finite dynamics. On certified noncollapsed continuum branches, it converges to Palatini–Holst gravity and yields the vacuum Einstein equation, while failed conditions produce explicit obstructions. The same framework also supports a cutoff-uniform nonsymmetric $3 + 1$ Einstein population with arbitrary three-dimensional dependence, stable across an open family of renewal laws sharing the same continuum limit.

Keywords. Emergent spacetime; Lorentzian geometry; predictive dynamics; renewal processes; variational dynamics; Palatini–Holst gravity; ADM decomposition.

1 Introduction

One of the oldest ambitions of fundamental physics is to explain why spacetime has the structure that it does, rather than to take that structure as the stage on which all other physics occurs. General relativity transformed gravity from a force propagating through spacetime into the dynamics of spacetime itself, but it still begins with a differentiable manifold carrying a Lorentzian metric. Most subsequent approaches to quantum gravity likewise replace some part of the classical geometric description while retaining another part as primitive. Causal-set theory begins with a locally finite causal order [1, 2]; graph, simplicial, and group-field approaches begin with combinatorial incidence data whose collective phases are expected to approximate geometry [3, 4, 5]; and noncommutative geometry replaces the underlying space by spectral and operator-algebraic data [6]. Other programmes seek gravitational dynamics from thermodynamics, entropy, or quantum information [7, 8, 9]. These approaches differ profoundly in their microscopic assumptions, yet they share a central difficulty: the mathematical language used to reconstruct spacetime often already contains some analogue of geometry, locality, causal order, dimension, or temporal structure.

The question pursued here is more primitive. We ask whether the structures normally used to define spacetime can instead arise from the organization of *predictive histories*. The microscopic object is not a point of a manifold, a simplex, a causal element, or a prescribed spectral triple. It is an operational renewal event: an intervention occurs, a finite outcome is registered, some

distinctions remain capable of influencing later operations, and the process may subsequently renew. At this level a history is simply a sequence of operational events. It has no assigned position, distance, metric, connection, curvature, or spacetime dimension, and its ordering is not yet interpreted as motion through a pre-existing geometry.

Prediction before geometry. This choice of starting point is motivated by a simple observation. Physical description is never sensitive to an arbitrary microscopic history in its full detail. What matters operationally is which distinctions in that history can still affect a future experiment. Two past histories that produce exactly the same statistics under every admissible continuation are therefore physically indistinguishable for all subsequent purposes. Identifying such histories is a predictive rather than a geometric reduction. It is closely related in spirit to causal-state constructions in computational mechanics, where pasts are quotiented by equality of their conditional futures [10, 11], and to continuous-time predictive representations of stochastic processes [12, 13]. The difference is that the present construction retains the complete operational intervention data and the positive process structure required for the subsequent algebraic and geometric reconstruction.

Because future equivalence is preserved under admissible continuation, the quotient is not merely a compressed description of the past. It carries a canonical successor structure and therefore becomes the minimum persistent predictive state of the process. Persistence is thus reconstructed from the organization of future distinguishability rather than supplied as an independent state space, graph adjacency, or background coordinate system.

Renewal as a microscopic principle. This perspective leads naturally to renewal theory. Renewal processes are usually introduced as stochastic models of systems that repeatedly reset, regenerate, or lose part of their past. They underlie classical results in probability and statistical mechanics [14], while non-Markovian renewal processes include heavy-tailed and infinite-mean regimes with qualitatively different long-time behavior [15, 16, 17]. In conventional applications, however, renewal occurs *inside* an already given time and state space. Waiting times are measured by an external clock, and the process evolves on a previously specified collection of states.

Here we reverse that logical order. Renewal is not placed in spacetime; renewal belongs to the microscopic organization from which spacetime is reconstructed. The microscopic law may retain history through waiting-time, record, or return statistics and need not be Markovian. A local Markovian evolution can nevertheless arise after predictive compression and coarse graining, once unresolved renewal memory has collapsed onto a sufficient future state. In this sense the familiar locality of continuum evolution is naturally interpreted as a *Markovian scaling regime of a more general renewal dynamics*, rather than as the fundamental form of microscopic evolution.

There is a useful conceptual analogy with statistical mechanics and renormalization. Thermodynamic variables emerge after most microscopic information has been discarded, while renormalization retains degrees of freedom that remain relevant at longer scales. Here the reduction is instead organized by *future distinguishability*: microscopic information is discarded precisely when no admissible future can read it. What survives is therefore not an arbitrary coarse graining but a canonical predictive quotient. The geometry developed below is built on that quotient. Loosely speaking, spacetime is the stable organization of those relational distinctions that prediction cannot forget.

Main result. The central result is a finite-to-continuum reconstruction and certification theorem. Complete conditional future statistics first determine a canonical minimum predictive carrier and a positive history algebra, whose intrinsic word-Gram hierarchy together form the *Renewal Grand*

Tensor. The finite relational dimension is selected in two independent ways. Exact endpoint–cycle source–rank balance among connected simple regular relations selects the tetrahedral cell K_4 without an alternating-form hypothesis; within the stronger pair-homogeneous simplex category, minimal faithful alternating completion selects the same cell. Its standard module is the three-dimensional A_3 root space. Calibrated root-race statistics then reconstruct the ten ADM variables. The positive Grand Tensor cannot itself supply a negative direction: an independently occurring protected binary orientation is an irreducible signed datum, and adjoining one signed direction to the positive three-dimensional spatial form fixes Lorentzian inertia $(1, 3)$ before a Clifford realization is chosen.

The dynamical part is reconstructed from the complete accepted physical cylinder rather than imposed as a continuum action. A marked renewal pressure recovers the full action slope, finite curl and period tests determine whether the increments integrate to one scalar common action, and a positive Fisher–Bregman gap, with its physical test-lift and cutoff constants retained, certifies stationarity of the accepted field dynamics for that independently reconstructed action. Finite Gibbs/Kullback–Leibler identities separately compare the accepted probability rows with their reconstructed costs. After resolved memory is removed, Lorentz covariance and the finite bivector commutant classify the first-order gravitational score as Palatini–Holst plus determinant volume. A positive spatial cut margin supplies regulator-uniform Sobolev, Poincaré, counting, and compact-screen bounds. Continuous and fully discrete propagation estimates give a spacetime L^2 curvature bound from initial energy and explicit growth, transfer and differentiated-source budgets; a sharp action-current comparison can supply the source budget. Connection convergence and interface-complete Cartan comparisons identify the resulting curvature. On the branch where these finite and cofinal certificates pass, the weak–strong Palatini limit gives

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \tag{1}$$

distributionally on the determining test core. If a required condition fails, the construction instead returns the corresponding finite or cofinal obstruction. The result is therefore an Einstein-or-obstruction theorem, rather than a claim that every microscopic renewal process flows to general relativity. A failed sufficient certificate does not exclude an Einstein subsequence obtainable by a different argument.

The positive branch is nonempty and is not confined to symmetry-reduced kinematics. In addition to the homogeneous and Gowdy sectors described below, a full three-dimensional local harmonic renewal writer admits a cutoff-independent nonlinear Cauchy theorem in a fixed Sobolev ball for all ten metric components and all spatial lattice modes. More strongly, this is not an isolated engineered update: the same estimates and Einstein limit hold uniformly on a fixed open 55-dimensional family of local differential laws, with no convergence of the law parameter required along the cutoff sequence. Around finite realizations of these laws, conditional physical residual moments and a profile-transport metric define relatively open neighborhoods of history-dependent transition rows whose successful histories have the same Einstein limit in probability and, along the declared odd refinement, eventually almost surely. The compatible initial records can also be produced at finite cutoff by solving the original reconstructed action’s complete lapse–shift constraint rows from unconstrained small free tensor records; no continuum vacuum metric is required as an input to that preparation.

These population results and the general certification theorem have different logical roles. [Theorems 4.13](#) and [4.15](#) construct nonsymmetric $3 + 1$ classes and reach the Einstein equation by cutoff-uniform evolution, consistency and finite preparation estimates, whereas [Theorem 4.19](#) is a recognition theorem for arbitrary complete gravitational renewal cylinders. Its five certificates are not the definition of the constructive population. Nor is the stable law family silently identified

with the raw exact finite Euler flow of the reconstructed action. That stronger route is analyzed independently in [Theorem 4.17](#) and [Proposition 4.18](#) and [Appendix F](#). The exact-action analysis now contains an all-cutoff regular transverse source block, an exact scalar–trace response decomposition, source-neutral finite preparations, controlled literal initial curvature on a nontrivial branch, and a hierarchy of temporal compatibility tests. A natural exactly prepared branch nevertheless develops a higher tangency defect under its own flow; this defect cannot be removed by changing only the leading multiplier tilt at fixed first jet, but it changes nontrivially under admissible canonical first-jet deformations. The remaining direct exact-action problem is therefore a nonlinear cofinal tangency and propagation problem, not merely a finite determinant test.

The homogeneous isotropic sector gives a complementary exact variational realization: arbitrary positive renewal variables yield the full lapse-varied vacuum Friedmann equations, so the flat and de Sitter rate profiles arise as stationary solutions rather than being prescribed in advance. A finite positive accept/reject process reconstructs the same reduced action directly from measured trial probabilities and converges quantitatively to the stationary branch. Exact conservative time blocking of this quadratic class preserves the scale-free action–law defect, so it does not generate an open basin toward the matched surface. Beyond homogeneity, a nonlinear unpolarized Gowdy T^3 regulator preserves an exact finite momentum invariant and, on its finite offset alphabet, has pathwise $O(h)$ convergence of the full Riemann tensor on smooth bounded slabs. These constructions provide complementary open, homogeneous, and nonlinear symmetry-reduced populations without asserting generic microscopic attraction to the certified branch.

The precise operational entrance is intentionally narrower than a claim that all geometric information is generated from positive probabilities alone. No predictive state space or persistent relational carrier is postulated independently: histories are identified exactly when every admissible continuation has the same conditional law, and the resulting future-equivalence quotient is the unique minimum persistent predictive carrier through which all future distinctions factor. The geometric branch also retains two kinds of information that this positive predictive quotient cannot manufacture. A protected binary orientation is an actual two-valued operational distinction, but no interpretation as past and future, positive and negative norm, or Lorentzian signature is assumed microscopically. Likewise, normalized race probabilities determine only relative rates, so absolute waiting-time and measure calibrations are required when those rates are interpreted geometrically.

A further distinction is essential. The future-equivalence class is derived as an abstract predictive state, but spacetime reconstruction must physically read that class at a cut and correlate it on the same history with the occurring root event, scheduler-separated fast path, orientation, calibration, and other declared marks. These data form a minimal faithful readable relation packet. The packet does not introduce a second predictive state: its state coordinate is the already derived future-equivalence class. Its additional content is physical occurrence and same-cut readability. The enrichment is conservative: forgetting the added marks recovers the original event cylinders exactly, while future minimization removes every unread refinement. Thus predictive persistence is derived, the physical information needed for the geometric reconstruction must actually occur together, and metric, connection, curvature, and spacetime dimension remain downstream outputs.

The Renewal Grand Tensor and Lorentzian reconstruction. The first major structural output is the object we call the *Renewal Grand Tensor*. The name does not refer to a tensor field on a pre-existing manifold. Predictively indistinguishable histories are first identified, producing the unique minimum persistent predictive carrier. Tomographically complete intervention statistics then reconstruct the finite positive process tensors, while support-minimal completely positive realization removes unobservable environmental degrees of freedom [\[18, 19\]](#). The surviving mixed histories

generate a finite C^* -algebra, and their intrinsic positive overlaps form a word-Gram hierarchy. Together these structures record which distinctions survive into the future, how renewal operations compose, and how the resulting histories overlap. The Grand Tensor is therefore not geometry imposed on a renewal process; it is the canonical algebraic and positive structure retained by the process after every future-invisible distinction has been removed.

This places the construction near noncommutative and spectral geometry, but with a different starting point. In ordinary spectral geometry an algebra, Hilbert-space representation, and Dirac operator are supplied as geometric data [6]. Lorentzian variants enrich this structure with an indefinite inner product, fundamental symmetry, temporal datum, or twist [20, 21, 22, 23, 24, 25, 26]. Recent work has also examined the emergence of temporal and Lorentzian structures within twisted almost-commutative settings [27, 28]. The present construction operates one level upstream. Neither spectral geometry nor a Krein structure is taken as the initial object. The positive history algebra is reconstructed from operational renewal statistics, while the protected binary orientation becomes an indefinite/Krein structure only after it acts on the predictively reduced relational carrier. Its Clifford realization then distinguishes one timelike direction.

Predictive collapse does more than compress a finite history. Because future equivalence is preserved under admissible continuation, the minimum predictive state propagates canonically from one event to the next and therefore supplies a persistent relational carrier. The local dimension is then tested by two logically distinct finite selectors. The first uses no alternating structure: for a connected simple regular relation, exact equality of the faithful endpoint-source and cycle-source ranks has the unique nontrivial solution K_4 . The second works in the more restrictive pair-homogeneous simplex category and asks for a nondegenerate alternating temporal-spatial coefficient pairing together with minimum faithful source innovation; it independently selects the same K_4 . Thus the alternating form is not the sole route to three spatial dimensions. It remains a branch axiom of the second selector and supplies only an antisymmetric coefficient pairing, not a symplectic phase space, Hamiltonian flow, metric, or Lorentzian sign. Neither selector is a universality theorem over arbitrary higher-arity microscopic systems.

The metric appears one layer later. The relational carrier supports a finite family of elementary root transitions. Their occurrence rates and waiting times provide observable first- and second-order statistics. The second-order statistics determine the spatial kinetic form, the directed first moment determines the shift, and a physical speed density fixes the remaining scale. These quantities invert uniquely to the ADM spatial metric, lapse, and shift. The metric coefficients are therefore not microscopic labels but collective statistical quantities reconstructed from the renewal process.

The geometric clock is likewise not identified with the slower renewal clock. A nondegenerate continuum bracket requires a scheduler-separated root race with total rate of order h^{-2} . Its winner and waiting-time statistics supply the absolute kinetic scale, whereas root-direction probabilities alone leave the common rate scale undetermined. The protected binary orientation then acquires its geometric meaning. The positive Grand-Tensor and root-race forms remain positive and therefore cannot produce an indefinite direction. The orientation Read supplies one independent signed line; leaving the reconstructed three-dimensional spatial form positive, the minimal nondegenerate signed extension has inertia $(-, +, +, +)$, up to overall sign convention. Only after this inertia is fixed do we choose a minimal Clifford representation. The gamma matrices therefore realize a signature already determined by the one-sign extension; they do not select the signature. Thus the Lorentzian sign is neither obtained by analytic continuation of a positive diffusion tensor nor inserted through a Dirac-matrix ansatz.

From kinematics to Einstein dynamics. Producing Lorentzian kinematics is not yet a derivation of gravitational dynamics. In particular, prescribing a renewal-rate history whose reconstructed metric is de Sitter would establish only realizability of one Einstein solution. The dynamical part therefore begins instead from the complete accepted physical cylinder. Future refinement reconstructs the minimum determining-field quotient and the accepted transition kernel acting on it. On the same resolved history, a protected vector reward together with physical duration determines an analytic renewal pressure. Its first derivative restores the affine action anchor lost by a centered probability score, while its higher jets reconstruct the mixed action Gram. Finite face-curl and period tests then determine whether these increments integrate to a single scalar common action and reconstruct that action when they do.

The actually accepted transition is next compared with the update generated by this independently reconstructed action. On the interior variational branch, the Fisher–Bregman gap controls the squared represented first variation, and explicit physical test-lift bounds retain the cutoff dependence needed in the continuum passage. Exact Gibbs/Kullback–Leibler identities provide a separate comparison of finite probability rows. The field-action comparison, rather than a cost fitted after observing the transition law, supplies the stationarity test.

The gravitational handoff is likewise formulated as a reconstruction and certification problem. The protected cycle reward entering this step is initially a generic measured face reward: no Palatini, Holst, Einstein–Hilbert, or cosmological Lagrangian is named in the entrance data. After resolved cut memory is removed, binary renewal subdivision and stable additivity produce a first-order face functional without assuming a derivative at a zero face. Lorentz covariance and the residual bivector commutant then leave only the identity and internal Hodge star, so only at this stage is the local first-order gravitational score classified as Palatini–Holst plus determinant volume.

Connection stationarity yields zero torsion on the spinless branch. Global spatial compactness is not introduced as a separate geometric axiom: the physical endpoint/capacity record determines an exact cut constant, and a uniform positive margin gives Sobolev, Poincaré, low-energy-counting, and finite-rank compact-screen bounds. Curvature control is supplied by a separate positive propagation estimate. A bounded initial curvature energy together with regulator-uniform observer/transport and differentiated-source budgets yields a spacetime L^2 curvature bound, also for a fully discrete curvature record. Strong coframe convergence supplies the nonlinear product passage; strong connection convergence and interface-complete consistency separately identify its curvature. These are the two analytic inputs to the weak–strong Palatini passage. On the positive certificate branch, the limiting metric variation is

$$\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad (2)$$

and determining stationarity therefore gives the vacuum Einstein equation distributionally.

These finite reconstructions assemble into a certification-or-obstruction alternative. For a complete gravitational renewal cylinder, the determining-field quotient, accepted kernel, renewal pressure, common-action exactness, action gap, spatial cut constant, face linearization, memory short, connection residual, curvature-propagation budgets, and Cartan compatibility are finite or cofinally testable. If the corresponding positive margins persist, the Einstein continuum follows. If they do not, the construction returns the failed object itself: a nonzero common-action curl or period, a positive Bregman/KL row, a macroscopic cut/potential witness, a face-linearity or memory residual, a torsion/connection defect, an unbounded curvature-propagation budget, a Cartan mismatch, or a cutoff compactness defect.

The homogeneous isotropic sector addresses a narrower question not settled by the abstract Einstein handoff. Instead of prescribing a de Sitter rate profile, we begin with arbitrary positive

root-symmetric variables $\kappa(t)$ and $\varrho(t)$. The reconstructed ADM map converts them bijectively to the scale factor and lapse. Rewriting the classified vacuum action in (κ, ϱ) and varying both variables before fixing the lapse yields the Hamiltonian constraint and scale equation, equivalently the full spatially flat vacuum Friedmann system. The exponential de Sitter profile therefore appears only after the stationary equations have been solved.

The same distinction persists at finite cutoff. A midpoint variational action retains independent lapse-weighted intervals, and its stationary recurrence converges at second order to the continuum flat/de Sitter branch. A finite positive accept/reject law reconstructs this action directly from its proposal distribution, accepted row, and total acceptance probability. A measured rank defect determines whether the local low-noise update and the cosmological parameter of the reconstructed action agree. On the compatible branch, finite stochastic cylinders have vanishing Euler residual and converge quantitatively to the stationary de Sitter solution.

The finite evolution mechanism is not confined to homogeneous trajectories. In the unpolarized Gowdy T^3 vacuum class both wave polarizations and a spatially varying nonlinear frame are retained. A source/characteristic/source splitting preserves an exact staggered momentum invariant, while the finite offset support and shared Hermite readout give pathwise second-jet control and hence $O(h)$ convergence of the full Riemann tensor on fixed smooth slabs. The independently varied unreduced action has the same vanishing continuum first variations. This symmetry-reduced theorem does not imply generic attraction to de Sitter or unrestricted four-dimensional regularity, but it provides a nonlinear inhomogeneous class in which full curvature compactness is derived rather than assumed.

Scope and organization. The emergence claim is relative to a precise operational entrance. Actual occurrence, complete contextual prediction, protected orientation, and absolute physical calibration are irreducible rows of the declared finite language. For gravitational dynamics one additionally needs the complete accepted field/action cylinder to occur physically. Once these records are present, however, the predictive carrier, determining field, accepted kernel, action slope, common scalar action, ADM geometry, compact spatial screen, and stationarity residual are reconstructed or certified rather than supplied as independent continuum geometry. Vanishing of the action gap and persistence of the positive noncollapse and curvature-propagation margins remain genuine physical conditions that can fail.

The dimensional claim has the same controlled scope. The K_4/A_3 result does not state that every possible microscopic theory must have three spatial dimensions. Rather, two independent finite classifications select the same cell: endpoint-cycle source-rank balance does so without an alternating-form assumption, while the alternating/minimal-source simplex selector does so in a stronger category. Their agreement removes the alternating form from the role of sole dimensional selector, while still falling short of universality over arbitrary higher-arity systems. Likewise the protected binary orientation is not a microscopic Lorentzian metric. The positive operational rows cannot reconstruct it, as shown by explicit separating models; once the A_3 spatial form is positive, this single independent sign fixes the minimal indefinite extension to Lorentzian inertia $(1, 3)$. The subsequent Clifford matrices are a representation of that quadratic form, not the origin of its signature.

The conceptual proposal is therefore that renewal may be more fundamental than its conventional role in probability theory suggests. From the usual point of view, renewal describes memory and resetting *within* time. From the present point of view, retention and forgetting in operational history determine which distinctions survive at all, while predictive equivalence reconstructs the persistent state on which relational geometry is built. Effective Markovian locality then appears not as the microscopic starting principle but as a limiting organization of predictive memory. In

this sense, the usual hierarchy

$$\boxed{\text{spacetime} \longrightarrow \text{stochastic dynamics}} \quad (3)$$

is replaced by

$$\boxed{\text{renewal dynamics} \longrightarrow \text{predictive structure} \longrightarrow \text{relational geometry} \longrightarrow \text{Lorentzian spacetime.}} \quad (4)$$

The claim is not that renewal theory in its classical form already contains general relativity. It is that when renewal is treated operationally, predictively, and with a minimal protected orientation, persistence and relational structure can be reconstructed before ordinary spacetime geometry is defined.

Section 2 constructs the minimum predictive carrier and the Renewal Grand Tensor from finite operational data. Section 3 identifies the minimal faithful relational branch and reconstructs the ADM and Lorentzian structures from the calibrated root process. Section 4 reconstructs the determining-field kernel and common action, introduces the accepted/action stationarity criterion, derives compact spatial screens from a finite cut margin, propagates curvature control from finite initial-energy and scaled source budgets in continuous or discrete time, and identifies connection curvature including interfaces. It then gives a cutoff-uniform nonsymmetric 3 + 1 population theorem, enlarges it to an open family of differential laws and relatively open finite transition-row neighborhoods with exact finite-action initial preparation, develops the exact-action response and dynamical-tangency hierarchy, and only afterward states the maximally general Palatini–Holst certification and Einstein-or-obstruction alternative. Section 5 studies complementary explicit vacuum sectors: it derives the full homogeneous Friedmann system and its positive operational realization, proves that conservative homogeneous blocking preserves the relative matching defect and characterizes matching by optimizer composition, develops a nonlinear inhomogeneous unpolarized Gowdy regulator with exact discrete momentum propagation and pathwise full-curvature convergence, and recovers the flat and de Sitter regulators as stationary solutions rather than prescribed rate profiles. Section 6 discusses the exact scope of the result and the remaining selection, exact-action tangency/propagation, and universality questions. Technical reconstruction, variational, representation-theoretic, operational-dynamical, and continuum estimates are collected in Appendices A–I.

2 Predictive reconstruction and the Renewal Grand Tensor

2.1 Operational histories and future equivalence

At a finite cutoff X , the operational entrance is a finite typed event protocol with finite admissible histories, finite intervention and outcome alphabets, and normalized conditional probabilities for every reachable history. The physical intervention letters include tomographically complete frames on each exposed finite operator port; their multitime probabilities are affine under randomization, additive under coarse graining, causal under deterministic discard, and compatible on common prefixes. A distinguished binary orientation is retained as an actual protected operational distinction. No predictive state space, persistent relational carrier, metric, scheduler, connection, or spacetime dimension is part of this entrance.

For a reachable history h ending at cut type x , let $\mathbf{F}_{X,x}$ be the set of all finite admissible future words starting at x . Write $p_X(w \mid h)$ for the corresponding conditional cylinder probability. Define

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \quad \text{for every } w \in \mathbf{F}_{X,x}. \quad (5)$$

Histories of different cut type are never identified. If additional actual terminal Reads or bounded readout queries are present, they are included in the future word alphabet; thus (5) means equality of the complete declared future signature.

Future equivalence is a typed right congruence. Indeed, if $h \sim_X h'$ and an event $a : x \rightarrow y$ is admissible, then the one-letter future gives $p_X(a \mid h) = p_X(a \mid h')$. On a positive-probability branch, the chain rule then gives

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a)$$

for every continuation v . Hence $ha \sim_X h'a$.

2.2 The minimum persistent predictive carrier

Define

$$\boxed{Z_{X,x}^{\min} := \mathbf{H}_{X,x} / \sim_X, \quad q_{X,x} : \mathbf{H}_{X,x} \twoheadrightarrow Z_{X,x}^{\min}.} \quad (6)$$

For an admissible event $a : x \rightarrow y$, the right-congruence property defines

$$\boxed{T_{X,a}([h]) = [ha], \quad \kappa_{X,a}([h]) = p_X(a \mid h).} \quad (7)$$

Thus the event law itself reconstructs a deterministic predictive state machine. If $A_1, A_2, \dots$ is the realized event sequence, $H_n = A_1 \cdots A_n$, and $Z_n = q_X(H_n)$, then

$$\boxed{Z_{n+1} = T_{X,A_{n+1}}(Z_n), \quad \mathbf{p}_{X,n}^{\min} = (Z_n, A_{n+1}, Z_{n+1}).} \quad (8)$$

The persistent relation is therefore the transport of a future-equivalence class under event composition, not an independently supplied state variable.

The quotient is universal. Every reachable deterministic predictive realization $S_{X,x}$ of the same conditional event law admits a unique surjection

$$\pi_{X,x} : S_{X,x} \twoheadrightarrow Z_{X,x}^{\min} \quad (9)$$

intertwining branch probabilities and updates. Consequently every declared downstream readout depending only on the complete future signature factors uniquely through $Z_X^{\min}$. Generated ledgers, hidden-state labels, and other deterministic predictive presentations are therefore representations of the canonical quotient rather than additional foundational coordinates.

The physical geometric branch considered below requires more than this predictive reduction. In particular, same-cut readability of the predictive class, the protected orientation, and the absolute winner-waiting calibration of the fast root race are actual physical rows. They are not determined by the positive predictive quotient. The purpose of (6)–(8) is to remove independently postulated persistence and every future-invisible state refinement.

2.3 Conservative readable relational completion

On a spacetime-capable branch, let $O_{X,n}$ denote the occurring root-bearing event at the n th slow cut, let $\Gamma_{X,n}$ denote the scheduler-separated fast root path between slow cuts, and let $\Xi_{X,n}$ collect the remaining same-history physical marks used by the geometric reconstruction, including the protected orientation and absolute calibrations. A raw readable relation packet is

$$\boxed{\mathbf{p}_{X,n}^{\text{rel}} = (Z_{X,n}^{\min}, \Gamma_{X,n}, O_{X,n}, \Xi_{X,n}, Z_{X,n+1}^{\min}).} \quad (10)$$

The state coordinates in (10) are not new hidden variables: they are same-cut physical realizations of the predictive classes already reconstructed in (6). Two raw packets are identified when every admitted continuation, readout query, and terminal Read has the same conditional law. Denote the resulting finite future-minimal quotient by $\mathfrak{P}_X^{\text{rel},\min}$.

Proposition 2.1 (Conservative readable completion). *Suppose the additional readable marks in (10) are attached to the underlying finite operational process by normalized same-history conditional kernels. Then:*

- (i) *forgetting Γ, Ξ , and the explicit same-cut realization of the predictive class recovers every cylinder probability and branch map of the underlying operational process exactly;*
- (ii) *future-signature refinement terminates at $\mathfrak{P}_X^{\text{rel},\min}$, and every other finite readable packet sufficient for the same geometric reconstruction has a unique surjection onto it;*
- (iii) *unread refinements add neither predictive content nor source dimension; metric, coframe, connection, curvature, and Einstein data remain downstream readout outputs and are not coordinates of the relation packet.*

Thus physical readability is a conservative enrichment of a derived predictive carrier, not a second postulated state space.

The proof is given in Appendix A.3. This proposition is the interface between predictive reconstruction and the geometric branch: the first determines which state distinctions exist, while the second records which of those distinctions and physical calibrations actually occur together on one history.

2.4 The Grand Tensor

The operational multitime table on the physical intervention frames reconstructs a positive causal process tensor. Its support-minimal square-root realization supplies canonical sequential memories, unique up to record-preserving memory unitaries. The represented minimal-state projections, mixed forward maps, and their Hilbert–Schmidt reverses generate a finite history algebra $\mathcal{A}_X^{\text{hist}}$. If $[w]$ denotes the represented history word and $\widehat{\tau}_{\text{reg}}$ is the normalized regular trace, define

$$\mathbb{K}_{X,r}(v, w) = \widehat{\tau}_{\text{reg}}([v]^*[w]), \quad |v|, |w| \leq r. \quad (11)$$

The positive structural Renewal Grand Tensor is

$$\mathbb{T}_{\text{Grand},X} = \left(\mathcal{A}_X^{\text{hist}}, \{ \mathbb{K}_{X,r} \}_{r \geq 0}, Z_X^{\min} \right). \quad (12)$$

It is a tensor in the operational sense of a complete positive moment object: the word-Gram hierarchy records all represented mixed moments, while the history algebra records the multiplication and involution those moments represent. Finite dimensionality forces the represented word hierarchy to become flat after a finite depth, at which point one further word layer reconstructs the complete represented algebra.

For the spacetime branch, (12) is used together with the minimal faithful readable packet $\mathfrak{P}_X^{\text{rel},\min}$ of Proposition 2.1. The distinction is deliberate: $\mathbb{T}_{\text{Grand},X}$ records the canonical positive predictive-history structure, whereas the readable packet records the actual same-history occurrence of orientation, calibration, scheduler, and root-path data required to interpret that structure geometrically. Forgetting the latter returns the positive structural tensor exactly.

Theorem 2.2 (Canonical predictive Grand-Tensor reconstruction). *Let a finite typed operational renewal process have a finite event grammar, normalized prefix-compatible conditional probabilities, tomographically complete physical intervention frames, finite operational positivity, causal trace recursion, and a protected binary orientation as described above. Then:*

- (i) *future equivalence (5) is a typed right congruence, and $Z_X^{\min}$ is the unique coarsest reachable deterministic predictive realization of the event law;*
- (ii) *the successor maps (7) are canonical and the persistent event packet (8) is derived from composition rather than independently postulated;*
- (iii) *the state-conditioned Hankel quotient is the unique reachable, future-separated minimum linear predictor and determines the executable forward algebra;*
- (iv) *the physical probability table reconstructs unique positive causal prefix combs whose square-root supports determine a canonical support-minimal sequential CP class, unique up to minimal-state-preserving unitaries;*
- (v) *the contextual history quotient, normalized regular trace, and positive word-Gram hierarchy are canonical, and finite flatness reconstructs the complete represented monoidal $*$ -algebra;*
- (vi) *every deterministic predictive presentation and every future-operational compiler factors uniquely through $Z_X^{\min}$; consequently (12) is canonical without an independently supplied ledger, hidden state, persistent state space, preferred Kraus frame, primitive metric, or preselected history algebra;*
- (vii) *any finite same-history readable enrichment by normalized conditional kernels admits the minimal faithful quotient $\mathfrak{P}_X^{\text{rel}, \min}$ of (10); its forgetful image is the underlying operational law, so same-cut readability does not alter the predictive reconstruction or introduce new geometric coordinates.*

The detailed proof is given in Appendix A. Three minimizations remain logically distinct: predictive state minimization, linear Hankel minimization, and contextual C^* -history separation. Conflating them either retains future-invisible spectator structure or erases distinctions visible to an actual operational continuation.

The foundational status of the retained operational rows is sharpened in Theorem A.18. Relative to the fixed typed audit language used here, actual occurrence, complete contextual prediction, protected orientation, and absolute physical calibration form a minimal known independent operational basis for the paper-level reconstruction. The remaining calibration freedom is itself classified in Theorem A.19: physically established log-affine incidence relations leave an exact kernel quotient, and one scalar anchor per independent physical quotient direction is necessary and sufficient. These are relative minimality statements for the declared language, not claims of absolute metaphysical minimality.

Remark 2.3 (What the Grand Tensor represents). The Grand Tensor is not an array attached to an already existing geometry. It is the finite positive history object from which the geometry is subsequently read. Its predictive component says which past distinctions remain effective and how they persist under continuation; its algebra says how represented operations compose; and its Gram hierarchy says how those histories overlap in the canonical intrinsic positive form. Geometry becomes possible only after these questions have been answered on one common, future-minimal carrier.

3 Minimal relational geometry and Lorentzian ADM reconstruction

3.1 Two independent finite selectors for the tetrahedral cell

The local dimension is not inferred from a single structural axiom. We use instead two finite selectors with different hypotheses. The first is the logically weaker one and requires no alternating form: exact balance between faithful endpoint and cycle source ranks inside connected simple regular relations uniquely gives K_4 . The second works in the stronger pair-homogeneous simplex category and asks for a nondegenerate alternating coefficient pairing together with minimum faithful source innovation; it gives the same cell. Their agreement is the dimensional statement used below.

Proposition 3.1 (Endpoint–cycle balance selects K_4 without an alternating form). *Let G be a connected simple k -regular relation graph on $v > 1$ endpoints. If its faithful endpoint source and cycle source have equal rank, then*

$$v - 1 = \frac{kv}{2} - v + 1, \quad \text{so} \quad (4 - k)v = 4. \quad (13)$$

The unique nontrivial connected simple regular solution is

$$\boxed{v = 4, \quad k = 3, \quad G \cong K_4.} \quad (14)$$

Thus the associated faithful endpoint module has dimension $v - 1 = 3$. This selector contains no temporal–spatial alternating-form hypothesis.

The proof is included in Appendix B, where the same balance is derived from the cut–cycle complex before the stronger simplex selector is introduced.

For the simplex selector, consider a local relation event on the minimum persistent predictive carrier of Theorem 2.2, with one future-readable source and one future-readable target. Physical reversal turns it into a signed edge. Future minimization removes parallel labels that no future can distinguish, while pair-homogeneous recurrence completes the orbit of one nonzero relation. The relation-colour-free skeleton is therefore K_N . Writing $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$, with $u_0 = N^{-1/2}\mathbf{1}_N$, the signed edge space decomposes as

$$C_1(K_N) = u_0 \wedge W_N \oplus \Lambda^2 W_N. \quad (15)$$

The first summand is the endpoint or cut source and the second the complete cycle or interference source, of dimensions $N - 1$ and $\binom{N-1}{2}$ respectively.

Proposition 3.2 (Alternating/minimal-source simplex selector). *Assume in addition that the temporal–spatial coefficient space $\mathbb{R}t \oplus W_N$ carries a nondegenerate alternating form, that the endpoint and complete signed-edge sources are faithful, that the cycle source is nonzero, and that the future-readable source innovation is minimal among conservative completions satisfying these queries. Then*

$$\boxed{N = 4, \quad G_{\text{rel}} = K_4, \quad W_{\text{sp}} = W_4 \cong \mathbb{R}^3.} \quad (16)$$

Moreover, after complexification,

$$\text{Hom}_{S_N}(\Lambda^2 W_N, W_N \otimes \text{sgn}) = \begin{cases} \mathbb{C}, & N = 4, \\ 0, & N \neq 4, \end{cases} \quad (17)$$

so only at $N = 4$ is the complete cycle representation one oriented copy of the spatial standard module rather than an additional inequivalent source type.

The proof is given in Appendix B. The first conclusion uses only parity, nonzero interference, and source minimality; the representation-theoretic statement is independent evidence for the exceptional economy of $N = 4$ inside the simplex family.

Remark 3.3 (Exact scope of the dimensional claim). The endpoint-cycle balance theorem and the alternating/minimal-source theorem are independent finite classifications with different hypotheses. The first shows that the alternating form is not the sole origin of the three-dimensional spatial module. The second does not become a hidden symplectic or Hamiltonian assumption: its alternating form supplies only an antisymmetric coefficient pairing and no metric, lapse, shift, connection, curvature, or Lorentzian sign. Neither theorem is a classification of arbitrary graphs, hypergraphs, or higher-arity microscopic relation systems.

3.2 The A_3 root race

Choose an anchored basis of W_4 in which the six unoriented A_3 roots are

$$r_1 = e_1, \quad r_2 = e_2, \quad r_3 = e_3, \quad r_4 = e_2 - e_1, \quad r_5 = e_3 - e_1, \quad r_6 = e_3 - e_2. \quad (18)$$

Let $k_a^\pm$ be the two directed rates associated with $\pm r_a$, and let h be the mesh scale. The predictable second and first moments are

$$\mathcal{B}_h = h^2 \sum_{a=1}^6 (k_a^+ + k_a^-) r_a r_a^\top, \quad \beta_h = h \sum_{a=1}^6 (k_a^+ - k_a^-) r_a. \quad (19)$$

The six dyadics $r_a r_a^\top$ form a basis of Sym_3 , so the rate sums reconstruct all six spatial inverse-metric coefficients. The rate differences map with rank three onto β_h ; their remaining three coordinates are circulating root currents. Winner probabilities alone lose the common rate scale. Observing both the winning root and its waiting time gives a faithful score frame for all directed rates.

Remark 3.4 (The missing scale is an exact calibration fibre). More generally, write all positive scales and additive origins in log-affine coordinates $x \in \mathbb{R}^N$. If the physically established relative incidences are $Lx = b$, the remaining calibration class is an affine translate of $\text{Ker } L$; after quotienting conventional changes of base units it is $\text{Ker } L / \text{Im } U$, where $LU = 0$. One scalar anchor per independent quotient direction is necessary and sufficient. The winner-only root race is the simplest instance: common rescaling of all rates is invisible to winner probabilities, and the waiting-time observation supplies the missing rate anchor. The speed density ϱ_h below is a separate physical measure calibration. The complete finite statement is [Theorem A.19](#).

Let $\varrho_h > 0$ be the physical speed density. The spatial metric and lapse are reconstructed by

$$\boxed{g_h = \varrho_h^{2/3} (\det \mathcal{B}_h)^{1/3} \mathcal{B}_h^{-1}, \quad N_h = \varrho_h^{1/3} (\det \mathcal{B}_h)^{1/6}.} \quad (20)$$

Equivalently,

$$\sqrt{\det g_h} = \varrho_h, \quad \mathcal{B}_h = N_h^2 g_h^{-1}. \quad (21)$$

Together with the shift β_h , these are exactly the ten ADM coefficients. The formula is an analytic bijection on the positive cone. A nondegenerate bracket requires total root rate of order h^{-2} ; identifying the geometric scheduler with a slower $O(h^{-1})$ renewal clock would make $\mathcal{B}_h \rightarrow 0$ and collapse the metric.

3.3 From positive predictive geometry to one irreducible signed direction

The Grand Tensor reconstructed in the previous section is positive: its canonical word-Gram hierarchy is a Hilbert-space Gram structure, and the root-race bracket is a positive quadratic form on the three-dimensional spatial module. Neither can contain a negative direction. The additional physical datum is a protected two-valued operational Read

$$\epsilon_X \in \{+1, -1\}, \quad (22)$$

with both values separated by the declared Read policy. In the support-minimal Hilbert realization it has orthogonal readout projections P_+ and P_- , $P_+ + P_- = I$, and therefore

$$\boxed{J := P_+ - P_-, \quad J = J^* = J^{-1},} \quad [u, v]_J := \langle u, Jv \rangle. \quad (23)$$

This is the operator representation of an already occurring $\mathbb{Z}_2$ mark. The microscopic record itself carries no norm, cone, lapse, temporal coordinate, or indefinite metric.

The sign is not reconstructible from the positive packet. The separating models of [Theorem A.18](#) keep the complete positive operational data fixed while reversing the protected orientation, and the obstruction survives common conservative refinement. The signed row is therefore irreducible relative to the declared operational language rather than a reparametrization of the positive Grand Tensor.

Proposition 3.5 (Minimal one-sign extension has Lorentzian inertia). *Let (W, g) be the positive three-dimensional spatial quadratic space reconstructed from the K_4/A_3 root race. Adjoin one independent real line $\mathbb{R}t$ whose protected orientation supplies the opposite sign and leave $g|_W$ unchanged. Then every minimal nondegenerate orthogonal signed extension is, up to multiplication of the whole quadratic form by -1 ,*

$$\boxed{q(\tau t + w) = -N^{-2}\tau^2 + g(w, w), \quad \text{inertia}(q) = (1, 3).} \quad (24)$$

No choice of Clifford representation enters this conclusion.

The proof and the corresponding independence statement are given in [Appendix B](#). The point is that the inertia conclusion precedes the Clifford realization.

Thus the conceptual chain is

$$\boxed{\begin{aligned} \text{predictive collapse} &\longrightarrow \text{positive Grand-Tensor geometry} \longrightarrow K_4/A_3 \text{ spatial geometry} \\ &\longrightarrow \text{one irreducible signed direction} \longrightarrow \text{Lorentzian inertia } (1, 3) \\ &\longrightarrow \text{Clifford realization.} \end{aligned}} \quad (25)$$

The Krein/Clifford layer therefore realizes an already determined signed quadratic structure; it is not the device that selects its inertia.

3.4 Clifford realization of the already determined Lorentzian form

The matrix representation is now introduced only after [Proposition 3.5](#). A minimal depth-two realization uses binary factors A, B to represent the four Clifford axes. The physical sign content remains the single Read [\(22\)](#); the second matrix factor is representation-theoretic multiplicity and is not a second operational time-orientation choice. Set

$$\gamma^0 = -iX_A \otimes I_B, \quad \gamma^1 = Z_A \otimes Z_B, \quad \gamma^2 = Z_A \otimes Y_B, \quad \gamma^3 = Y_A \otimes I_B. \quad (26)$$

They satisfy $(\gamma^0)^2 = -I$, $(\gamma^a)^2 = I$, and $\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 0$ for $\mu \neq \nu$; together they generate $M_4(\mathbb{C}) \cong \text{Cl}_{1,3}(\mathbb{C})$. If $E_h E_h^\top = g_h^{-1}$, the principal symbol

$$\sigma_h(\xi) = \gamma^0 N_h^{-1}(\xi_0 - \beta_h \cdot \vec{\xi}) + \gamma^a (E_h)_a{}^i \xi_i \quad (27)$$

has square

$$\sigma_h(\xi)^2 = \left[-N_h^{-2}(\xi_0 - \beta_h \cdot \vec{\xi})^2 + g_h^{ij} \xi_i \xi_j \right] I. \quad (28)$$

The Clifford cell therefore represents the Lorentzian form fixed in (24); it does not create its one-minus-three-plus signature.

Theorem 3.6 (Minimal faithful 3 + 1 ADM reconstruction from either finite selector). *Let the minimum persistent predictive carrier of Theorem 2.2 admit a geometric branch with one source and one target, physical reversal, faithful endpoint data, and a local relation selected as K_4 by either Proposition 3.1 or the alternating/minimal-source simplex criterion of Proposition 3.2. Assume an occurring scheduler-separated root-race calibration supplies directed winners, their waiting times, and a positive physical speed density. Assume also that the protected orientation packet supplies the Read (22), independent of the positive packet, with one nontrivial signed direction. Then:*

- (i) *the selected local relation cell is K_4 and its spatial module is the A_3 root space $W_4 \cong \mathbb{R}^3$;*
- (ii) *the winner–waiting statistics of the scheduler-separated root race form a faithful coordinate system for all directed rates;*
- (iii) *the moments (19) and the speed density invert uniquely to the ADM spatial metric, lapse, and shift through (20);*
- (iv) *a nondegenerate continuum bracket requires an $O(h^{-2})$ fast scheduler;*
- (v) *positivity of the reconstructed spatial form together with exactly one independent signed direction fixes Lorentzian inertia (1, 3) up to overall sign, before any Clifford representation is chosen;*
- (vi) *the Clifford cell (26) faithfully realizes that already determined quadratic form, and the principal symbol has square (28);*
- (vii) *under summably correctable mixed-Gram and refinement defects, the dimension, source ranks, pair gap, ADM packet, and signed inertia persist on the canonical cofinal limit quotient.*

No persistent predictive state, metric, coframe, connection, curvature tensor, or spacetime dimension is independently supplied. Persistence is reconstructed by Theorem 2.2; the protected orientation and absolute winner–waiting calibration remain irreducible physical input rows on this branch. The alternating form is required only on the second dimensional selector and is not a prerequisite of the first.

The residual status of the last two rows is not merely a limitation of the construction: Theorem A.18 gives explicit separating models showing that neither the protected orientation nor the absolute calibration is functorially reconstructed from the remaining positive operational data, even after common cofinal refinement.

The detailed proof is given in Appendix B. The local algebraic audit is finite: on the free A_3 accumulator, all equal-endpoint word relations are generated by inverse digons, root triangles, and commutator squares of depth at most four. Global topology enters separately through a finite period bank.

4 Renewal action reconstruction and the Einstein equation

The ADM reconstruction of the preceding section is kinematical: it identifies which Lorentzian geometry is encoded by a calibrated renewal packet. We now show that, once the corresponding physical history records actually occur, the dynamical state, action, stationarity test, and spatial compactness screen can be reconstructed or certified from finite renewal data. The only residual hypotheses in the final handoff are therefore genuine positive-branch and continuum-control conditions rather than an independently supplied Einstein action.

4.1 The complete accepted cylinder determines the dynamical state

At cutoff X , let $\tilde{\Theta}_X$ denote the finite same-history record carrying the reconstructed geometric variables used by the gravitational sector, generic protected reward and duration coordinates, and the resolved accepted-branch labels. At this stage the reward coordinates are not named as Palatini, Holst, Einstein–Hilbert, or cosmological terms; they are simply actual finite Reads (or fixed functions of actual protected record coordinates) whose variation content will be reconstructed and classified below. Two values $x, y \in \tilde{\Theta}_X$ are equivalent when every admitted future word and every declared determining coordinate has the same value from x and y , including branch-domain indicators. Put

$$\Theta_{\text{fld},X}^{\min} := \tilde{\Theta}_X / \sim_{\text{fld},X} . \quad (29)$$

For resolved accepted subbranches a , let $p_{a,X}(\theta)$ be their physical probabilities and $u_a^{\min}$ their descended updates. On the positive acceptance domain define

$$K_X^{\text{acc}}(\theta' \mid \theta) = \frac{1}{p_{\text{acc},X}(\theta)} \sum_{a: u_a^{\min}(\theta) = \theta'} p_{a,X}(\theta) . \quad (30)$$

Theorem 4.1 (Determining-field and accepted-kernel reconstruction). *The quotient (29) is the unique coarsest finite record through which every declared determining-field future and every primitive update factors. Finite partition refinement terminates, and (30) is the unique table-native accepted transition reproducing all selected post-accepted field futures.*

Thus the accepted field dynamics is not selected after the fact once the complete accepted cylinder is present. What cannot be reconstructed from the marginal accepted/not-accepted bit is the complete field cylinder itself; this is a physical occurrence requirement rather than an implicit state-space assumption. The proof and the corresponding non-identifiability boundary are given in Appendix C.

4.2 Renewal pressure and reconstruction of a common action

Let (Ω_X, p_X) be a finite faithful marked first-return law, let $\tau_X : \Omega_X \rightarrow (0, \infty)$ be the protected physical duration, and let E_X be the finite variation space. A protected cycle-reward synthesis is a linear map

$$\mathcal{G}_X : E_X \longrightarrow L^2(\Omega_X, p_X), \quad v \longmapsto g_{v,X}, \quad (31)$$

whose values are actual Reads or fixed functions of protected record coordinates. Define

$$\mathcal{L}_X(q, r) = \mathbb{E}_X \left[e^{-g_{q,X} - r\tau_X} \right] . \quad (32)$$

Since $\partial_r \mathcal{L}_X(0,0) = -\mathbb{E}_X \tau_X < 0$, there is a unique analytic pressure $\mathcal{P}_X$ near the origin with

$$\boxed{\mathcal{P}_X(0) = 0, \quad \mathcal{L}_X(q, \mathcal{P}_X(q)) = 1.} \quad (33)$$

Let $S_X v = D \log p_{X,q}|_{q=0}[v]$, let $\pi_X = D\mathcal{P}_X(0)$, and let $T_X c = c\tau_X$.

Theorem 4.2 (Renewal-pressure action reconstruction). *The marked renewal table reconstructs $\mathcal{P}_X$ and all of its finite jets. If $\bar{\tau}_X = \mathbb{E}_X \tau_X$, then*

$$\boxed{\pi_X(v) = -\frac{\mathbb{E}_X g_{v,X}}{\bar{\tau}_X}, \quad \mathcal{G}_X = -S_X - T_X \pi_X.} \quad (34)$$

Hence the centered probability score together with physical duration recovers the complete uncentered action slope, including its affine anchor. The pressure Hessian and duration cross-jets reconstruct the full marginal and mixed action Gram whenever all reward sectors occur on the same marked cycle record.

The remaining question is whether the reconstructed increments belong to one scalar action. Let $\mathcal{V}_X$ be the finite protected variation complex and let $\alpha_X \in C^1(\mathcal{V}_X; \mathbb{R})$ be the reconstructed action increment.

Theorem 4.3 (Finite common-action exactness). *There is a scalar common action $\mathcal{S}_X$ satisfying*

$$\boxed{\alpha_X = d_0 \mathcal{S}_X} \quad (35)$$

if and only if $d_1 \alpha_X = 0$ on every declared 2-cell and the periods of α_X vanish on one integral basis of $H_1(\mathcal{V}_X; \mathbb{Z})$. After one additive anchor is fixed, $\mathcal{S}_X$ is unique and is reconstructed by path integration. Moreover, if P_{har} is the harmonic projection and $\sigma_{\text{coex}} > 0$ is the least singular value of d_1 on the coexact range, then

$$\boxed{\text{dist}(\alpha_X, \text{Ran } d_0)^2 \leq \|P_{\text{har}} \alpha_X\|^2 + \sigma_{\text{coex}}^{-2} \|d_1 \alpha_X\|^2.} \quad (36)$$

Thus failure of a common action is itself a finite positive obstruction, not an unseen modeling choice.

Remark 4.4 (Non-circularity of the gravitational reward). The synthesis $\mathcal{G}_X$ is fixed by the protected cycle record before it is compared with the accepted kernel K_X^{acc} . Neither the transition probabilities nor a desired continuum equation are used to fit a gravitational cost after the fact. Moreover, no Palatini or Holst decomposition is assumed at this stage. That decomposition appears only after derivative-free face linearization, complete resolved-memory subtraction, and the Lorentz-equivariant commutant classification below. The subsequent Bregman/KL test is therefore an independent question: it asks whether the actually accepted dynamics is stationary for the previously reconstructed action.

4.3 Accepted transitions and common-action stationarity

On a compact convex chart $\Theta \subset \mathbb{R}^d$ of the reconstructed determining field, let Ψ and the reconstructed common action $\mathcal{A}$ be C^2 on an open neighbourhood of Θ . The operational potential is strictly convex, with Fisher metric

$$G(\theta) = \nabla^2 \Psi(\theta), \quad mI \preceq G(\theta) \preceq MI, \quad (37)$$

and, for $\lambda, L \geq 0$ and $\eta > 0$,

$$-\lambda G(\theta) \preceq D^2 \mathcal{A}(\theta) \preceq LG(\theta), \quad \eta\lambda < 1. \quad (38)$$

For the Bregman divergence [29]

$$D_\Psi(y, \theta) = \Psi(y) - \Psi(\theta) - \langle \nabla \Psi(\theta), y - \theta \rangle, \quad (39)$$

put

$$\Phi_\theta(y) = \mathcal{A}(y) + \eta^{-1} D_\Psi(y, \theta), \quad y_\theta^* = \arg \min_{y \in \Theta} \Phi_\theta(y). \quad (40)$$

The minimum is unique. On the branch used for unrestricted first variations, y_θ^* lies in the interior of Θ for every supported source θ . This interiority is important: a constrained minimum at a chart boundary need not have zero ordinary action gradient.

For an actually accepted transition $\theta \rightarrow y$, define

$$\Delta_\Psi(\theta, y) = \Phi_\theta(y) - \Phi_\theta(y_\theta^*) \geq 0. \quad (41)$$

If $K = (1 - \vartheta)I + \vartheta K^{\text{acc}}$, with state-independent $0 < \vartheta \leq 1$, has stationary law ν , set

$$\begin{aligned} \bar{\Delta}_\Psi &= \int \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \Delta_\Psi(\theta, y), \\ \mathfrak{S}(\theta) &= \nabla \mathcal{A}(\theta)^* G(\theta)^{-1} \nabla \mathcal{A}(\theta). \end{aligned} \quad (42)$$

Theorem 4.5 (Accepted action gap controls interior stationarity). *Under the preceding hypotheses, including interiority of the supported proximal minimizers, an explicit finite constant $C_{\Psi, \mathcal{A}, \eta}$ satisfies*

$$\int \mathfrak{S}(\theta) \nu(d\theta) \leq C_{\Psi, \mathcal{A}, \eta} \bar{\Delta}_\Psi. \quad (43)$$

In particular, zero gap forces the represented first variation of the common action to vanish ν -almost surely.

The proof in Appendix C keeps the convexity and chart constants explicit. To pass to physical metric tests at varying cutoffs, let $v_{X, K}(k; \theta)$ be the actual represented lift of a fixed determining test k on a compact cylinder K , and put

$$B_{X, K}(k)^2 := \sup_{\theta \in \text{supp } \nu_X} \|v_{X, K}(k; \theta)\|_{G_X(\theta)}^2. \quad (44)$$

Duality then gives the quantitative physical-test estimate

$$\mathbb{E}_{\nu_X} |\delta \mathcal{A}_X[v_{X, K}(k)]|^2 \leq B_{X, K}(k)^2 C_X \bar{\Delta}_{\Psi, X}, \quad C_X = C_{\Psi_X, \mathcal{A}_X, \eta_X}. \quad (45)$$

Thus the right-hand side, rather than the unweighted gap alone, is the quantity required to vanish. On a countable determining core this gives convergence of the first variations in mean square and an almost-sure vanishing subsequence under any common coupling; the deterministic continuum handoff is applied to the resulting realized sequences. The construction and proof are given in Proposition C.9.

For a finite probability row with reference law $p_\theta(y) > 0$ and same-history action cost $c_\theta(y)$, the entropy-proximal row is

$$q_\theta^*(y) = \frac{p_\theta(y) e^{-\eta c_\theta(y)}}{\sum_z p_\theta(z) e^{-\eta c_\theta(z)}}, \quad (46)$$

and its row variational gap is exactly $\eta^{-1} D_{\text{KL}}(K^{\text{acc}}(\cdot \mid \theta) \| q_\theta^*)$. This is an independent likelihood test against the reconstructed cost. Equality of two finite-temperature probability rows does not by itself make every sampled field configuration a critical point of $\mathcal{A}$; the field-action estimate and its physical test lifts remain the stationarity criterion used in the Einstein passage.

4.4 Global noncollapse and compact spatial screens

Let the physical endpoint quotient of a global regulator carry masses $m_{v,X}$, symmetric spatial conductances $c_{e,X}$, physical cell scale h_X , and provenance-safe efficient capacities $u_{e,X}$. The Dirichlet form and its nonnegative self-adjoint Laplacian are

$$\mathcal{E}_X^{\text{sp}}(f) = \sum_{\{u,v\} \in E_X} c_{uv,X} |f(u) - f(v)|^2, \quad \mathcal{E}_X^{\text{sp}}(f) = \langle f, \Delta_X^{\text{sp}} f \rangle_{L^2(\mu_X)}. \quad (47)$$

The endpoint/action packet also contains a finite capacity–conductance comparison. On the positive comparison branch there are cutoff-independent constants $0 < a_- \leq a_+ < \infty$ such that

$$\boxed{a_- h_X c_{e,X} \leq u_{e,X} \leq a_+ h_X c_{e,X} \quad (e \in E_X).} \quad (48)$$

This is an explicit same-history comparison between two reconstructed physical rows; it is not an identification made after taking the continuum limit. Define the efficient and analytic cut constants

$$\boxed{I_X^{\text{eff}} := \min_{0 < \mu_X(A) \leq \mu_X(V_X)/2} \frac{u_X(\partial A)}{\mu_X(A)^{2/3}}, \quad I_X^{\text{sp}} := \min_{0 < \mu_X(A) \leq \mu_X(V_X)/2} \frac{h_X c_X(\partial A)}{\mu_X(A)^{2/3}}.} \quad (49)$$

Then

$$a_- I_X^{\text{sp}} \leq I_X^{\text{eff}} \leq a_+ I_X^{\text{sp}}. \quad (50)$$

Thus a positive efficient-capacity margin is exactly the operational input needed to obtain a positive spatial isoperimetric margin on this branch.

For a nonempty half-volume cut A , write $A^c = V_X \setminus A$ and define the mean-zero demand

$$b_{A,X}(v) = \mu_X(A)^{2/3} \left[\frac{m_{v,X} \mathbf{1}_A(v)}{\mu_X(A)} - \frac{m_{v,X} \mathbf{1}_{A^c}(v)}{\mu_X(A^c)} \right]. \quad (51)$$

This fixes the normalization of the dual cut witness below.

Theorem 4.6 (Cut margin and compact-screen alternative). *Assume the total physical volume is uniformly bounded, $\mu_X(V_X) \leq V_*$, the weighted degree satisfies $\sum_u c_{uv,X} \leq D_* h_X^{-2} m_{v,X}$, and (48) holds. If $\inf_X I_X^{\text{eff}} = I_*^{\text{eff}} > 0$, then $I_X^{\text{sp}} \geq I_*^{\text{eff}}/a_+ =: I_*^{\text{sp}} > 0$ and every mean-zero f obeys regulator-uniform Sobolev and Poincaré bounds*

$$\|f\|_{L^6(\mu_X)}^2 \leq C_S \mathcal{E}_X^{\text{sp}}(f), \quad \|f\|_{L^2(\mu_X)}^2 \leq C_P \mathcal{E}_X^{\text{sp}}(f), \quad (52)$$

with

$$C_S = \frac{128 D_*}{(I_*^{\text{sp}})^2}, \quad C_P = V_*^{2/3} C_S. \quad (53)$$

The low-energy counting function satisfies

$$\boxed{N_X(R) \leq 1 + 4e^{3/2} V_* (C_S R)^{3/2}} \quad (R > 0), \quad (54)$$

and for $P_{X,R} = \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}})$,

$$\boxed{\|(I - P_{X,R})f\|_{L^2(\mu_X)}^2 \leq R^{-1} \mathcal{E}_X^{\text{sp}}(f).} \quad (55)$$

Hence the spectral projections form uniformly exhausted finite-rank compact screens. If instead $I_{X_j}^{\text{eff}} > 0$ tends to zero, the minimizing cuts A_j give normalized potentials ϕ_j with

$$\sum_e u_{e,X_j} |\mathrm{d}\phi_j(e)| = 1, \quad \left| \sum_v b_{A_j,X_j}(v) \phi_j(v) \right| = (I_{X_j}^{\text{eff}})^{-1} \longrightarrow \infty, \quad (56)$$

which is an explicit macroscopic-neck obstruction. If a cut has zero capacity already at finite cutoff, that disconnected-support cut is retained directly, without dividing by its capacity.

The discrete Sobolev–Nash–heat-kernel mechanism behind this theorem is standard in spirit; see, for example, Chung [30, Chs. 10–11] and Saloff-Coste [31]. Because the regulator normalization is part of the present result, Appendix C gives the complete finite proof, including the constants, the $L^1 \rightarrow L^\infty$ heat-semigroup bound, and the derivation of the $R^{3/2}$ counting law.

Thus compact spatial control is derived on the positive cut branch and is replaced by a concrete obstruction on the collapsed branch. Uniform spatial ranks and tails alone do not establish spacetime compactness. The common-cylinder Hodge/action packet also retains convergence of transported finite screens and temporal control of the field coefficients. Appendix C gives the compact-screen upgrade from this packet to the strong convergence needed for quadratic Palatini insertions.

4.5 Lorentz-natural classification of the gravitational score

Let e^I be the reconstructed coframe, ω^{IJ} a metric-compatible reconstructed Cartan connection, $B^{IJ} = e^I \wedge e^J$, and $T^I = De^I$. Before any continuum gravitational Lagrangian is named, the generic protected face reward is subjected to two finite operations described in Appendix C: binary renewal subdivision, which produces a unique first-order linear functional on its exact or summably stable branch, and subtraction of the complete resolved cut-memory log-partition. Denote the resulting generic first-order gravitational score by $A_{X,\text{sh}}^{\text{grav}}$. For later reference, write the three Lorentz-natural first-order densities (up to the overall normalization convention) as

$$L_{\text{Palatini}} = \frac{1}{2} \epsilon_{IJKL} e^I \wedge e^J \wedge R^{KL}, \quad (57)$$

$$L_{\text{Holst}} = e^I \wedge e^J \wedge R_{IJ}, \quad (58)$$

$$L_{\text{vol}} = \frac{1}{4!} \epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L. \quad (59)$$

Theorem 4.7 (Lorentz-natural classification of the generic gravitational score). *Assume $A_{X,\text{sh}}^{\text{grav}}$ is real and Lorentz natural on the represented coframe-bivector/curvature variables, its residual typed commutant on the bivector carrier is scalar, and determinant coframe volume is represented. Then, up to coefficient normalization, its only first-order Lorentz-natural bulk terms are*

$$A_{X,\text{sh}}^{\text{grav}} \in \text{span}_{\mathbb{R}}\{L_{\text{Holst}}, L_{\text{Palatini}}, L_{\text{vol}}\}. \quad (60)$$

Equivalently, there are real coefficients $\alpha_X, \beta_X, \lambda_X$ such that

$$A_{X,\text{sh}}^{\text{grav}} = \alpha_X L_{\text{Holst}} + \beta_X L_{\text{Palatini}} + \lambda_X L_{\text{vol}}. \quad (61)$$

No Palatini, Holst, or Einstein action is part of the hypothesis.

The classification follows because both the coframe bivector and curvature transform in $\Lambda^2 \mathbb{R}^{1,3}$ and $\text{End}_{SO^+(1,3)}(\Lambda^2 \mathbb{R}^{1,3}) = \text{span}_{\mathbb{R}}\{I, \star\}$; the scalar residual commutant excludes any additional internal tensor, and determinant volume is the only independent alternating four-coframe scalar at this order. The full proof is in Appendix C.

4.6 Initial-energy propagation and identified curvature

The remaining analytic issue is curvature compactness. Rather than assume a spacetime-wide curvature bound as an independent hypothesis of the Einstein handoff, we use a positive curvature-energy propagation certificate. This is the finite analogue of the standard Einstein–Bianchi energy method; compare Anderson–Choquet–Bruhat–York [32] and the bounded L^2 curvature framework of Klainerman–Rodnianski–Szeftel [33].

On a compact foliated cylinder $K = [0, T_*] \times \Sigma$, let $y_X(t) = (E_X(t), B_X(t))$ be the actual retained electric/magnetic curvature coefficients in the reconstructed observer frame, with positive physical Gram $M_X(t) = \text{diag}(M_{E,X}, M_{B,X}) \succ 0$. Put

$$z_X(t)^2 := y_X(t)^T M_X(t) y_X(t). \quad (62)$$

The finite Bianchi/transport propagation writer is required to occur in the form

$$\dot{y}_X = (K_X + L_X)y_X + f_X, \quad M_X K_X + K_X^T M_X = 0, \quad (63)$$

where K_X is the positive-Gram skew principal curl block, L_X contains lower-order observer, lapse, connection, and transport terms, and f_X is the complete differentiated curvature source, including any incoming boundary-energy contribution when the cylinder has a timelike spatial boundary. Define

$$a_X(t) := \frac{1}{2} \max \left\{ 0, \lambda_{\max} \left[M_X^{-1/2} (\dot{M}_X + M_X L_X + L_X^T M_X) M_X^{-1/2} \right] \right\}. \quad (64)$$

On the gravitational shared-link branch, f_X contains the physically transferred, volume-normalized Palatini metric row together with the differentiated Cartan and finite Bianchi defects; Appendix C keeps these terms separate.

The principal cancellation is tested on the actual record. For example, a recorded curl $C_X : V_{B,X} \rightarrow V_{E,X}$ has positive-Gram adjoint $C_X^\dagger = M_{B,X}^{-1} C_X^T M_{E,X}$, and the block $\begin{pmatrix} 0 & C_X \\ -C_X^\dagger & 0 \end{pmatrix}$ is exactly M_X -skew. Defining this adjoint does not prove that an actual geometric second curl equals $-C_X^\dagger$: their discrepancy is retained in the source or symmetric-growth record. Consequently no hidden h_X^{-1} principal norm is discarded in the estimate.

Theorem 4.8 (Initial-energy curvature propagation certificate). *Assume (63). Then*

$$\boxed{z_X(t) \leq e^{\int_0^t a_X} \left[z_X(0) + \int_0^t e^{-\int_0^s a_X} \|f_X(s)\|_{M_X(s)} ds \right]}. \quad (65)$$

Suppose moreover that the actual curvature field entering the Palatini test has a uniformly stable interpolation

$$R_X = \mathcal{I}_X y_X + \zeta_X, \quad \|\mathcal{I}_X y_X(t)\|_{L^2(\Sigma_t)} \leq C_I z_X(t), \quad \|\zeta_X\|_{L^2(K)} \leq C_\zeta, \quad (66)$$

and that the lapse is bounded by N_ . If, uniformly in X ,*

$$z_X(0) \leq Z_*, \quad \int_0^{T_*} a_X(t) dt \leq A_*, \quad \int_0^{T_*} \|f_X(t)\|_{M_X(t)} dt \leq F_*, \quad (67)$$

then

$$\boxed{\|R_X\|_{L^2(K)} \leq C_I \sqrt{N_* T_*} e^{A_*} (Z_* + F_*) + C_\zeta}. \quad (68)$$

Thus the curvature-size requirement needed below follows with $p = 2 > 3/2$.

Propagation on a finite time alphabet. The preceding estimate has a fully discrete counterpart, so a differentiable curvature path is not needed merely to propagate its size. At one spatial cutoff, take a partition of the selected time interval with lengths $s_j > 0$. Let $Y_j = M_j^{1/2} y_j$ be the normalized actual curvature records and let T_j be their recorded physical transfer in these coordinates. Suppose

$$Y_{j+1} - T_j Y_j = s_j (K_j + L_j) \frac{Y_{j+1} + T_j Y_j}{2} + \varepsilon_j, \quad (69)$$

where $K_j^T = -K_j$, $(L_j + L_j^T)/2 \preceq a_j I$, $\|T_j\|_{\text{op}} \leq e^{d_j}$, and $a_j, d_j \geq 0$. The residual ε_j includes the complete source, solver and coordinate-transfer errors of the actual step. In particular, nonlinear changes of field coordinates are not assumed to commute with midpoint discretization.

Theorem 4.9 (Fully discrete initial-energy curvature certificate). *Assume $s_j a_j / 2 \leq b < 1$, and set*

$$D_X = \sum_j d_j, \quad A_X^d = \sum_j s_j a_j, \quad \mathcal{D}_X = \sum_j \frac{\|\varepsilon_j\|^2}{s_j}, \quad T_X = \sum_j s_j. \quad (70)$$

Each recorded linear step is uniquely solvable, and

$$\boxed{\max_j \|Y_j\| \leq e^{D_X + A_X^d / (1-b)} \left(\|Y_0\| + \frac{\sqrt{T_X \mathcal{D}_X}}{1-b} \right)}. \quad (71)$$

There is no restriction involving $s_j \|K_j\|_{\text{op}}$. With a uniformly stable interval interpolation of these actual curvature records, a uniform lapse bound, and a bounded reconstruction error as in (66), uniform bounds on the quantities in (71) imply $\sup_X \|R_X\|_{L^2(K)} < \infty$.

The statement concerns the represented linear step with its coefficients fixed. For an implicit nonlinear evolution, existence of that evolution is not inferred from this matrix inverse; its actual coefficient and residual records must satisfy (69). The proof in Appendix C uses the skew cancellation before taking norms and retains the physical transfer between successive Grams.

The source norm is determined by the action-current transfer. The forcing budget can itself be checked against the reconstructed action, provided the actual differentiated current has the required incidence. Write $\mathbf{a}_X = D\mathcal{A}_X$ for its action-gradient covector and suppose

$$f_X = \mathsf{T}_X \mathbf{a}_X + r_X^{\text{src}}. \quad (72)$$

The transfer T_X includes the physical metric-test lift, the nonzero Palatini normalization, trace reversal, the differentiated Bianchi current, and its frame comparison. Torsion, Cartan, boundary and discretization contributions not in this map remain in r_X^{src} .

Proposition 4.10 (Sharp differentiated action-current certificate). *Let $H_X \succ 0$ be the actual source Gram, transferred to the curvature-energy space. The least squared amplification from the action dual norm is*

$$\boxed{\Gamma_X = \|H_X^{1/2} \mathsf{T}_X G_X^{1/2}\|_{\text{op}}^2, \quad \|\mathsf{T}_X \mathbf{a}\|_{H_X}^2 \leq \Gamma_X \|\mathbf{a}\|_{G_X^{-1}}^2}. \quad (73)$$

Consequently a pointwise certificate $\|\mathbf{a}_X(t)\|_{G_X^{-1}}^2 \leq C_X(t) \Delta_X(t)$ gives

$$\int_0^{T_*} \|f_X\|_{H_X} dt \leq \sqrt{2T_* \int_0^{T_*} (\Gamma_X C_X \Delta_X + \|r_X^{\text{src}}\|_{H_X}^2) dt}. \quad (74)$$

For the normalized discrete source $\varepsilon_j/s_j = \mathsf{T}_j \mathbf{a}_j + r_j^{\text{src}}$, the corresponding squared-energy estimate is $\mathcal{D}_X \leq 2 \sum_j s_j (\Gamma_j C_j \Delta_j + \|r_j^{\text{src}}\|_{H_j}^2)$.

This replaces a separately supplied source-size bound by an explicitly scaled action-side bound on the verified transfer branch. It does not turn a small undifferentiated Euler residual into a small Bianchi current: a single centered grid derivative can have $\Gamma_X = h_X^{-2}$. The sharpness example and the expected-energy version of the certificate are proved in Appendix C. An estimate only in expectation yields a bound in probability; almost-sure uniform curvature control additionally uses the pathwise or summable source conditions stated there.

Identification with the actual connection curvature. Curvature size and curvature identification are separate questions. Let $\mathcal{T}_{X,K}$ be the finite determining curvature-test bank, and let $\mathcal{R}_X(\omega_X)$ denote the finite Cartan writer on the common-cylinder interpolation. Define

$$\mathfrak{C}_X(K) := \sup_{\substack{\Phi \in \mathcal{T}_{X,K} \\ \|\Phi\|_{L^2(K)} \leq 1}} \left| \int_K \langle R_X - \mathcal{R}_X(\omega_X), \Phi \rangle \right|. \quad (75)$$

The banks exhaust a fixed countable determining smooth core, with compatible test restrictions and vanishing test-interpolation errors. Their geometric consistency is recorded separately by

$$\mathfrak{I}_X(\Phi) := \int_K \langle \mathcal{R}_X(\omega_X), \Phi \rangle - \langle d_{\text{dist}}\omega_X + \omega_X \wedge \omega_X, \Phi \rangle. \quad (76)$$

The distributional derivative includes every interface contribution of a piecewise connection. Thus matching two finite writers alone is not used to identify the nonlinear curvature of a weakly convergent connection.

Theorem 4.11 (Identified curvature compactness after propagation). *Suppose either Theorem 4.8 or Theorem 4.9 supplies the uniform curvature bound on every compact K . Assume also*

$$\omega_X \rightarrow \omega \text{ strongly in } L_{\text{loc}}^2, \quad \mathfrak{C}_X(K) \rightarrow 0, \quad \mathfrak{I}_X(\Phi) \rightarrow 0 \quad (77)$$

for the determining smooth core. Then every cofinal sequence has a subsequence such that

$$R_X \rightharpoonup R \text{ in } L^2(K), \quad R = d\omega + \omega \wedge \omega \text{ distributionally.} \quad (78)$$

A common subsequence works on a countable compact exhaustion.

For discontinuous connection reconstructions, the identification has a concrete sufficient realization. On a uniformly regular cubical chart put $J_f = n_f^b \wedge [\omega_h]_f$ at each interior face and

$$\mathcal{J}_h^2 = \sum_f h^{-1} \|J_f\|_{L^2(f)}^2. \quad (79)$$

A bounded-overlap volume lifting $\mathcal{L}_h J$ of the face measures gives

$$\begin{aligned} R_h^{\text{lift}} &= d_b \omega_h + \omega_h \wedge \omega_h + \mathcal{L}_h J, \\ \|R_h^{\text{lift}}\|_{L^2(K)} &\leq \|d_b \omega_h\|_2 + C \|\omega_h\|_4^2 + C \mathcal{J}_h. \end{aligned} \quad (80)$$

Its discrepancy from the full distributional curvature is at most $C_K h \mathcal{J}_h \|\nabla \Phi\|_\infty$ on a fixed smooth test. The construction and proof are given in Theorem C.28. These are explicit additional interface budgets, not consequences of a small connection Euler residual. The lifting is used only when the action's actual curvature writer agrees with it on the determining tests, up to a vanishing recorded discrepancy.

Remark 4.12 (Scope of the propagated curvature estimate). The curvature bounds are obtained before the limiting Einstein equation is invoked. They replace an all-time L^p curvature assumption by initial energy, finite propagation and transfer budgets, and a separately identified connection limit. The action-current certificate can discharge the source budget when its scaled errors are controlled; it does not establish an unrestricted nonlinear evolution bound for the remaining coefficients. Flat and de Sitter regulators satisfy the required bounds directly. In the nonlinear Gowdy class below, the actual full-curvature bound and its identification instead follow pathwise from the second-jet estimate on the finite history alphabet.

Two theorem levels. The curvature and compactness machinery above is needed for the most general renewal-to-Einstein handoff, but the Einstein branch is not established only by postulating its eventual certification conditions. We first give a direct nonlinear population theorem. For one explicitly specified local renewal writer, cutoff-uniform energy, consistency, subsidiary propagation, and finite-word realization are proved on a fixed Sobolev neighbourhood. The Einstein limit in this subsection is therefore obtained without taking (E1)–(E5) below as independent hypotheses.

4.7 A constructive open nonsymmetric $3 + 1$ Einstein population

The general certification theorem stated below deliberately separates reconstruction from dynamical propagation. We now exhibit a concrete population on which the continuum conclusion can be proved directly from a cutoff-uniform nonlinear energy, without assuming the five handoff certificates as independent analytic hypotheses. The construction uses the same reconstructed Lorentzian coefficient packet but a declared local harmonic writer. It is therefore a positive population theorem, not a claim that every accepted kernel or the raw finite Euler flow of the reconstructed action must choose this writer.

Let $N = 2m + 1$, $h = N^{-1}$, and $\Lambda_h = (h\mathbb{Z}/\mathbb{Z})^3$. With shifts S_i set

$$D_i^+ = (S_i - I)/h, \quad D_i^- = (I - S_i^{-1})/h, \quad D_i^0 = (D_i^+ + D_i^-)/2. \quad (81)$$

For $g = \eta + q$ in a small Lorentzian chart, write the harmonic normal row as

$$a(g)q_{tt} + 2b^i(g)\partial_i q_t - c^{ij}(g)\partial_i \partial_j q - F(g, \partial g) = 0, \quad a = -g^{00}, \quad b^i = -g^{0i}, \quad c^{ij} = g^{ij}, \quad (82)$$

where F is the analytic first-jet source obtained from the reduced vacuum Einstein tensor. Put

$$\mathsf{K}_b = \sum_i \{M_{bi} D_i^0 + D_i^0 M_{bi}\}, \quad (83)$$

and, with $Y_h = (v, D_1^+ q, D_2^+ q, D_3^+ q)$,

$$\mathsf{G}_h(q, v) = F(g, Y_h) - \sum_{i,j} D c^{ij}(g) [D_i^+ q] D_j^+ q + \sum_i D b^i(g) [D_i^+ q] v. \quad (84)$$

The local evolution law is

$$\boxed{q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^- (c^{ij}(g) D_j^+ q) - \mathsf{K}_b v + \mathsf{G}_h(q, v).} \quad (85)$$

The divergence and skew placements are fixed by the positive quadratic principal energy. The compensator removes exactly the first-derivative terms introduced by these placements, so the

smooth continuum limit is (82). A parity obstruction of the bare $A_3 \cong D_3$ execution graph requires one additional odd-parity execution incidence for a cutoff-uniform Cartesian Sobolev realization; this incidence is used only to execute the writer and is not added to the root race entering (19). The complete proof is in Appendix D. In the main theorem we use the discrete Sobolev norms

$$\|u\|_{H_h^r}^2 = \sum_{|\alpha| \leq r} \|D^\alpha u\|_h^2, \quad \|X\|_{\mathcal{X}_h^r}^2 = \|q\|_{H_h^{r+1}}^2 + \|v\|_{H_h^r}^2, \quad (86)$$

where D^α is formed from the commuting forward differences.

For an integer $s \geq 11$ define the compatible continuum data class

$$\begin{aligned} \mathcal{D}_s(\varepsilon) = \{(\gamma, K) : \gamma \in H^{s+1}(\mathbb{T}^3), K \in H^s(\mathbb{T}^3), \\ R(\gamma) + (\text{tr}_\gamma K)^2 - |K|_\gamma^2 = 0, \quad \nabla_\gamma^i K_{ij} - \partial_j \text{tr}_\gamma K = 0, \\ \|\gamma - I\|_{H^{s+1}} + \|K\|_{H^s} < \varepsilon\}. \end{aligned} \quad (87)$$

This is relative openness in the vacuum constraint set; it is not an ambient open ball of fixed nonzero constraint violations.

Theorem 4.13 (Open nonsymmetric 3+1 renewal–Einstein population). *There are $\varepsilon_s, T_s, h_s, C_s > 0$, independent of the cutoff, with the following properties.*

- (O1) *For every $h < h_s$, every real ten-component initial record $X_h(0) = (q_h(0), v_h(0))$ in the fixed ball $\|q_h(0)\|_{H_h^{s+1}} + \|v_h(0)\|_{H_h^s} \leq \varepsilon_s$ has a unique solution of (85) on $[0, T_s]$, and*

$$\sup_{t \leq T_s} (\|q_h(t)\|_{H_h^{s+1}} + \|v_h(t)\|_{H_h^s}) \leq C_s \|X_h(0)\|_{\mathcal{X}_h^s}. \quad (88)$$

No symmetry, Fourier-support restriction, analyticity, or cutoff-shrinking amplitude is imposed.

- (O2) *Every sufficiently small $(\gamma, K) \in \mathcal{D}_s(\varepsilon_s)$ has a harmonic finite preparation for which the whole cutoff sequence converges on $[0, T_s] \times \mathbb{T}^3$ to a unique metric g with the prescribed initial data. Writing $\varepsilon_0 = \|\gamma - I\|_{H^{s+1}} + \|K\|_{H^s}$, one has*

$$\boxed{G(g) = 0, \quad \sup_{t \leq T_s} (\|g_h - g\|_{H^{s-1}} + \|\partial_t g_h - \partial_t g\|_{H^{s-2}} + \|\partial_t^2 g_h - \partial_t^2 g\|_{H^{s-3}}) \leq C_s h \varepsilon_0.} \quad (89)$$

Here g_h denotes the canonical interpolation of $\eta + q_h$.

- (O3) *The harmonic subsidiary field and Einstein residual obey, at the corresponding lower orders,*

$$\sup_{t \leq T_s} (\|c(g_h)\|_{H^{s-2}} + \|\partial_t c(g_h)\|_{H^{s-3}} + \|G(g_h)\|_{H^{s-3}}) \leq C_s h \varepsilon_s, \quad (90)$$

while the positive A_3 rate lift remains in an interior cone and its spatial cut margin stays uniformly positive. In particular the reconstructed metric is uniformly noncollapsed and its metric curvature is uniformly bounded on the common slab and converges to the curvature of g at the lower Sobolev order supplied by (89).

- (O4) *The differential writer has a genuinely finite renewal realization. With segment duration $\tau_h \asymp h^2$, fixed-radius local residual corrections and finite precision can produce committed successor words whose complete acceleration defect is $O(\varepsilon h^4)$ and whose differentiated defect is $O(\varepsilon h^2)$ on the common slab. Every such certified history inherits the bounds above. This statement concerns finite-range information flow; it does not identify the arithmetic service time with the physical root clock.*

Consequently the positive Einstein population is not confined to homogeneous or one-dimensional inhomogeneous reductions: it contains a relatively open small-data class with arbitrary three-dimensional dependence. The next result shows that the differential writer itself is also not isolated.

Remark 4.14 (Relation to the certification theorem and to exact-action flow). Theorem 4.13 removes the symmetry-reduced and cutoff-shrinking-lifespan limitations for the declared writer, but it is not a universal attraction theorem. Its evolution law is an energy-compatible local realization of the harmonic Palatini envelope; it is not asserted to be the exact finite Euler flow of every reconstructed action cylinder. Appendix F studies that stronger identification separately. On a regular finite exact-action branch the constraint quotient need not already have the local first-class structure of continuum general relativity. Nevertheless a large transverse multiplier block is exactly regular at every odd cutoff, the complete signed response admits an exact positive-reproduction plus scalar–trace decomposition, and selected original- action preparations eliminate the leading weak target while retaining nontrivial three-dimensional data. On a further prepared branch the leading active source is trace-free and uniformly rank-controlled, so the 2/3 trace resonance is absent there at leading order; the actual transverse geometric response is correspondingly suppressed. What remains is dynamical: the prepared exact-action manifold must be tangent to its own flow, the resulting higher signed system must remain regular under refinement, and the estimates must persist on a common physical interval. None of these stronger exact-action conditions is assumed in Theorem 4.13.

4.8 An open renewal-law basin and action-prepared finite seeds

The fixed writer of Theorem 4.13 is the centre of a larger finite law class. Let

$$\Lambda_h = - \sum_{i=1}^3 D_i^- D_i^+, \quad \mathcal{B} = \{B = B^T \in \mathbb{R}^{10 \times 10} : \|B\|_{\text{op}} < b_0\}, \quad (91)$$

where $b_0 > 0$ is fixed below the spatial coercivity margin of the common small metric chart. Thus $\mathcal{B}$ is a 55-dimensional open ball. For $B \in \mathcal{B}$ define

$$\boxed{q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^-(c^{ij}(g)D_j^+ q) - K_b v + G_h(q, v) - h^2 B \Lambda_h^2 q.} \quad (92)$$

The added fourth-order stencil has fixed execution radius. It need not be small in the top ultraviolet vector-field norm; its incidence and sign are instead controlled by the same positive shifted energy.

For a finite chronological realization of (92), let I_j have duration $\tau_j \asymp h^2$. A candidate physical letter e at a live prefix π carries a piecewise-Hermite segment Q_e with actual acceleration defect

$$f_e = Q_e'' - V_{B,h}(Q_e, Q_e'), \quad (93)$$

and, with unit cost assigned to a grammar or execution failure, define

$$c_{h,j}(\pi, e) = \varepsilon^{-2} \int_{I_j} (\|f_e\|_{H_h^s}^2 + \|\partial_t f_e\|_{H_h^{s-3}}^2) dt, \quad C_h = \sum_j c_{h,j}. \quad (94)$$

The cost is determined by the physical letter and its prefix, not by its probability.

Finally let $\mathcal{H}_h(\gamma, \pi, N, \beta)$ be the canonical Hamiltonian obtained from the reconstructed phase-compatible action after stationary connection and Legendre elimination, and set

$$\boxed{\mathcal{C}_h(X) = (-\nabla_N \mathcal{H}_h, \nabla_\beta \mathcal{H}_h) \big|_{N=1, \beta=0}.} \quad (95)$$

This is the original four-row finite initial constraint map; it is not the residual of the local evolution writer.

Theorem 4.15 (Open renewal-law basin with action-prepared finite seeds). *After decreasing the common small-data radius if necessary, the following statements hold with constants independent of the cutoff.*

- (L1) *For every $B \in \overline{\mathcal{B}}$ the law (92) has the same fixed-radius nonlinear lifespan as Theorem 4.13. There is an energy $\mathcal{E}_{s,h}^B$ uniformly equivalent to $\|X\|_{\mathcal{X}_h^s}^2$ such that*

$$(\mathcal{E}_{s,h}^B)' \leq C_s \mathcal{E}_{s,h}^B + C_s (\mathcal{E}_{s,h}^B)^{3/2}. \quad (96)$$

- (L2) *Let $B_h \in \overline{\mathcal{B}}$ be arbitrary, with no convergence of B_h required. For the compatible data of (87), the whole sequence of solutions of (92) converges to the same harmonic vacuum development g as the central writer, and*

$$\sum_{j=0}^3 \|q_{B_h,h}^{(j)} - q_{0,h}^{(j)}\|_{C_t H_h^{s-1-j}} \leq C_s h \|B_h\|_{\text{op}} \varepsilon. \quad (97)$$

Hence the metric, subsidiary, noncollapse and curvature conclusions of Theorem 4.13 are uniform on the entire coefficient ball.

- (L3) *Let $K_{h,j}(e \mid \pi)$ be arbitrary history-dependent finite transition rows satisfying, at every live prefix,*

$$\boxed{\sum_e K_{h,j}(e \mid \pi) c_{h,j}(\pi, e) < M \tau_j h^4.} \quad (98)$$

No independence, invariant starting law, or per-letter small-defect condition is imposed. Then

$$\boxed{\mathbb{E} C_h \leq M T h^4, \quad \mathbb{P}\{C_h > h^2\} \leq M T h^2.} \quad (99)$$

On the complementary event the finite history shadows a member of the differential-law family closely enough to have the same Einstein limit. At each fixed cutoff and grammar the strict conditions (98) form nonempty relatively open convex subsets of the product of row simplexes, and may contain strictly positive rows. Along the odd refinement sequence the exceptional probabilities are summable, so under any common coupling the Einstein conclusion holds eventually almost surely.

- (L4) *For every fixed sufficiently small nonzero preparation amplitude t , the exact map (95) has a finite, nonsymmetric analytic zero-set chart generated from arbitrary small free tensor records W after removal of four fixed balancing modes. Its prepared records $\mathcal{I}_h(t, W)$ satisfy*

$$\boxed{\mathcal{C}_h(\mathcal{I}_h(t, W)) = 0, \quad \|\mathcal{I}_h(t, W)\| \leq C_s |t|.} \quad (100)$$

The chart is relatively open in the complete finite zero set, has codimension $4N^3$, and the linearized constraint map has a right inverse obeying

$$\|\mathcal{R}_{h,t} f\| \leq C_s (\|P_\perp f\| + |t|^{-1} |P_0 f|). \quad (101)$$

No CMC, conformal-flatness, spatial symmetry, finite Fourier support, or continuum-compatible metric is imposed on the free record W . Harmonic completion of these exact finite

records feeds every sequence $B_h \in \overline{\mathcal{B}}$ into the common Einstein population, and the original four lapse-shift rows vanish identically at the initial cut independently of the subsequently chosen acceleration law.

The law-space statement is deliberately quantitative. It gives a fixed open ball of differential laws and relatively open finite probability-row neighborhoods in a physically scaled residual topology; it does not assert a fixed unscaled total-variation ball of arbitrary microscopic kernels or attraction from arbitrary leading-order forces.

Remark 4.16 (The robustness class is nontrivial but not arbitrary). The family in [Theorem 4.15](#) can differ from the central writer by a nonvanishing amount at the highest lattice frequencies. The law parameter is, however, retained during a physical history. Unrestricted rapid switching between two individually stable coefficient marks can parametrically amplify a resolved high-frequency mode. Likewise, a row may be close in ordinary total variation while accumulating order-one failure probability over $O(h^{-2})$ stages. These boundaries explain why the neighborhood is formulated by energy-compatible coefficients and complete physical residual profiles rather than by an unscaled norm on arbitrary kernels. The proofs are in [Appendix E](#).

4.9 Exact-action response, preparation, and dynamical tangency

The constructive laws above are not identified with the raw finite Euler flow of the reconstructed phase-compatible action. That stronger route can be analyzed directly. Let the finite canonical phase have dimension $12V$, let $\lambda = (N, \beta) \in \mathbb{R}^{4V}$ denote lapse-shift multipliers, and let $P_h = \nabla_\lambda H_h$ be the exact multiplier row of the retained finite Hamiltonian. Write

$$F_h^{\text{can}} = \mathbb{J} \nabla_X H_h, \quad B_h = D_X P_h F_h^{\text{can}}, \quad M_h = D_\lambda P_h. \quad (102)$$

The exact constrained evolution satisfies $X_t = F_h^{\text{can}}$, $P_h = 0$, and $B_h + M_h \dot{\lambda} = 0$.

For the flat Cartan tangent let $\mathbf{P}_{\text{tr}}$ denote the orthogonal projection onto scalar spatial matrices and put

$$K_0 = 2I - 6\mathbf{P}_{\text{tr}}, \quad K_0^{-1} = \frac{1}{2}I - \frac{3}{4}\mathbf{P}_{\text{tr}}. \quad (103)$$

For any full-rank finite source inclusion F and actual target b , define

$$\Pi = F(F^*F)^{-1}F^*, \quad w = F(F^*F)^{-1}b, \quad (104)$$

and let E be the isometric inclusion of scalar matrices, $E\tau = \tau I / \sqrt{3}$. Set

$$C = E^* \Pi E, \quad H = 2I - 3C, \quad \zeta = E^* w. \quad (105)$$

These objects distinguish the positive source-reproduction problem from the indefinite Cartan correction.

Theorem 4.17 (Exact-action response and prepared-branch structure). *The exact finite-action system has the following properties.*

- (X1) *On a regular one-null branch the extended constraint manifold has restricted symplectic characteristic quotient dimension*

$$\boxed{12V - 2}. \quad (106)$$

Thus the finite branch is not regularly canonically equivalent to a $12V$ -dimensional system with $4V$ independent local first-class constraints and no additional second-class constraints. A continuum GR identification, if present, must involve a selected or singular cofinal mechanism rather than a finite-cutoff relabeling.

- (X2) On the scalar-momentum branch of Appendix F, the transverse-shift source A_h is injective on the $2(V-1)$ -dimensional transverse space at every odd cutoff and obeys

$$ch^2 \|\Omega_h \beta\|_h^2 \leq \|A_h \beta\|_h^2 \leq Ch^2 \|\Omega_h \beta\|_h^2, \quad A_h^* K_0^{-1} A_h = \frac{1}{2} A_h^* A_h. \quad (107)$$

Consequently these columns can be eliminated exactly from the signed response system without discarding any remaining source equation.

- (X3) For every full-rank remaining source F , the signed Gram $F^* K_0^{-1} F$ is invertible if and only if H in (105) is invertible. On that branch, with $\chi = H^{-1} \zeta$, the unchanged physical response is

$$\boxed{v = w - 3(I - \Pi)E\chi, \quad E^* v = -\chi,} \quad (108)$$

and

$$\boxed{\|v\|^2 = \|w\|^2 + 9\langle \chi, (I - C)\chi \rangle.} \quad (109)$$

Hence $v_h \rightarrow 0$ is equivalent to the two separately measurable conditions $w_h \rightarrow 0$ and $H_h^{-1} E^* w_h \rightarrow 0$. Diagonalizing C recovers the unique trace-angle resonance $\mu = 2/3$ as a corollary, rather than as a separate physical postulate.

- (X4) There are nontrivial exactly prepared original-action families with arbitrary sufficiently fine odd cutoff for which the complete weak quadratic target vanishes identically while the three spatial axial modes remain nonzero. The exact finite constraints admit analytic preparations

$$\boxed{\mathcal{C}_h(X_h(a)) = 0, \quad X_h(a) = aY_h + O(a^2).} \quad (110)$$

A distinct rationally detuned family has, at every sufficiently fine odd cutoff, complete leading active rank $3V - 3$, trace-free active range, and

$$\boxed{(\mathcal{A}_h^{\text{act}})^* K_0^{-1} \mathcal{A}_h^{\text{act}} = \frac{1}{2} (\mathcal{A}_h^{\text{act}})^* \mathcal{A}_h^{\text{act}} \succeq ch^{16} I.} \quad (111)$$

Thus the leading $2/3$ signed trace resonance is absent on that active range. At a verified regular source-neutral seed the exact normalized original-action continuation has literal temporal and spatial link curvature $O_h(|a|)$ at the initial cut.

- (X5) On the selected weak-neutral active family treated in the supplement, the true transverse normal operator has a cutoff-independent spectral interval and the actual direct transverse solution satisfies

$$\boxed{\|\mathfrak{E}_h(\beta_{0,h})\|_{H_h^s} \leq C_s h.} \quad (112)$$

The remaining scalar lapse equation is exactly positive:

$$\boxed{\mathcal{S}_h = \frac{1}{2} \mathcal{D}_h^* \mathcal{D}_h + \frac{h^2}{2} \sigma_h^* K_{Q,h}^{-1} \sigma_h, \quad cI \preceq K_{Q,h} \preceq CI.} \quad (113)$$

The retained flux lifts the bare diagonal soft sector; a large response predicted by the diagonal compression alone is therefore not a theorem about the original full scalar solve.

- (X6) Static response suppression and metric-jet shadowing are distinct. For a C^r family of full-rank sources, differentiation of $H\chi = \zeta$ gives

$$\boxed{\chi^{(j)} = H^{-1} \left[\zeta^{(j)} + 3 \sum_{\ell=1}^j \binom{j}{\ell} C^{(\ell)} \chi^{(j-\ell)} \right].} \quad (114)$$

In particular a small instantaneous electric response does not control its time derivative. In the active/weak amplitude grading, harmonic first-jet shadowing additionally requires the normalized lapse-shift rate to satisfy

$$a^{-1}\|(u_a)_A\|_h \rightarrow 0, \quad a^{-2}\|(u_a)_W\|_h \rightarrow 0. \quad (115)$$

The theorem gives exact finite response structure and nonempty prepared branches, but it does not identify those branches with a cutoff-uniform Einstein evolution. Dynamical invariance of the prepared set is a separate question.

Proposition 4.18 (Exact-action tangency obstruction and upstream deformability). *There exists an exactly prepared nontrivial original-action branch at a fixed finite cutoff for which the complete weak multiplier velocity vanishes at the initial cut and the literal curvature is $O_h(a)$, but the derivative of an explicit weak Poisson-null tangency functional is nonzero. The corresponding exact histories are controlled on every interval $0 \leq t \leq a^3 S$ for fixed S , while no bound of the form*

$$\sup_{0 \leq t \leq T} \|R_{0i0j}(g_h)(a, t)\| \leq Ca \quad (116)$$

can hold uniformly for all sufficiently small a on one fixed positive physical interval T for that family. Altering only the admissible leading multiplier tilt at the same canonical first jet does not remove the leading tangency defect. However, inside the finite canonical first-jet chart there is an exact lower-incidence tangent (Z, m) for which

$$\boxed{D\mathcal{L}[(Z, m)] = 0, \quad D\Delta_\circ[(Z, m)] \neq 0.} \quad (117)$$

Thus the obstruction is attached to the chosen first jet rather than an invariant no-go property of the unchanged finite action. What remains open is an exact regular nonlinear root of the complete transported tangency system, its cofinal continuation, higher signed-response propagation, differentiated curvature control, and a cutoff-independent exact-action lifespan.

The proofs and constructions are collected in Appendix F. The logical boundary is therefore sharper than a comparison between two discretizations. A stable open family of renewal laws is already known to converge to Einstein evolution, while the raw action has an independently understood finite response geometry and a concrete hierarchy of compatibility conditions whose nonlinear cofinal closure remains to be proved.

4.10 General renewal-to-Einstein certification

Theorem 4.19 (General renewal-to-Einstein certification). *The preceding constructive theorem treats one explicit nonlinear writer. For a general complete gravitational renewal cylinder, consider a cofinal sequence of Lorentzian relational regulators of Theorem 3.6. On every compact spacetime cylinder K , define the following five certification conditions on compatible restrictions of the same operational process.*

- (E1) *The complete accepted cylinder and generic protected reward-duration record reconstruct the determining kernel and scalar common action. Its actual first variations tend to zero on a fixed determining metric-test core. The scaled bound (45), with the realized subsequence of Proposition C.9, is a sufficient operational route to this condition. The physical coframe lifts of these tests have uniformly bounded, convergent coefficient tensors in the curvature and volume insertions.*

- (E2) *Complete resolved-memory subtraction and exact or summably stable binary face subdivision give the first-order score classified by [Theorem 4.7](#).*
- (E3) *In a fixed physical action calibration the spacetime-constant coefficients converge, $(\alpha_X, \beta_X, \lambda_X) \rightarrow (\alpha, \beta, \lambda)$, with $\beta \neq 0$ the coefficient of L_{Palatini} . The spinless connection Euler residual tends to zero and the Cartan map is uniformly nondegenerate on the retained coframe chart.*
- (E4) *The endpoint packet satisfies the hypotheses of [Theorem 4.6](#), with $\liminf_X I_X^{\text{eff}} > 0$. Its common-cylinder compact-screen packet, including transported-screen convergence and temporal control, gives a nondegenerate limit with $e_X \rightarrow e$ strongly in local L^2 and $\sup_X \|e_X\|_{L^6(K)} < \infty$.*
- (E5) *The actual curvature records satisfy either the continuous propagation certificate of [Theorem 4.8](#) or the fully discrete certificate of [Theorem 4.9](#), with uniform initial, growth, transfer and source budgets and stable curvature interpolation. The scaled action-current estimate (74) may supply the source budget. The connection and its actual Cartan writer satisfy (77).*

If conditions (E1)–(E5) hold on a cofinal tail, then a subsequence has zero limiting torsion and, for each determining compactly supported inverse-metric test $k^{\mu\nu}$, the metric first variation converges to

$$\boxed{\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad \chi \neq 0.} \quad (118)$$

Consequently

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \quad (119)$$

holds distributionally on the determining test core, and on its closure in the declared test topology.

Remark 4.20 (Certification is not the definition of the constructive branch). Conditions (E1)–(E5) are sufficient recognition conditions for a general complete cylinder: they control action stationarity, first-order gravitational provenance, coupling/torsion, noncollapse/strong coframe compactness, and curvature propagation/identification. They do not define the local writer of [Theorem 4.13](#), and that theorem does not use them as independent assumptions. Conversely, the certification theorem is more general than the constructive population: it applies whenever those finite and cofinal quantities can be verified, regardless of how the microscopic writer is generated.

The nonzero Palatini coefficient is distinct from invertibility of the Palatini–Holst connection operator. A nonzero pure Holst coefficient can eliminate torsion, but its torsion-free metric variation does not supply the Einstein equation. Likewise the propagated curvature bound supplies size, whereas (77) identifies the nonlinear connection curvature. These distinctions prevent either condition from being hidden inside a generic “positive branch.”

4.11 General vacuum Einstein-or-obstruction alternative

Call a cofinal regulator a *complete gravitational renewal cylinder* when the geometric packet, complete accepted determining-field cylinder, generic protected reward-duration record, and physical endpoint, propagation and Cartan records occur on compatible restrictions of one operational process. The records include the physical test lifts, action calibration, actual curvature interpolation and common-cylinder comparison maps used above. The word *gravitational* labels the sector being tested; it does not mean that an Einstein Lagrangian is stored in the entrance record.

Theorem 4.21 (General vacuum Einstein-or-obstruction certification). *For a complete gravitational renewal cylinder, perform the finite reconstructions and residual tests of this section at each cutoff. On a selected cofinal sequence, the following alternative holds.*

- (A1) *If the certificates (E1)–(E5) of Theorem 4.19 pass on a cofinal tail, the sequence has a subsequence with a nontrivial Lorentzian 3 + 1 continuum satisfying*

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \quad (120)$$

distributionally on the determining test core. Curvature size is an output of the initial-energy propagation certificate, not an additional all-time hypothesis.

- (A2) *Otherwise a failed certificate is retained on the same physical records: a common-action curl or period; a nonvanishing realized metric variation or failure of its scaled action-gap bound; a vanishing Palatini normalization or failed coupling calibration; a cut/potential witness; a face-linearity, resolved-memory or torsion defect; an unbounded initial, growth, transfer or differentiated-source budget; an unstable curvature interpolation; a Cartan or interface-identification discrepancy; or a common-cylinder compactness failure. A cofinal failure is witnessed by a subsequence of residuals bounded away from zero, diverging budgets, vanishing positive margins, or escaping compactness tails, as appropriate.*

The alternative is unconditional relative to the complete cylinder and the specified certification procedure. Its second branch is an obstruction to that certificate; it is not a theorem that no Einstein subsequence or other continuum description can exist.

5 Vacuum renewal dynamics and explicit Einstein regulators

The general certification theorem and the open nonsymmetric population of Section 4 establish the field equation and a fixed-radius 3 + 1 realization, respectively. We now study complementary solution-level sectors without inserting an Einstein metric into the operational entrance. On the root-symmetric homogeneous branch we leave the renewal variables arbitrary, derive their full lapse-varied stationarity equations, and only afterwards identify the flat and de Sitter solutions. We then treat a nonlinear spatially inhomogeneous unpolarized Gowdy sector in which the finite evolution preserves a discrete momentum constraint and admits quantitative residual, metric, and action tests. Finally, the homogeneous branch is populated by a finite positive acceptance law whose measured probabilities reconstruct its finite action and whose low-noise refinement converges to the stationary solution.

5.1 Homogeneous renewal coordinates and the full lapse constraint

On the A_3 root-symmetric branch write

$$k_{\alpha,h}(t) = \frac{\kappa(t)}{8h^2}, \quad m_h(t) = \varrho(t)h^3, \quad \kappa(t) > 0, \quad \varrho(t) > 0. \quad (121)$$

The root identity gives $\mathcal{B} = \kappa I_3$ and reversal symmetry gives $\beta = 0$. If $g_{ij} = a^2 \delta_{ij}$, the ADM inversion (20) is therefore exactly

$$\boxed{a = \varrho^{1/3}, \quad N = \varrho^{1/3} \kappa^{1/2}, \quad \varrho = a^3, \quad \kappa = \frac{N^2}{a^2}.} \quad (122)$$

Thus κ is not the scale factor: together with the physical density it contains the lapse information that must remain independent until after variation.

On a compact interval and torus of coordinate volume V_0 , the torsion-free Palatini–volume branch of [Theorem 4.19](#), equivalently its Einstein–Hilbert representative, reduces after the standard Gibbons–Hawking–York Dirichlet boundary completion [\[34, 35\]](#) and division by the nonzero constant χV_0 , to

$$S[a, N] = \int \left[-\frac{6a\dot{a}^2}{N} - 2\Lambda Na^3 \right] dt. \quad (123)$$

Here the induced spatial metric is fixed at the initial and final slices, so $\delta a(t_0) = \delta a(t_1) = 0$. The lapse remains an independent bulk variation and is not gauge-fixed before the Euler equations are taken. The four-dimensional reduction and the exact cancellation of the second-derivative endpoint term by the boundary completion are written out in [Appendix G](#).

Theorem 5.1 (Vacuum Friedmann equations in renewal coordinates). *Let $u = \log \kappa$, $v = \log \varrho$, and $\mathcal{H} = \dot{a}/(Na)$, with $D_\tau = N^{-1}d/dt$. Under [\(122\)](#), the action [\(123\)](#) becomes*

$$S[\kappa, \varrho] = \int \left[-\frac{2\dot{\varrho}^2}{3\varrho^{4/3}\sqrt{\kappa}} - 2\Lambda\varrho^{4/3}\sqrt{\kappa} \right] dt. \quad (124)$$

Its Euler expressions, defined by $\delta S = \int (E_u \delta u + E_v \delta v) dt$, are

$$E_u = Na^3(3\mathcal{H}^2 - \Lambda), \quad E_v = Na^3 \left[4D_\tau \mathcal{H} + \frac{8}{3}(3\mathcal{H}^2 - \Lambda) \right]. \quad (125)$$

Consequently stationarity in the two independent positive renewal variables is equivalent to

$$3\mathcal{H}^2 = \Lambda, \quad D_\tau \mathcal{H} = 0, \quad (126)$$

which is the complete vacuum Einstein equation within the spatially flat homogeneous isotropic ansatz, including the lapse constraint.

Proof. The map $(u, v) \mapsto (\log a, \log N) = (v/3, u/2 + v/3)$ has nonzero Jacobian, so no homogeneous variation is lost. Varying [\(123\)](#) before fixing N gives

$$E_N = 2a^3(3\mathcal{H}^2 - \Lambda), \quad E_a = 6Na^2(2D_\tau \mathcal{H} + 3\mathcal{H}^2 - \Lambda).$$

The chain rule gives $E_u = (N/2)E_N$ and $E_v = (a/3)E_a + (N/3)E_N$, proving [\(125\)](#). The two equations are therefore equivalent to [\(126\)](#). [Appendix G](#) checks directly that these are exactly the 00 and spatial Einstein equations for the reconstructed metric, rather than relying on a symmetry-reduction principle. $\square$

Corollary 5.2 (Flat and de Sitter rates are stationary renewal solutions). *For $\Lambda > 0$ put $H_0 = \sqrt{\Lambda/3}$ and let $\tau(t) = \int^t N(s)ds$. Every positive connected stationary solution of [\(124\)](#) has*

$$a(\tau) = a_0 e^{\sigma H_0(\tau - \tau_0)}, \quad \varrho = a^3, \quad \kappa = N^2/a^2, \quad \sigma \in \{-1, 1\}. \quad (127)$$

After variation, choose proper-time gauge $N = 1$. On the expanding branch,

$$k_{\alpha, h}(t) = \frac{a_0^{-2} e^{-2H_0(t-t_0)}}{8h^2}, \quad m_h(t) = a_0^3 e^{3H_0(t-t_0)} h^3. \quad (128)$$

For $\Lambda = 0$, a is constant; for $\Lambda < 0$ there is no positive real solution in this spatially flat vacuum sector.

The exponential factor in (128) is therefore an output of the stationary renewal equations on this branch, not an input used to obtain those equations. The result does not select Λ , the expanding versus contracting sign, or the initial scale a_0 .

5.2 A finite lapse-varied renewal action

The same distinction can be kept at finite cutoff. The construction is a midpoint variational discretization in the standard sense of discrete mechanics [36], but every equation needed below is derived explicitly. Put $q = a^{3/2} = \sqrt{\varrho}$ and let $s_j = N_j \Delta t_j > 0$ be independent lapse-weighted intervals. With $\bar{q}_j = (q_j + q_{j+1})/2$, $A_0 = 8/3$, and $B_0 = 2\Lambda$, define

$$S_d(q, s) = - \sum_{j=0}^{M-1} \left[A_0 \frac{(q_{j+1} - q_j)^2}{s_j} + B_0 s_j \bar{q}_j^2 \right]. \quad (129)$$

This is the midpoint discretization of the already Gibbons–Hawking–York-completed reduced functional (123), not of the bare second-derivative Einstein–Hilbert expression. The Dirichlet endpoint data are therefore q_0, q_M fixed (equivalently the induced endpoint three-metrics are fixed), while every interior q_j and every lapse-weighted interval s_j are varied independently. Consequently

$$\delta S_d = \sum_{j=1}^{M-1} E_{q_j} \delta q_j + \sum_{j=0}^{M-1} E_{s_j} \delta s_j, \quad \delta q_0 = \delta q_M = 0, \quad (130)$$

with no omitted endpoint acceleration term. If one discretized the uncompleted Einstein–Hilbert expression instead, a matching discrete gravitational boundary term would have to be retained explicitly. Through

$$\varrho_j = q_j^2, \quad N_{j+1/2} = s_j / \Delta t_j, \quad \kappa_{j+1/2} = \frac{(s_j / \Delta t_j)^2}{q_j^{4/3}}, \quad (131)$$

this is a smooth finite-dimensional action in positive renewal coordinates; no exponential history appears in its definition.

Theorem 5.3 (Finite stationary classification and de Sitter convergence). *Let $v_j = (q_{j+1} - q_j)/s_j$. The full Euler equations of (129) are*

$$A_0 v_j^2 - B_0 \bar{q}_j^2 = 0, \quad (132)$$

$$2A_0(v_j - v_{j-1}) - B_0(s_{j-1} \bar{q}_{j-1} + s_j \bar{q}_j) = 0. \quad (133)$$

For $\Lambda > 0$, put $\omega = \sqrt{B_0/A_0} = \sqrt{3\Lambda/4}$. Every positive stationary solution has one common sign $\sigma \in \{-1, 1\}$ and

$$q_{j+1} = q_j \frac{1 + \sigma \omega s_j/2}{1 - \sigma \omega s_j/2}, \quad 0 < \omega s_j/2 < 1. \quad (134)$$

Conversely every such recurrence is stationary. If $\tau_j = \sum_{i < j} s_i$, $T = \sum_j s_j$, $h_t = \max_j s_j$, and $\omega h_t/2 \leq b < 1$, then

$$\max_j \left| \log \frac{q_j}{q_0} - \sigma \omega \tau_j \right| \leq \frac{\omega^3 T}{12(1 - b^2)} h_t^2. \quad (135)$$

Hence $a_j = q_j^{2/3}$, $\varrho_j = q_j^2$, and in proper-time gauge $\kappa_j = q_j^{-4/3}$ converge with second-order logarithmic error to the stationary flat/de Sitter profiles of Corollary 5.2.

The proof, including the lapse variation and the exact rational recurrence, is in Appendix G. In particular, fixing $N = 1$ before variation would lose (132) and admit spurious non-Einstein homogeneous trajectories.

5.3 Measured acceptance law and action–dynamics matching

To relate the finite variational action directly to an operational probability law, we next exhibit a finite positive class in which the action is recovered from the trial probabilities themselves. For $A, B > 0$, $C \in \mathbb{R}$, and $s > 0$, consider

$$E_s(x, y) = A \frac{(y - x)^2}{s} + Bs \left(\frac{x + y}{2} \right)^2 + C(y^2 - x^2), \quad \begin{pmatrix} A & C \\ C & B \end{pmatrix} \succeq 0. \quad (136)$$

Within this declared finite parametric class, three measured costs identify A, B, C . Define the directly measurable rank defect

$$\mathfrak{D} = AB - C^2 \geq 0. \quad (137)$$

For a finite proposal alphabet F , let $p_s(y | x) > 0$ be a normalized proposal row and accept a proposal with probability

$$a_s(x, y) = e^{-E_s(x, y)/\varepsilon}. \quad (138)$$

Write

$$Z_s(x) = \sum_{y \in F} p_s(y | x) a_s(x, y), \quad Q_s(y | x) = \frac{p_s(y | x) a_s(x, y)}{Z_s(x)}. \quad (139)$$

Both Q_s and the total acceptance probability Z_s are entries of the same complete trial table.

Theorem 5.4 (Measured acceptance reconstructs the finite action). *The complete trial table reconstructs the unnormalized cost exactly:*

$$E_s(x, y) = -\varepsilon \log \frac{Z_s(x) Q_s(y | x)}{p_s(y | x)}. \quad (140)$$

For a path with fixed endpoints, the observed writer

$$\mathcal{A}_{\text{obs}} = \varepsilon \sum_j \left[\log \frac{Q_{s_j}(q_{j+1} | q_j)}{p_{s_j}(q_{j+1} | q_j)} + \log Z_{s_j}(q_j) \right] + C(q_M^2 - q_0^2) \quad (141)$$

is identically

$$\mathcal{A}_{\text{obs}} = - \sum_j \left[A \frac{(q_{j+1} - q_j)^2}{s_j} + Bs_j q_j^2 \right] \quad (142)$$

off shell on the parametric domain. Thus every interior-amplitude and lapse first variation agrees exactly with the reconstructed finite action.

The one-step minimizing skeleton has proper-time amplitude drift $c = -C/A$. The bulk action in (142) has $\Lambda_{\text{act}} = 4B/(3A)$, whereas the limiting constant-Hubble geometry of the minimizing skeleton has $\Lambda_{\text{geom}} = 4C^2/(3A^2)$. Consequently

$$G[g] + \Lambda_{\text{act}} g = \frac{4\mathfrak{D}}{3A^2} g. \quad (143)$$

The local operational update and the vacuum parameter of its measured bulk action therefore agree exactly if and only if $\mathfrak{D} = 0$.

The total acceptance probability in (140) is essential: the accepted row Q_s alone is invariant under addition of a source-dependent cost and therefore does not identify the lapse action. The complete trial record removes this ambiguity. On the rank-one branch with $A = A_0$, $B = A_0\omega^2$, and $C = -\sigma A_0\omega$, the cost is the square

$$E_s(x, y) = \frac{A_0}{s} \left[y - x - \sigma\omega s \frac{x+y}{2} \right]^2, \quad (144)$$

so its zero-cost recurrence is exactly (134).

Theorem 5.5 (Finite positive operational family with an Einstein limit). *Fix $T, \omega, q_0 > 0$ and a sign σ . There exists a sequence of genuinely finite accept/reject cylinders with M time stages, finite positive proposal alphabets, calibrated step $s = T/M$, rank-one costs (144), and a finite retry cap such that, for constants independent of M ,*

$$\mathbb{E}\mathcal{E}_M \leq CM^{-4}, \quad \mathbb{E} \max_j |q_j - q_0 e^{\sigma\omega\tau_j}| \leq CM^{-2}, \quad (145)$$

where $\mathcal{E}_M = A_0 \sum_j R_j^2/s$ and $R_j = q_{j+1} - q_j - \sigma\omega s \bar{q}_j$. The expected absolute first variation of (129) against every fixed smooth scale/lapse test is $O(M^{-2})$, including the lapse constraint. The retry-failure probability is at most M^{-6} . Under any common coupling of cutoffs, failures occur only finitely often almost surely and the successful piecewise-affine paths converge strongly in H^1 and uniformly to the stationary solution. Attaching the A_3 root race with

$$\varrho_j = q_j^2, \quad \kappa_j = q_j^{-4/3}, \quad k_{\alpha, h_x, j} = \frac{\kappa_j}{8h_x^2}, \quad m_{h_x, j} = \varrho_j h_x^3 \quad (146)$$

therefore gives positive renewal regulators converging to a vacuum metric with $\Lambda = 4\omega^2/3$.

Appendix G gives the finite grids, temperature schedule, entropy–energy estimate, retry bound, and deterministic residual estimates used in this theorem. The construction proves nonemptiness of a matched operational class; it does not assert that an arbitrary Grand Tensor satisfies the quadratic duration law or the rank-one condition.

5.4 Conservative blocking preserves the action–law defect

The preceding construction raises a distinct stability question: can ordinary time coarse graining drive a neighboring quadratic law toward the matched surface? For the conservative interval elimination of the measured quadratic cost, the answer is no. Assume $0 < s\sqrt{B/A} < 2$ and set

$$\mu = \sqrt{AB}, \quad \theta = 2 \operatorname{artanh}\left(\frac{s}{2}\sqrt{B/A}\right), \quad \rho = \frac{C}{\mu}, \quad \Delta_{\text{rel}} = 1 - \rho^2 = \frac{AB - C^2}{AB}. \quad (147)$$

Then the same cost can be written

$$\mathcal{E}_{\mu, \theta, C}(x, y) = \mu \coth \theta (x^2 + y^2) - 2\mu \operatorname{csch} \theta xy + C(y^2 - x^2). \quad (148)$$

For two successive intervals define conservative blocking by

$$(E_1 \star_0 E_2)(x, y) = \inf_{z \geq 0} \{E_1(x, z) + E_2(z, y)\}. \quad (149)$$

Theorem 5.6 (No transverse matching attraction under conservative blocking). *For every $\theta_1, \theta_2 > 0$,*

$$\boxed{\mathcal{E}_{\mu, \theta_1, C} \star_0 \mathcal{E}_{\mu, \theta_2, C} = \mathcal{E}_{\mu, \theta_1 + \theta_2, C}.} \quad (150)$$

Hence $AB = \mu^2$, C , $\mathfrak{D} = AB - C^2$, and the scale-free mismatch Δ_{rel} are invariant under every finite sequence of homogeneous blockings. For two identical midpoint intervals of duration s , rewritten as one interval of duration $2s$,

$$\boxed{A' = A + \frac{Bs^2}{4}, \quad B' = \frac{AB}{A + Bs^2/4}, \quad C' = C.} \quad (151)$$

Consequently $\Delta_{\text{rel}} > 0$ remains strictly positive: the matched surface $\Delta_{\text{rel}} = 0$ has no open basin of attraction for this conservative blocking operation, even after changes of time/amplitude units or common positive rescaling of the action. On finite positive amplitude alphabets with a symmetric untwisted trial kernel, the endpoint twist C is retained exactly under finite blocking, and a controlled low-temperature/grid limit recovers the same nonzero relative defect.

This is a no-go result for a specified natural coarse-graining operation, not a no-go theorem for every possible renewal renormalization. In particular, a driven or conditioned law may define a different physical flow. The theorem shows that generic Einstein matching cannot be inferred merely from conservative elimination of intermediate amplitudes.

Matching is an exact cross-scale consistency condition. The absence of an attracting flow does not leave the matched surface without an intrinsic test. The same blocking operation characterizes it by asking whether local minimization commutes with the elimination of an intermediate amplitude.

Proposition 5.7 (Optimizer composition characterizes action-law matching). *Let $m_\theta(x)$ minimize $y \mapsto \mathcal{E}_{\mu, \theta, C}(x, y)$ over $y > 0$ for $x > 0$, and use $\rho = C/\mu$ and Δ_{rel} from (147). Then*

$$m_\theta(x) = r_\theta x, \quad r_\theta = (\cosh \theta + \rho \sinh \theta)^{-1}, \quad (152)$$

and, for every $\theta, \phi > 0$,

$$\boxed{r_{\theta+\phi}^{-1} - (r_\theta r_\phi)^{-1} = \Delta_{\text{rel}} \sinh \theta \sinh \phi.} \quad (153)$$

Thus $AB = C^2$ is equivalent to $m_{\theta+\phi} = m_\phi \circ m_\theta$ for every positive pair, and already to that equality for one positive pair at one $x > 0$. On this branch $r_\theta = e^{-\rho\theta}$ with $\rho = \pm 1$.

The proof is in Appendix G. Equation (153) gives a second finite diagnostic of the relative defect, independent of its reconstruction from three cost entries. On compact calibrated ranges away from zero denominators it is stable under small measurement errors. It concerns the minimizing maps of the reconstructed cost; identifying these with observed low-noise updates is a further test of the accepted law. It is a characterization of matching, not a mechanism which makes an arbitrary law approach it.

5.5 A nonlinear inhomogeneous two-polarization vacuum sector

The homogeneous calculation leaves open whether the finite evolution and constraint mechanism survives genuinely spatially varying nonlinear dynamics. A controlled test is provided by the unpolarized Gowdy T^3 vacuum class [37]. On a smooth slab $[t_0, t_1] \times \mathbb{T}^3$, $t_0 > 0$, write

$$g = t^{-1/2} e^{\lambda/2} (-dt^2 + dy^2) + t [e^P (dx + Q dy)^2 + e^{-P} dy^2]. \quad (154)$$

Set

$$a = P_t, \quad b = P_\theta, \quad c = e^P Q_t, \quad d = e^P Q_\theta, \quad (155)$$

and

$$e = \frac{1}{2}(a^2 + b^2 + c^2 + d^2), \quad f = ab + cd. \quad (156)$$

The first-order vacuum system is

$$\begin{aligned} a_t &= b_\theta - a/t + c^2 - d^2, & b_t &= a_\theta, \\ c_t &= d_\theta - c/t - ac + bd, & d_t &= c_\theta + ad - bc, \\ P_t &= a, & Q_t &= e^{-P} c, & \lambda_t &= 2te, \end{aligned} \quad (157)$$

with coordinate and momentum constraints

$$P_\theta = b, \quad e^P Q_\theta = d, \quad \lambda_\theta = 2tf. \quad (158)$$

The identities

$$e_t - f_\theta = -\frac{a^2 + c^2}{t}, \quad f_t - e_\theta = -\frac{f}{t} \quad (159)$$

propagate the last constraint and preserve the zero mean of tf required for periodic λ .

For the variational test the areal gauge is imposed only after variation. With independent fields $(R, \lambda, P, Q, N, \beta)$, $R, N > 0$, and $v_u = \dot{u} - \beta u'$, use the boundary-marked ADM density

$$\begin{aligned} \mathcal{L}_G &= -\frac{v_R(v_\lambda - 4\beta')}{2N} + \frac{R}{2N}(v_P^2 + e^{2P}v_Q^2) \\ &+ N \left[-2R'' + \frac{R'\lambda'}{2} - \frac{R}{2}\{(P')^2 + e^{2P}(Q')^2\} \right]. \end{aligned} \quad (160)$$

Its lapse and shift equations are independent constraints. The solution of (157)–(158) lifts to those equations with $R = t$, $N = 1$, $\beta = 0$; hence the finite action test below is not obtained by first gauge-fixing away the lapse and shift variations.

Theorem 5.8 (Nonlinear two-polarization renewal regulator). *Let $X_*(t)$ be a smooth solution of (157)–(158) on a fixed slab $[t_0, t_1] \times \mathbb{T}^1$, with the spatial derivatives required below uniformly bounded and with the reconstructed metric uniformly nondegenerate. Let $h = \ell$ be the synchronized temporal and spatial mesh. Then there is a nearest-neighbour symmetric source/transport/source splitting with the following properties.*

(G1) *With $U = (u_1, u_2, u_3, u_4) = \sqrt{t}(a, b, c, d)$, $\mathcal{J}(U) = u_1 u_2 + u_3 u_4$, $D_+ = (S_+ - I)/\ell$, and $A_+ = (I + S_+)/2$, the quantity*

$$\boxed{\mathcal{K}_\ell = D_+ \lambda - 2A_+ \mathcal{J}(U)} \quad (161)$$

is preserved exactly by every deterministic full step. The map is second-order consistent on the stated smooth chart.

(G2) *If $X_{j+1} = \Phi_h(t_j, X_j) + r_j$ and*

$$\mathcal{E}_h = \sum_j \frac{\|r_j\|_{2,\ell}^2}{2h}, \quad \mathcal{F}_h = \ell^{-(2r+1)} \mathcal{E}_h, \quad (162)$$

then, under the stated smooth-chart bootstrap or an enforced offset envelope,

$$\boxed{\max_j \|X_j - S_\ell X_*(t_j)\|_{r,\infty,\ell} \leq C(\epsilon_0 + h^2 + \sqrt{\mathcal{F}_h})}. \quad (163)$$

For $r \geq 1$, the accumulated staggered-constraint error is bounded by its initial value plus $C\sqrt{\mathcal{F}_h}$.

- (G3) On the finite offset alphabet with $\|\epsilon_j\|_\infty \leq c_0 h^4$ and initial C_h^1 error $O(h^2)$, the shared tensor-product cubic Hermite readout is pathwise $O(h^2)$ in $W^{1,\infty}$ and $O(h)$ in its second derivatives. Consequently, on compact subslabs,

$$\boxed{\|g_h - g_*\|_{W^{1,\infty}} + \|D^2 g_h - D^2 g_*\|_{L^\infty} + \|\text{Riem}(g_h) - \text{Riem}(g_*)\|_{L^\infty} \leq Ch}, \quad (164)$$

where the first term is in fact $O(h^2)$. Thus $\sup_h \|\text{Riem}(g_h)\|_{L^2(K)} < \infty$ and $\|\text{Ric}(g_h)\|_{L^\infty(K)} \leq Ch$ pathwise. The Levi-Civita curvature of the limiting metric is therefore identified directly.

- (G4) An independent midpoint/centered discretization of (160), with N and β varied independently before restriction to areal gauge, has first-variation residual tending to zero against every fixed smooth local test. A finite positive offset/acceptance realization can be chosen with the support cap in item (G3) and obeys the exact Gibbs entropy-energy bound; its successful histories therefore inherit the pathwise curvature estimate.

The construction and proof are given in Appendix H. It is a nonlinear and spatially inhomogeneous population result, but its scope remains symmetry reduced and local to a smooth positive slab. Unlike the general certificate of Theorem 4.8, the curvature bound in (164) is obtained directly from pathwise second-jet control and requires no separately assumed all-time curvature budget on this class. It does not imply an unrestricted four-dimensional nonlinear Cauchy theorem.

Measured access to the curvature-controlled histories. A finite alphabet with admissible offsets establishes realizability, but does not by itself give a uniform probability of proposing those offsets. The following result separates this operational issue from the deterministic curvature estimate.

Proposition 5.9 (Finite proposal-access certificate). *At stage j , let p_j be the measured finite proposal row and let $\mathcal{G}_j$ contain only proposals satisfying the offset and solver bounds of Theorem 5.8. Suppose $p_j(\mathcal{G}_j) \geq \mu_j > 0$ and $0 \leq E_j \leq c_j$ on this set, uniformly over the supported source histories. Accept with the actual rule $\mathbf{1}_{\mathcal{G}_j} \exp(-E_j/\epsilon_j)$. Then*

$$\mu_j e^{-c_j/\epsilon_j} \leq Z_j \leq p_j(\mathcal{G}_j), \quad D_{\text{KL}}(Q_j \| p_j) \geq -\log p_j(\mathcal{G}_j). \quad (165)$$

With a genuine reset to the same source between retries, caps R_j give

$$\Pr(\text{some stage fails}) \leq \sum_j \exp[-R_j \mu_j e^{-c_j/\epsilon_j}]. \quad (166)$$

Every successful history inherits the pathwise curvature and independent action-test estimates of Theorem 5.8. If the displayed failure bounds are summable along a countable refinement, these conclusions hold on every sufficiently fine successful refinement almost surely.

The proof in Appendix H also gives a sufficient proposal floor from finite launch-share and completed-writer records. A missing safe proposal mass is a genuine obstruction which changing temperature cannot remove. These are bounds on trial counts; an imposed physical deadline must also accommodate execution of the capped trials. The selection rule and its normalizer remain physical records, and its positive deviation cost is not identified with the signed gravitational action.

5.6 Flat and de Sitter geometric regulators

For $\Lambda = 0$, choose the constant stationary branch with $a_0 = 1$. Then

$$\mathcal{B}_h = I_3, \quad \varrho_h = 1, \quad g_h = I_3, \quad N_h = 1, \quad \beta_h = 0, \quad (167)$$

and the graph generators converge to $\frac{1}{2}\Delta_{\mathbb{T}^3}$. The Lorentzian limit is

$$\boxed{(\mathbb{R} \times \mathbb{T}^3, -dt^2 + dx_1^2 + dx_2^2 + dx_3^2).} \quad (168)$$

Its curvature and vacuum Einstein tensor vanish.

For $\Lambda > 0$, choose the expanding stationary branch of Corollary 5.2 with $a_0 = 1$, $t_0 = 0$, and $H = H_0 = \sqrt{\Lambda/3}$. Then

$$\mathcal{B}_h(t) = e^{-2Ht}I_3, \quad \varrho_h(t) = e^{3Ht}, \quad g_h(t) = e^{2Ht}I_3, \quad N_h = 1, \quad \beta_h = 0. \quad (169)$$

The continuum coframe and connection are

$$\vartheta^0 = dt, \quad \vartheta^a = e^{Ht}dx^a, \quad \omega^a{}_0 = H\vartheta^a, \quad \omega^a{}_b = 0. \quad (170)$$

Cartan's equations give zero torsion and constant sectional curvature H^2 , so

$$\boxed{g_H = -dt^2 + e^{2Ht}(dx_1^2 + dx_2^2 + dx_3^2), \quad G_{\mu\nu} + 3H^2(g_H)_{\mu\nu} = 0.} \quad (171)$$

The spatial quotient is globally hyperbolic and is the compact spatial quotient of the flat de Sitter slicing.

Theorem 5.10 (Stationary flat and de Sitter renewal regulators). *For the operational predictive carrier and geometric branch of Theorems 2.2 and 3.6, the stationary homogeneous families above have the following properties.*

- (i) *their rate and density profiles solve the lapse-varied homogeneous renewal Euler system of Theorem 5.1 and arise as the continuum limits of the finite stationary system in Theorem 5.3;*
- (ii) *each spatial cutoff defines a normalized finite A_3 renewal law whose root moments and speed densities reconstruct the displayed ADM variables without a primitive metric;*
- (iii) *the flat sequence converges to (168), with $\Lambda = 0$, while the curved sequence converges to (171), with $\Lambda = 3H^2$;*
- (iv) *on every compact time slab the cut, Poincaré, low-energy counting, and compact-screen margins are uniform in h ;*
- (v) *the flat curvature vanishes identically, while on each compact de Sitter slab the reconstructed curvature and observer/transport coefficients are uniformly bounded; the initial-energy and integrated propagation/source budgets of Theorem 4.8 therefore pass directly, and in fact the curvature is uniformly bounded in L^∞ ;*

- (vi) along $h_n = 2^{-n^2}$ with time mesh h_n and screen $[-n, n]$, exact parallel transport of the reconstructed Levi-Civita connection gives metric-unitary links, while the normalized torsion, plaquette-curvature, and Palatini first-variation defects are $O_T(h_n)$ and summable.

In addition, [Theorem 5.5](#) supplies a nontrivial finite positive accept/reject realization whose measured action and accepted dynamics converge to the same expanding stationary branch.

The detailed homogeneous variational and operational proofs are in [Appendix G](#); the spatial generator, compact-screen, Cartan-curvature, and exact-link estimates are in [Appendix I](#). The result is not an attractor theorem: it does not select Λ , the time orientation, or the quadratic operational class from arbitrary microscopic renewal data.

6 Outlook

The construction developed here suggests a different order for the relation between dynamics and spacetime. In the conventional picture, stochastic or quantum processes take place within an already defined spacetime. Here the logical order is reversed. The microscopic description is a renewal process of operational histories; predictive equivalence determines which distinctions persist; stable relational statistics reconstruct geometry; and only at that stage does a Lorentzian spacetime description become available.

The key conceptual object is therefore not a microscopic position or a pre-existing geometric state, but a distinction that remains effective under future continuation. Two histories are identified when no admissible future can distinguish them. What survives this quotient is the persistent predictive content of the process. In this sense, identity is determined by future consequence rather than by microscopic origin. The relational structure on which geometry is eventually built consists precisely of those distinctions that prediction cannot forget.

This changes the interpretation of the metric. The metric is not attached to the microscopic events as a primitive label, nor is the renewal process assumed to explore an underlying manifold whose geometry already exists. Instead, the ADM variables are reconstructed from collective statistics of the renewal process. The continuum metric is therefore naturally interpreted as a stable large-scale organization of those statistics. Spacetime geometry is not the arena sampled by the microscopic dynamics; it is an effective structure acquired by the dynamics when its predictive relations become sufficiently regular.

The Lorentzian character of this geometry has a correspondingly precise status. The positive predictive structure and the root-race bracket determine a positive spatial geometry, and a positive Gram construction cannot create a negative direction. The protected binary orientation is therefore not a hidden metric assumption but an irreducible signed datum: explicit finite separating models show that it cannot be reconstructed from the other positive operational rows, even after common refinement. Once the spatial module has dimension three, adjoining exactly this one signed direction gives Lorentzian inertia $(1, 3)$ up to global sign. The Clifford cell used in the construction then realizes that already fixed quadratic form; it does not manufacture the signature by representation design.

A second consequence is that emergent spacetime need not stop at kinematics. The constructive and certification results play complementary roles. The open $3 + 1$ theorem proves a cutoff-uniform nonsymmetric Einstein population from a specified finite writer and its local energy estimates; the general certification theorem tests arbitrary complete cylinders by finite/cofinal action, noncollapse, and curvature records. A common action is reconstructed independently from the renewal record, and the observed evolution can then be tested for stationarity with respect to that action. On the

gravitational branch, this variational structure converges to the vacuum Einstein equation. The significance is that Einstein dynamics is not introduced merely because a reconstructed metric happens to resemble a known solution. Geometry and dynamics arise from different layers of the same operational structure and are linked only after their consistency has been tested. The curvature estimate has the same operational character: it propagates an initial energy budget and records how action errors are amplified when they enter the differentiated geometric current. Failure is attached to that explicit comparison, rather than hidden in an unspecified regular continuum limit.

This perspective suggests a relational interpretation of spacetime. Spacetime is not treated as a container that exists before physical events and provides them with positions, distances, and causal relations. The more primitive notion is continuation: some distinctions in a history remain capable of affecting what can happen next, while others become irrelevant. The stable organization of the surviving distinctions defines the relational carrier, and its coarse-grained statistical structure becomes geometric. Smooth Lorentzian spacetime is then the macroscopic form taken by a sufficiently regular predictive renewal process.

The framework should not be read as claiming that every renewal process generates general relativity, or that the present operational starting point is necessarily the deepest possible description. The construction identifies a particular class in which predictive persistence, Lorentzian geometry, and vacuum gravitational dynamics can all be reconstructed without assuming a background spacetime. Other renewal processes may fail to enter this class, or may give rise to different effective structures. Moreover, the conservative homogeneous blocking tested in [Theorem 5.6](#) leaves the scale-free matching defect invariant, so the matched Einstein surface is not an attractor of that operation. Composition of the associated local minimizing maps nevertheless gives an exact test for matching. This separates a consistency principle from a dynamical attraction mechanism: broader universality would require a genuinely different physical coarse-graining or conditioning flow.

The dimensional result should likewise be read as a pair of structural classifications rather than as a universal consequence of positivity. The logically weaker endpoint–cycle balance theorem already singles out K_4 among connected simple regular relations and uses no alternating-form parity condition. The alternating/minimal-source simplex theorem independently selects the same cell, while the representation-theoretic identity $\Lambda^2 W_4 \cong W_4 \otimes \text{sgn}$ explains why only at four endpoints does the complete cycle source become one oriented copy of the spatial source. The agreement of these criteria is the dimensional statement of the paper; no claim is made that arbitrary higher-arity pregeometric categories must satisfy either criterion.

The dynamical evidence is no longer restricted to the homogeneous and Gowdy sectors or to a single local update. The writer of [Theorem 4.13](#) gives a cutoff-independent nonlinear Cauchy interval for all ten metric components and arbitrary three-dimensional dependence in a fixed Sobolev ball, while [Theorem 4.15](#) places that writer inside a fixed open 55-dimensional family with the same whole-sequence Einstein limit. The same result gives relatively open neighborhoods of history-dependent finite transition rows in a complete residual-profile topology and exact finite action-prepared initial constraint charts from free tensor records. Thus the constructive population is robust in differential-law, probability-row and initial-constraint directions, although not under arbitrary leading-order forcing or unrestricted rapid law switching. The homogeneous sector remains useful because its complete lapse variation and finite acceptance law can be written in closed form, while the Gowdy sector supplies a nonlinear large-amplitude symmetry-reduced test with pathwise full-curvature convergence and exact staggered momentum propagation.

A sharper distinction concerns dynamical selection. The stable local writer is an explicitly specified action-compatible realization; it is not identified with the raw exact finite Euler flow.

The exact-action results [Theorem 4.17](#) and [Proposition 4.18](#) and [Appendix F](#) now go substantially beyond a static rank audit. A transverse source block is exactly regular at every odd cutoff, the complete signed response separates into positive source reproduction and a scalar–trace correction, and selected original-action preparations have zero leading weak target with small literal initial curvature. On a further prepared family the active trace resonance is absent and the actual transverse geometric response decays with refinement. The remaining obstruction is dynamical: an exactly prepared branch can fail a higher tangency condition under its own Euler flow, producing a provable initial layer. That defect is not removable by leading multiplier retuning at fixed first jet, but it changes under admissible canonical first-jet deformation. The direct exact-action problem is therefore to find a regular nonlinear root of the complete transported tangency system and propagate its full multiplier/curvature estimates on a cutoff-independent physical interval.

Several broader questions now become natural. The local existence of an open law-space neighborhood is established here; the remaining universality question is how far that neighborhood can be enlarged, whether a physical coarse-graining or conditioning flow selects it from more general microscopic laws, and whether the independently reconstructed exact action admits a cofinal regular tangency branch whose own flow enters the same Einstein universality class. A second question is how matter and gauge degrees of freedom should enter the same predictive framework, and whether their effective actions and couplings can likewise be reconstructed rather than imposed. A third concerns residual memory: the microscopic theory need not be Markovian, while the continuum regimes considered here become effectively local after predictive compression. Understanding controlled departures from this limit may provide a way to characterize non-Markovian corrections to ordinary local field theory.

There is also a more basic question upstream of the present formalism. The construction begins from histories, but its canonical quotient suggests that histories themselves may not be the most fundamental objects. What matters is whether a distinction changes the space of possible futures. This points toward a more primitive notion of *effective difference under continuation*: distinctions are physically retained insofar as they continue to affect subsequent possibilities, and are identified once their future effects become identical. Renewal history is then a way of organizing these effective differences, while spacetime is the stable geometry they acquire at large scale.

Within the scope established here, the conceptual chain is therefore

$$\begin{array}{l} \text{renewal histories} \longrightarrow \text{predictive persistence} \longrightarrow \text{relational structure} \\ \hspace{10em} \longrightarrow \text{Lorentzian geometry} \longrightarrow \text{gravitational dynamics.} \end{array}$$

The central conclusion is not that spacetime has been eliminated from physics, but that it need not occupy the most fundamental level of description. Lorentzian geometry and vacuum Einstein dynamics can instead appear as the stable large-scale organization of a more primitive predictive process. Spacetime is then not the container in which renewal occurs, but the form that renewal acquires when persistent distinctions, orientation, and dynamics become coherent.

Data and formalization availability

Selected finite algebraic components of the construction have been formalized in [Lean](#). Source code is available at <https://github.com/AI-MathPhys/renewal-geometry>. The analytic continuum arguments are those given in the present article and its appendices.

References

- [1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* **59** (1987), 521–524.
- [2] S. Surya, The causal set approach to quantum gravity, *Living Reviews in Relativity* **22** (2019), 5.
- [3] T. Konopka, F. Markopoulou, and S. Severini, Quantum graphity: a model of emergent locality, *Physical Review D* **77** (2008), 104029.
- [4] J. Ambjørn, J. Jurkiewicz, and R. Loll, Quantum gravity, or the art of building spacetime, in *Approaches to Quantum Gravity*, D. Oriti, ed., Cambridge University Press, Cambridge, 2009, pp. 341–359.
- [5] D. Oriti, ed., *Approaches to Quantum Gravity: Toward a New Understanding of Space, Time and Matter*, Cambridge University Press, Cambridge, 2009.
- [6] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.
- [7] T. Jacobson, Thermodynamics of spacetime: the Einstein equation of state, *Physical Review Letters* **75** (1995), 1260–1263.
- [8] E. Verlinde, On the origin of gravity and the laws of Newton, *Journal of High Energy Physics* **2011** (2011), no. 4, 029.
- [9] M. Van Raamsdonk, Building up spacetime with quantum entanglement, *General Relativity and Gravitation* **42** (2010), 2323–2329.
- [10] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *Journal of Statistical Physics* **104** (2001), 817–879.
- [11] N. F. Travers and J. P. Crutchfield, Equivalence of history and generator ϵ -machines, *Physical Review E* **89** (2014), 062107.
- [12] S. E. Marzen and J. P. Crutchfield, Structure and randomness of continuous-time, discrete-event processes, *Journal of Statistical Physics* **169** (2017), 303–315.
- [13] A. M. Jurgens and J. P. Crutchfield, Divergent predictive states: the statistical complexity dimension of stationary, ergodic hidden Markov processes, *Chaos* **31** (2021), 083114.
- [14] W. Feller, *An Introduction to Probability Theory and Its Applications, Vol. II*, Wiley, New York, 1971.
- [15] A. M. Garsia and J. Lamperti, A discrete renewal theorem with infinite mean, *Commentarii Mathematici Helvetici* **37** (1962), 221–234.
- [16] K. B. Erickson, Strong renewal theorems with infinite mean, *Transactions of the American Mathematical Society* **151** (1970), 263–291.
- [17] F. Caravenna and R. Doney, Local large deviations and the strong renewal theorem, *Electronic Journal of Probability* **24** (2019), Paper No. 72, 1–48.

- [18] W. F. Stinespring, Positive functions on C^* -algebras, *Proceedings of the American Mathematical Society* **6** (1955), 211–216.
- [19] M.-D. Choi, Completely positive linear maps on complex matrices, *Linear Algebra and its Applications* **10** (1975), 285–290.
- [20] A. Strohmaier, On noncommutative and pseudo-Riemannian geometry, *Journal of Geometry and Physics* **56** (2006), 175–195.
- [21] N. Franco, Temporal Lorentzian spectral triples, *Reviews in Mathematical Physics* **26** (2014), 1430007.
- [22] N. Franco and M. Eckstein, An algebraic formulation of causality for noncommutative geometry, *Classical and Quantum Gravity* **30** (2013), 135007.
- [23] K. van den Dungen, M. Paschke, and A. Rennie, Pseudo-Riemannian spectral triples and the harmonic oscillator, *Journal of Geometry and Physics* **73** (2013), 37–55.
- [24] K. van den Dungen, Krein spectral triples and the fermionic action, *Mathematical Physics, Analysis and Geometry* **19** (2016), 4.
- [25] A. Devastato, S. Farnsworth, F. Lizzi, and P. Martinetti, Lorentz signature and twisted spectral triples, *Journal of High Energy Physics* **2018** (2018), no. 3, 089.
- [26] A. Devastato, M. Filaci, P. Martinetti, and D. Singh, Actions for twisted spectral triple and the transition from the Euclidean to the Lorentzian, *International Journal of Geometric Methods in Modern Physics* **17** (2020), suppl. 1, 2030001.
- [27] G. Nieuviarts, Emergence of time from a twisted spectral triple in almost-commutative geometry, *Journal of Noncommutative Geometry* (2026), to appear; arXiv:2512.15450.
- [28] G. Nieuviarts, Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple, *International Journal of Geometric Methods in Modern Physics* (2026), to appear; arXiv:2502.18105.
- [29] L. M. Bregman, The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming, *USSR Computational Mathematics and Mathematical Physics* **7** (1967), 200–217.
- [30] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, vol. 92, American Mathematical Society, Providence, RI, 1997.
- [31] L. Saloff-Coste, *Aspects of Sobolev-Type Inequalities*, London Mathematical Society Lecture Note Series, vol. 289, Cambridge University Press, Cambridge, 2002.
- [32] A. Anderson, Y. Choquet-Bruhat, and J. W. York, Jr., Einstein–Bianchi hyperbolic system for general relativity, arXiv:gr-qc/9710041, version 2 (1998).
- [33] S. Klainerman, I. Rodnianski, and J. Szeftel, The bounded L^2 curvature conjecture, *Inventiones Mathematicae* **202** (2015), 91–216.
- [34] J. W. York, Jr., Role of conformal three-geometry in the dynamics of gravitation, *Physical Review Letters* **28** (1972), 1082–1085.

- [35] G. W. Gibbons and S. W. Hawking, Action integrals and partition functions in quantum gravity, *Physical Review D* **15** (1977), 2752–2756.
- [36] J. E. Marsden and M. West, Discrete mechanics and variational integrators, *Acta Numerica* **10** (2001), 357–514.
- [37] V. Moncrief, Global properties of Gowdy spacetimes with $T^3 \times \mathbb{R}$ topology, *Annals of Physics* **132** (1981), 87–107.

A Operational reconstruction and positive history algebra

This section proves [Theorem 2.2](#). All operational alphabets and exposed operator spaces are finite-dimensional. The construction begins from histories and conditional future statistics; no generated ledger or persistent state space is supplied at the entrance.

A.1 Finite typed operational renewal datum

Definition A.1 (Finite typed event grammar). At cutoff X , a finite typed event grammar consists of:

- (E1) a finite set of cut types $\mathbf{O}_X$;
- (E2) finite sets $\Sigma_X(x, y)$ of primitive event letters $a : x \rightarrow y$;
- (E3) a prefix-closed set $\mathbf{H}_{X,x}$ of reachable finite typed histories ending at each cut type x , with typed concatenation and empty words;
- (E4) finite preparation, intervention-outcome, and terminal-Read alphabets, together with finite exposed operator ports;
- (E5) a distinguished protected binary orientation record whose two values are separated by the declared physical Read policy.

No state label, persistent relation object, metric, scheduler, connection, or spacetime semantic is assigned to a history.

For $h \in \mathbf{H}_{X,x}$ let $\Sigma_{X,h}$ be the event letters admissible after h .

Definition A.2 (Finite conditional operational renewal law). A finite conditional operational renewal law on [Definition A.1](#) assigns

$$p_X(a \mid h) \in [0, 1], \quad \sum_{a \in \Sigma_{X,h}} p_X(a \mid h) = 1, \quad (172)$$

for every reachable history. For an admissible future word $w = a_1 \cdots a_k$ define

$$p_X(w \mid h) = \prod_{j=1}^k p_X(a_j \mid ha_1 \cdots a_{j-1}), \quad p_X(\epsilon_x \mid h) = 1. \quad (173)$$

The physical intervention letters include, on every exposed Hermitian Choi-coordinate space, a finite physical CP frame whose real span is the full Hermitian space. The induced multitime frame probabilities are affine under randomization, additive under coarse graining, normalized on complete instruments, causal under deterministic discard, compatible on common prefixes, and satisfy the finite positivity conditions of [Theorem A.14](#). Actual bounded terminal Reads, marks, and readout queries may be included as terminal event outcomes; equivalently they may be appended to the future test bank.

The distinguished renewal semantics used in the geometric branch is carried by the admitted completion/return event family. No ordinary reset state is needed for the predictive construction below; if a stronger reset law is imposed, it is an additional property of the same event process rather than a state-space axiom.

A.2 Future equivalence and the derived persistent carrier

Definition A.3 (Complete predictive future signature). For $h \in \mathbf{H}_{X,x}$ define

$$\Sigma_X(h) = (p_X(w \mid h))_{w \in \mathbf{F}_{X,x}}, \quad (174)$$

where $\mathbf{F}_{X,x}$ is the complete finite future-test bank starting at x . Histories $h, h' \in \mathbf{H}_{X,x}$ are future equivalent when

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every } w \in \mathbf{F}_{X,x}. \quad (175)$$

Histories of different cut type are never identified.

Theorem A.4 (History-level right congruence). *Future equivalence is a typed right congruence. If $h \sim_X h'$ and $a : x \rightarrow y$ is admissible, then*

$$p_X(a \mid h) = p_X(a \mid h'), \quad (176)$$

and on the positive-probability branch

$$ha \sim_X h'a. \quad (177)$$

Proof. Use the one-letter future a in (175). If the common probability is positive, then for any continuation v the chain rule gives

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a).$$

Hence the complete future signatures agree after appending a . $\square$

Definition A.5 (Minimum predictive state). For every cut type x put

$$Z_{X,x}^{\min} := \mathbf{H}_{X,x} / \sim_X, \quad q_{X,x} : \mathbf{H}_{X,x} \rightarrow Z_{X,x}^{\min}. \quad (178)$$

For an admissible event $a : x \rightarrow y$ define

$$T_{X,a}([h]) = [ha], \quad \kappa_{X,a}([h]) = p_X(a \mid h). \quad (179)$$

Corollary A.6 (Derived predictive state machine). *The maps in (179) are well defined. If $w = a_1 \cdots a_k$, $z_0 = q_X(h)$, and $z_j = T_{X,a_j}(z_{j-1})$, then*

$$p_X(w \mid h) = \prod_{j=1}^k \kappa_{X,a_j}(z_{j-1}). \quad (180)$$

Thus the predictive state machine is reconstructed from the conditional event law.

Proof. Well-definedness is Theorem A.4; the product formula is (173) written on future-equivalence classes. $\square$

Definition A.7 (Deterministic predictive realization). A reachable deterministic predictive realization consists of finite sets $S_{X,x}$, history maps $s_x : \mathbf{H}_{X,x} \rightarrow S_{X,x}$, partial updates $u_a : S_{X,x} \rightarrow S_{X,y}$, and branch functions $r_a : S_{X,x} \rightarrow [0, 1]$ such that on every reachable branch

$$s_y(ha) = u_a(s_x(h)), \quad p_X(a \mid h) = r_a(s_x(h)). \quad (181)$$

Theorem A.8 (Canonical minimum predictive realization). *The quotient $Z_X^{\min}$ is the unique coarsest reachable deterministic predictive realization. For every reachable realization S_X there is a unique surjection*

$$\boxed{\pi_{X,x} : S_{X,x} \twoheadrightarrow Z_{X,x}^{\min}} \quad (182)$$

such that

$$q_x = \pi_x s_x, \quad \pi_y u_a = T_{X,a} \pi_x, \quad r_a = \kappa_{X,a} \pi_x. \quad (183)$$

Consequently a generated ledger or hidden-state presentation, when present, is only an intermediate predictive realization and has a unique quotient onto $Z_X^{\min}$.

Proof. If $s_x(h) = s_x(h')$, induction on future-word length using (181) gives $p_X(w | h) = p_X(w | h')$ for every future word w . Hence $h \sim_X h'$, so $\pi_x(s_x(h)) := [h]$ is well defined. Reachability makes π_x unique and surjective. The intertwining identities follow from the definitions, and the quotient itself is a realization by Corollary A.6. $\square$

Theorem A.9 (Persistence derived from composition). *Let $A_1, A_2, \dots$ be the realized event sequence, let $H_n = A_1 \cdots A_n$, and put $Z_n = q_X(H_n)$. Then on every realized branch*

$$\boxed{Z_{n+1} = T_{X,A_{n+1}}(Z_n).} \quad (184)$$

Consequently

$$\boxed{\mathfrak{p}_{X,n}^{\min} = (Z_n, A_{n+1}, Z_{n+1})} \quad (185)$$

is a persistent relational event packet derived from the event process. Every other deterministic persistent predictive packet maps uniquely onto (185).

Proof. By definition, $Z_{n+1} = q_X(H_n A_{n+1}) = T_{X,A_{n+1}} q_X(H_n)$. For any other predictive realization, apply the universal map of Theorem A.8 at consecutive cuts. $\square$

Corollary A.10 (Future-operational transfer). *Let $\mathcal{D}$ be any downstream finite readout whose value is unchanged whenever two candidate states have the same complete declared future signature. Then there is a unique $\overline{\mathcal{D}}$ on $Z_X^{\min}$ such that*

$$\boxed{\mathcal{D} = \overline{\mathcal{D}} \circ \pi_X.} \quad (186)$$

Thus deleting every future-null state coordinate preserves such downstream outputs exactly.

Proof. The readout is constant on the fibres of the universal surjection (182); define $\overline{\mathcal{D}}$ on a fibre by the common value. Surjectivity gives uniqueness. $\square$

A.3 Conservative readable relational completion

The predictive quotient above determines persistence but does not by itself assert that the quotient class, scheduler path, orientation, and calibration are jointly readable at one cut. We now isolate the finite conservative extension used by the geometric reconstruction.

Let $\Phi_{a,*}$ be the support-minimal Schrödinger-picture branch map for an underlying event letter a . Let $z \in Z_X^{\min}$ be its predictive source class. For a finite family of added same-history marks $m = (\gamma, \xi, z')$, where γ is a scheduler-path record, ξ contains orientation and calibration marks, and z' is the next readable predictive class, let

$$K_{X,a}(m | z) \geq 0, \quad \sum_m K_{X,a}(m | z) = 1. \quad (187)$$

On block states $\rho = (\rho_z)_z$, define

$$\boxed{[\tilde{\Phi}_{a,m,*}(\rho)]_{z'} = \sum_z K_{X,a}(m \mid z) \Phi_{a,*}(\rho_z).} \quad (188)$$

The symbol in (188) denotes a same-history conditional extension, not a tensor-product decomposition of the state space.

Theorem A.11 (Conservativity and future minimality of the readable packet). *For the finite extension (188):*

- (C1) *every refined branch is completely positive and the refined family is normalized;*
- (C2) *if $F_X(\rho) = \sum_z \rho_z$ forgets the explicit predictive block and all added marks, then*

$$\boxed{F_X \left(\sum_m \tilde{\Phi}_{a,m,*}(\rho) \right) = \Phi_{a,*}(F_X \rho);} \quad (189)$$

hence every underlying cylinder probability and branch map is recovered exactly after forgetting the readable enrichment;

- (C3) *quotienting raw readable packets by equality of all admitted future continuations, readout queries, and terminal Reads terminates at a unique coarsest finite right congruence $\mathfrak{P}_X^{\text{rel},\min}$;*
- (C4) *every other finite readable presentation sufficient for the same future readout has a unique surjection onto $\mathfrak{P}_X^{\text{rel},\min}$, so unread refinements add no predictive content or source dimension.*

Consequently the readable relation packet is a conservative physical realization of the already derived predictive carrier, not an independently postulated state space.

Proof. Complete positivity follows because every target block in (188) is a nonnegative scalar combination of completely positive branch maps. Summing (187) over the added marks gives one, and therefore summing the refined branches proves (189). Induction on word length gives exact preservation of every underlying finite cylinder.

For future minimality, define two raw readable packets to be equivalent when all declared future signatures agree. Prefixing the same admissible event to two equal future signatures leaves them equal, so the relation is a typed right congruence. Finite partition refinement terminates because every strict round increases the number of classes. The usual fibre argument then gives its universal property: any other sufficient readable presentation identifies only packets with equal complete future signatures and therefore surjects uniquely onto the stable quotient. $\square$

Remark A.12 (What the completion adds). Same-cut readability, scheduler records, protected orientation, and absolute calibration are actual occurrence rows. The predictive class appearing in the packet is nevertheless the quotient already derived from the operational law. The conservative completion therefore separates physical incidence from predictive persistence without inserting metric, coframe, connection, curvature, or spacetime dimension.

A.4 The state-conditioned Hankel quotient

Fix a minimum predictive state sector (x, z) . Let $\mathbf{P}_{x,z}$ and $\mathbf{F}_{x,z}$ denote the finite pasts and futures meeting at that sector, and set

$$\mathfrak{l}_{x,z}(f, p) = \Pr(f \circ p). \quad (190)$$

For each past p define $h_p(f) = \mathfrak{l}_{x,z}(f, p)$ and

$$M_{x,z} = \text{span}_{\mathbb{R}}\{h_p : p \in \mathbf{P}_{x,z}\}. \quad (191)$$

Theorem A.13 (Canonical typed Hankel realization). *For every primitive event $a : (x, z) \rightarrow (y, T_{X,a}z)$ there is a unique linear map*

$$A_a : M_{x,z} \longrightarrow M_{y,T_{X,a}z}, \quad A_a h_p = h_{a \circ p}. \quad (192)$$

Every future f defines $\lambda_f(m) = m(f)$ and

$$\mathfrak{l}(f, p) = \lambda_f(h_p), \quad \lambda_f A_a = \lambda_{f \circ a}. \quad (193)$$

The realization is reachable and future separated. For any other linear realization $N_{x,z}$,

$$N_{x,z}^{\text{reach}} / N_{x,z}^{\text{unobs}} \cong M_{x,z}; \quad (194)$$

therefore every linear realization has dimension at least $\dim M_{x,z}$, with equality exactly on the reachable future-separated branch.

Proof. If $\sum_j c_j h_{p_j} = 0$, then for every target future f ,

$$\sum_j c_j h_{a \circ p_j}(f) = \sum_j c_j \mathfrak{l}(f \circ a, p_j) = 0.$$

Hence appending a respects every column relation and (192) is well defined. Reachability is built into (191); every nonzero response function is detected by some future. For another realization, map its reachable state n to the function $f \mapsto \mu_f(n)$. The image is $M_{x,z}$ and the kernel is exactly the future-unobservable subspace, proving (194). $\square$

The maps A_a generate the table-native executable forward algebra. A Hilbert adjoint is not introduced at this stage; it is selected only after the positive process tensor has fixed the support-minimal Hilbert realization.

A.5 Comb tomography and support-minimal sequential realization

For each exposed Hermitian Choi space choose a real spanning physical frame $\{F_\alpha\}$ and a dual frame $\{D^\alpha\}$ satisfying $\text{Tr}(D^\alpha F_\beta) = \delta_\beta^\alpha$. For an N -step table define

$$R^{(N)} = \sum_{\alpha_1, \dots, \alpha_N} p(\alpha_1, \dots, \alpha_N) D^{\alpha_1} \otimes \dots \otimes D^{\alpha_N}. \quad (195)$$

Theorem A.14 (Finite process-comb tomography). *The tensor $R^{(N)}$ in (195) is independent of the chosen spanning physical frame and is the unique Hermitian tensor representing all multilinear intervention probabilities. It is a deterministic finite quantum process exactly when*

$$R^{(N)} \succeq 0, \quad \text{Tr}_{2k-1} R^{(k)} = I_{2k-2} \otimes R^{(k-1)}, \quad R^{(0)} = 1. \quad (196)$$

On this branch there are memories $A_k \cong \overline{\text{supp } R^{(k)}}$, $A_0 = \mathbb{C}$, and isometries

$$V_k : H_{2k-2} \otimes A_{k-1} \longrightarrow H_{2k-1} \otimes A_k \quad (197)$$

whose sequential link realizes $R^{(N)}$. Every other support-minimal sequential realization satisfies

$$V'_k = (I \otimes U_k) V_k (I \otimes U_{k-1}^*) \quad (198)$$

for unique memory unitaries U_k compatible with the minimum predictive state.

Proof. Products of the physical frames span the full Hermitian tensor space, so the dual-frame formula is the unique representative of the multilinear probability functional. Positivity and recursive partial-trace identities are necessary by the Choi theorem and deterministic causal normalization; conversely they are the finite process-comb conditions. For support minimality, vectorize $(R^{(k)})^{1/2}$. Consecutive causal trace identities make the two canonical vectors minimal purifications of the same reduced positive operator, so minimal-purification uniqueness supplies the isometries and the inter-realization memory unitaries. $\square$

Remark A.15 (Why support reduction is necessary). An arbitrary nonminimal positive factorization may contain spectator directions. Taking ambient Hilbert adjoints before removing them can create mixed history words absent from the operational process. The support-minimal square-root class is therefore the canonical boundary at which a history C^* -algebra becomes well defined.

A.6 History algebra, contextual separation, and flat word Grams

Choose one support-minimal realization. Let $\mathcal{A}_X^{\text{raw}}$ be the finite-dimensional $*$ -algebra generated by the represented projections onto Z_X^{min} , primitive branch maps, and their Hilbert–Schmidt adjoints. A represented element is *contextually null* when every admitted left and right history context and every terminal Read has zero matrix coefficient on it. Let $\mathcal{J}_X^{\text{ctx}}$ be the set of all such elements.

Lemma A.16 (Contextual nullity is a two-sided $*$ -ideal). $\mathcal{J}_X^{\text{ctx}}$ is the largest two-sided $*$ -ideal invisible in every admitted history context. Hence

$$\mathcal{A}_X^{\text{hist}} = \mathcal{A}_X^{\text{raw}} / \mathcal{J}_X^{\text{ctx}} \quad (199)$$

is the unique minimum context-preserving separated history envelope.

Proof. If a is null, inserting any represented word on either side only enlarges the tested context, and reversing the context shows that a^* is null. Thus the null set is a two-sided $*$ -ideal. Every context-invisible ideal is contained in it by definition, which gives the universal property. $\square$

For a finite-dimensional unital C^* -algebra $\mathcal{A}$, define the normalized regular trace

$$\widehat{\tau}_{\text{reg}}^{\mathcal{A}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}}, \quad L_a(b) = ab. \quad (200)$$

If $\mathcal{A} \cong \bigoplus_{\lambda} M_{n_{\lambda}}(\mathbb{C})$, then $\text{Tr}_{\mathbb{C}}(L_a) = \sum_{\lambda} n_{\lambda} \text{Tr}(a_{\lambda})$, so $\widehat{\tau}_{\text{reg}}$ is a faithful tracial state invariant under every $*$ -isomorphism.

Let $\mathcal{W}_{X, \leq r}$ have basis the composable represented history words of length at most r , and let $W_r e_w = [w]$ in $\mathcal{A}_X^{\text{hist}}$. Define

$$\mathbb{K}_{X,r}(v, w) = \widehat{\tau}_{\text{reg}}([v]^* [w]). \quad (201)$$

Theorem A.17 (Finite flat word reconstruction). *Every $\mathbb{K}_{X,r}$ is positive. There is a least r_X such that*

$$\text{rank } \mathbb{K}_{X,r_X} = \text{rank } \mathbb{K}_{X,r_X+1}. \quad (202)$$

At this depth the represented word span is the whole history algebra, and the panel through depth $r_X + 1$ reconstructs the object units, minimum predictive state algebra, multiplication, involution, predictive core algebras, and every represented mixed history.

Proof. Positivity follows from $\sum_{v,w} \bar{c}_v c_w \mathbb{K}(v, w) = \hat{\tau}_{\text{reg}}(a^* a) \geq 0$. Faithfulness makes the Gram kernel exactly the represented word-relation space. The nested word spans stabilize because the history algebra is finite-dimensional. Equality of two consecutive ranks therefore means equality of two consecutive spans; the stabilized span contains the units and is invariant under left multiplication by every generator, so it equals the full generated algebra. The one-step extension reconstructs the generator multiplication matrices and reversed words reconstruct the involution. $\square$

A.7 Proof of Theorem 2.2

Proof of Theorem 2.2. Item (i) is Theorems A.4 and A.8. Item (ii) is Theorem A.9. Item (iii) is Theorem A.13. Item (iv) is Theorem A.14. Item (v) is Lemma A.16 and Theorem A.17. Item (vi) follows from Theorem A.8 and Corollary A.10 and support-minimal unitary uniqueness. Item (vii) is Theorem A.11. None of these constructions introduces an independent predictive state space, metric, connection, or spacetime dimension beyond the actual operational rows explicitly retained. $\square$

A.8 Relative minimality of the operational entrance

For the purpose of the foundational audit, keep the typed potential event grammar fixed and distinguish four semantic rows. Let $\mathfrak{D}$ denote the actual-occurrence interface selecting which potential events, writers, and Reads are physically present; let $\mathbb{P}^+$ denote the complete contextual predictive law, including the finite ancillary/matrix-order positivity and causal-recursion conditions used in the comb reconstruction; let Θ denote the protected orientation section; and let Λ denote the absolute physical calibration needed when durations, rates, lengths, or measures are interpreted in physical units. This separation is an audit of information content, not an enlargement of the operational model above.

Theorem A.18 (Relative primitive-basis minimality for the paper-level entrance). *Within the fixed category of finite typed operational-predictive models used in this paper, the semantic entrance*

$$[\mathfrak{G}; \mathfrak{D}, \mathbb{P}^+, \Theta, \Lambda] \quad (203)$$

has the following status.

- (R1) *The rows already assumed in the paper are sufficient for the foundational reconstruction actually used below: $\mathbb{P}^+$ and composition give the minimum future quotient and persistent predictive carrier, the contextual positive table gives the support-minimal causal process and history algebra, while Θ and Λ supply precisely the signed and absolute-scale information required on the Lorentzian ADM branch.*
- (R2) *Each of $\mathfrak{D}$, $\mathbb{P}^+$, Θ , and Λ is independent of the other three relative to the fixed typed grammar: for each forgotten row there are two anchored finite models with identical reducts and inequivalent full models.*

- (R3) *These four independence obstructions survive arbitrary common conservative refinement and cofinal reindexing; hence no functorial cutoff limit can reconstruct a forgotten row from the other three.*
- (R4) *Protected orientation can be absorbed into occurrence exactly when the occurrence packet contains a named actual Read together with a physical decoder selecting one orientation section. This is a definitional bundling, not an informational reduction of the orientation choice.*

Consequently the displayed semantic entrance is a minimal known sufficient and independent basis relative to the declared operational language and target class. No claim of absolute minimality among all conceivable foundational languages is made.

Proof. Sufficiency for the predictive and positive-process part is exactly [Theorems A.4, A.8, A.14](#) and [A.17](#), [Corollary A.6](#), and [Lemma A.16](#). On the geometric branch, [Lemmas B.9](#) and [B.11](#) and [Theorem B.10](#) show how the physically calibrated root race supplies the absolute kinetic data used by the ADM inversion, while [Lemmas B.12](#) and [B.14](#) and [Corollary B.15](#) use the protected sign and do not derive it from positive data.

For row-irredundancy it is enough to give one anchored separating pair for each row. For $\mathbb{P}^+$, fix one complete binary instrument with named outcomes a, b , keep every other row fixed, and compare Bernoulli laws $\Pr(a) = p$ and $\Pr(a) = q$ with $0 < p \neq q < 1$. Forgetting the predictive row makes the models identical, whereas any full anchored isomorphism fixes the outcome names and therefore cannot identify the two probabilities.

For $\mathfrak{D}$, adjoin a potential deterministic writer $W = 0$ whose mathematical extension is unique. In one model retain only the base event as actual; in the other declare W to be an actual readable writer as well. The structural and probabilistic data agree after the occurrence incidence is forgotten, but the full models differ on whether W belongs to the physical interface. Thus structural realizability, even when unique, does not imply actual occurrence.

For Θ , take the same anchored real two-plane with named basis e_1, e_2 and positive Gram I_2 in both models, but choose the unit alternating forms $\omega(e_1, e_2) = +1$ and $\omega(e_1, e_2) = -1$. Every positive Gram datum agrees. An anchored isomorphism fixes the ordered basis and cannot reverse the alternating form, so the protected orientation is not selected by the positive packet.

For Λ , fix a projective race with winner law p_a and an internal waiting variable $S \sim \text{Exp}(1)$ independent of the winner. Two models with the same grammar, occurrence, predictive law, and orientation but physical durations $T = c_0 S$ and $T = c_1 S$, $c_0 \neq c_1$, have identical uncalibrated reducts and different absolute rates $k_a = p_a/c_i$. Hence the common physical scale is not determined by normalized/projective race data. This is the same scale direction whose loss is visible in [Lemma B.9](#) when waiting-time calibration is omitted.

To see cofinal stability, tensor either member of any separating pair with the same strictly refining finite spectator process and let every cutoff map append and later discard only spectator records. The row-forgotten theories are then naturally isomorphic at every cutoff. If a full natural isomorphism appeared on a cofinal tail, composing it with spectator discard would give an anchored isomorphism of the original finite separating pair, a contradiction. Thus common refinement or passage to a cofinal limit cannot create the missing row.

Finally, if occurrence itself contains a named actual orientation Read and a deterministic physical decoder into the two-element orientation torsor, the decoded section and a separately named Θ determine one another. If the decoder is omitted, the two opposite sections above remain compatible with the same positive occurrence statistics. The former is therefore only a change of packaging, while the latter retains genuine orientation independence. $\square$

Theorem A.19 (Exact calibration fibre and minimum anchor bank). *Write all positive physical scales and additive origins in log-affine coordinates $x \in \mathbb{R}^N$. Suppose the physically established incidence relations are*

$$Lx = b. \quad (204)$$

If the solution set is nonempty and x_0 is one solution, then:

(K1) *the complete calibration fibre is*

$$\boxed{x_0 + \mathcal{K}_L, \quad \mathcal{K}_L := \text{Ker } L,} \quad (205)$$

of dimension $N - \text{rank } L$;

(K2) *exactly $\dim \mathcal{K}_L$ independent scalar anchors are necessary and sufficient to determine one calibration in this affine fibre;*

(K3) *if $U : \mathbb{R}^d \rightarrow \mathbb{R}^N$ encodes conventional changes of base units with $LU = 0$, the intrinsic physical residual is*

$$\boxed{\mathcal{K}_L^{\text{phys}} = \mathcal{K}_L / \text{Im } U,} \quad (206)$$

and one anchor per independent quotient direction is necessary and sufficient;

(K4) *winner probabilities for a root race instantiate a one-dimensional common-rate direction of this type: multiplying all directed rates by the same positive constant leaves the winner law unchanged, whereas the waiting-time scale separates that direction.*

Proof. If $Lx = Lx_0 = b$, then $x - x_0 \in \text{Ker } L$, and conversely every $x_0 + k$, $k \in \text{Ker } L$, solves (204); this gives (205) and its dimension by rank-nullity. A basis of $\mathcal{K}_L$ and the corresponding dual linear functionals give a sufficient scalar anchor bank. Any smaller bank has nontrivial common kernel on $\mathcal{K}_L$ and is therefore insufficient. Quotienting the subgroup of pure unit conventions gives (206) and the same argument on the quotient gives the intrinsic anchor count. For the root race, the winner law is $p_b = k_b / \sum_c k_c$ and is invariant under $k_b \mapsto ck_b$; the exponential waiting time has total rate $\sum_c k_c$ and therefore fixes the missing common scale. $\square$

B Minimal relational dimension and ADM reconstruction

This section proves [Theorem 3.6](#). The argument has two logically separate parts. The first selects the local relational module. The second reconstructs ADM geometry from a scheduler-separated root race on that module.

B.1 From one serial relation to a complete cell

Assumption B.1 (Minimal faithful predictive relation branch). A local relation event has one future-readable source and one future-readable target. Physical reversal retains both orientations. Future-minimality identifies parallel relation labels exactly when every continuation identifies them. A nonempty relation set is invariant under a group acting transitively on unordered endpoint pairs.

Lemma B.2 (Pair-homogeneous completion). *Under [Assumption B.1](#), the relation-colour-free local skeleton is the complete graph K_N on the endpoint set.*

Proof. The selected branch of the derived persistent predictive carrier contains exactly one future-readable source and one future-readable target, so one occurring relation gives one unoriented pair after physical reversal. The orbit of that pair under a group transitive on $\binom{X}{2}$ is the complete set of unordered pairs. Future-minimality removes only unread parallel labels, not distinct endpoint pairs. Hence the skeleton is K_N . $\square$

Let $X = \{1, \dots, N\}$, let $u_0 = N^{-1/2} \mathbf{1}_N$, and write $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$. Identify the signed edge $[i \rightarrow j]$ with $e_i \wedge e_j \in \Lambda^2 \mathbb{R}^N$. The boundary map is $\partial(e_i \wedge e_j) = e_j - e_i$.

Theorem B.3 (Cut-cycle decomposition). *There is an orthogonal decomposition*

$$C_1(K_N) = u_0 \wedge W_N \oplus \Lambda^2 W_N, \quad (207)$$

with

$$\partial(u_0 \wedge w) = \sqrt{N} w, \quad \text{Ker } \partial = \Lambda^2 W_N. \quad (208)$$

Consequently

$$\dim \text{Ran } \partial = N - 1, \quad \dim \text{Ker } \partial = \binom{N-1}{2}. \quad (209)$$

Proof. The exterior square of $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$ is $u_0 \wedge W_N \oplus \Lambda^2 W_N$. For $x, y \in \mathbb{R}^N$,

$$\partial(x \wedge y) = \left(\sum_i x_i \right) y - \left(\sum_i y_i \right) x.$$

Taking $x = u_0$, $y = w \in W_N$ gives the first identity. If both vectors lie in W_N , both coordinate sums vanish, so the second summand is in the kernel. Dimension counting shows that it is the whole kernel and gives (209). $\square$

Remark B.4 (Metric tomography does not select N). The root dyadics of a complete simplex define an isomorphism

$$\mathbb{R}^{\binom{N}{2}} \longrightarrow \text{Sym}(W_N), \quad (a_{ij}) \longmapsto \sum_{i < j} a_{ij} (e_j - e_i)(e_j - e_i)^\top$$

for every N . Indeed, a vanishing weighted sum is a graph Laplacian whose off-diagonal entries are $-a_{ij}$. Thus the equality $\binom{N}{2} = \dim \text{Sym}(W_N)$ is universal and cannot by itself select three spatial dimensions. The paper uses two different non-tomographic selectors: exact endpoint-cycle source-rank balance, and the stronger alternating/minimal-source simplex criterion.

Remark B.5 (Conditional character of the selector). The alternating form in the next theorem is part of the definition of the selected relational category. It is not reconstructed from the positive word Grams and does not, by itself, specify a metric, a Hamiltonian, a symplectic phase space of continuum fields, or a Lorentzian sign. Accordingly the theorem asserts uniqueness only among faithful pair-homogeneous simplex cells satisfying this alternating/minimal-source query. It is the second dimensional selector, not the sole derivation of K_4 ; the regular-graph source-balance result below uses no alternating form.

Theorem B.6 (Minimal faithful K_4 selector on the alternating simplex branch). *Assume, in addition to Assumption B.1, that:*

- (D1) *the temporal-spatial coefficient space is $\mathbb{R}t \oplus W_N$ and carries a nondegenerate alternating form;*

- (D2) *the endpoint and complete signed-edge source Grams are faithful;*
- (D3) *the cycle source is nonzero;*
- (D4) *among conservative completions satisfying these queries, the future-readable source innovation is minimal.*

Then

$$N = 4, \quad G_{\text{rel}} = K_4, \quad W_{\text{sp}} = W_4 \cong \mathbb{R}^3. \quad (210)$$

The faithful endpoint and edge source ranks are $r_0(N) = N - 1$ and $r_1(N) = \binom{N}{2}$.

Proof. The alternating coefficient space has dimension $1 + \dim W_N = N$. A nondegenerate alternating form exists only in even dimension, so N is even. A nonzero cycle source requires $\Lambda^2 W_N \neq 0$, hence $N \geq 3$; together with evenness this gives $N \geq 4$. On the even integers $N \geq 4$, both $N - 1$ and $\binom{N}{2}$ are strictly increasing. The unique minimum future-readable source cost is therefore attained at $N = 4$. The graph conclusion follows from [Lemma B.2](#), and $\dim W_4 = 3$. $\square$

A representation-theoretic strengthening explains why the cycle packet is particularly economical at $N = 4$. After complexification, $W_N \cong S^{(N-1,1)}$, $\Lambda^2 W_N \cong S^{(N-2,1,1)}$, and $W_N \otimes \text{sgn} \cong S^{(2,1^{N-2})}$. These Young diagrams coincide exactly for $N = 4$. Hence

$$\text{Hom}_{S_N}(\Lambda^2 W_N, W_N \otimes \text{sgn}) = \begin{cases} \mathbb{R}, & N = 4, \\ 0, & N \neq 4, \end{cases} \quad (211)$$

on the nontrivial simplex branch. Thus only K_4 identifies the entire cycle representation with one oriented spatial copy rather than an additional inequivalent source type.

Theorem B.7 (Alternating-free endpoint–cycle selector). *Let G be a connected simple k -regular relation graph on v endpoints. If its complete endpoint and cycle sources are faithful and have equal rank, then*

$$v - 1 = \frac{kv}{2} - v + 1, \quad \text{hence} \quad (4 - k)v = 4. \quad (212)$$

Apart from the one-point graph, the unique solution is

$$\boxed{v = 4, \quad k = 3, \quad G \cong K_4.} \quad (213)$$

Thus K_4 is also the unique nontrivial connected regular simple relation graph with exact faithful endpoint–cycle rank balance. This corroborates the simplex selector but does not extend it to arbitrary higher-arity boundary systems.

Proof. For a connected graph the endpoint-source rank is $v - 1$, while the cycle rank is $|E| - v + 1 = kv/2 - v + 1$. Equality gives $2(v - 1) = kv/2$, equivalently $(4 - k)v = 4$. With $v > 1$ and integer $k \geq 1$, the only connected simple regular solution is $v = 4$, $k = 3$, which is K_4 . $\square$

B.2 The A_3 source frame

In the anchored basis of [\(18\)](#), define $Q_a = r_a r_a^\top$.

Lemma B.8 (Root dyadics span Sym_3). *The six matrices $Q_1, \dots, Q_6$ form a basis of Sym_3 .*

Proof. Q_1, Q_2, Q_3 are the three diagonal matrix units. For $i < j$,

$$(e_j - e_i)(e_j - e_i)^\top - e_i e_i^\top - e_j e_j^\top = -(e_i e_j^\top + e_j e_i^\top).$$

The three difference roots therefore recover the three symmetric off-diagonal matrix units. The six matrices are linearly independent and span the six-dimensional space Sym_3 . $\square$

Let the twelve oriented roots race with positive rates $k_a^\pm$. Put $K = \sum_a (k_a^+ + k_a^-)$, let A be the winning oriented root, and let T be the first waiting time.

Lemma B.9 (Winner–waiting score is faithful). *For each oriented branch b , the log-rate score is*

$$s_b = \mathbf{1}_{\{A=b\}} - k_b T. \quad (214)$$

Its Gram matrix is diagonal and positive:

$$\mathbb{E}[s_b s_c] = \delta_{bc} \frac{k_b}{K}. \quad (215)$$

Winner probabilities alone have covariance $\text{diag}(p) - pp^\top$ and therefore lose the common rate scale.

Proof. For independent exponential clocks, $T \sim \text{Exp}(K)$, the winner has probability $p_b = k_b/K$, and A and T are independent. Therefore

$$\begin{aligned} \mathbb{E}[s_b s_c] &= \delta_{bc} p_b - p_b k_c \mathbb{E}T - k_b p_c \mathbb{E}T + k_b k_c \mathbb{E}T^2 \\ &= \delta_{bc} \frac{k_b}{K} - 2 \frac{k_b k_c}{K^2} + 2 \frac{k_b k_c}{K^2}, \end{aligned}$$

which proves (215). The covariance of the winner indicator is singular in the all-ones rate direction because probabilities are unchanged by common rescaling. $\square$

B.3 ADM inversion and scheduler scale

Define $\mathcal{B}_h$ and β_h by (19).

Theorem B.10 (Complete finite ADM frame). *The six rate sums map isomorphically to $\mathcal{B}_h \in \text{Sym}_3$. The six rate differences map with rank three onto $\beta_h \in \mathbb{R}^3$ and have a three-dimensional kernel, the tetrahedral circulation space. For every $\mathcal{B}_h \succ 0$ and $\varrho_h > 0$, the formula (20) is the unique solution of*

$$\mathcal{B}_h = N_h^2 g_h^{-1}, \quad \varrho_h = \sqrt{\det g_h}. \quad (216)$$

Hence the root rates, waiting-time scale, and speed density reconstruct the six spatial metric components, three shift components, and lapse.

Proof. The first assertion is Lemma B.8. The difference map has image containing r_1, r_2, r_3 , hence has rank three; rank–nullity gives a three-dimensional kernel. For the inversion, take determinants in $\mathcal{B} = N^2 g^{-1}$: $\det \mathcal{B} = N^6 / (\det g) = N^6 / \varrho^2$. Thus $N = \varrho^{1/3} (\det \mathcal{B})^{1/6}$. Substituting back gives $g = N^2 \mathcal{B}^{-1}$, which is exactly (20). Positivity makes the solution unique. $\square$

Lemma B.11 (Fast-clock lower bound). *Suppose $\|r_a\| \leq R$ and $\mathcal{B}_h \succeq cI_3$. Then*

$$\sum_a (k_a^+ + k_a^-) \geq \frac{3c}{R^2 h^2}. \quad (217)$$

In particular, a total rate $O(h^{-1})$ produces $\mathcal{B}_h = O(h)$ and a collapsed continuum bracket.

Proof. Taking traces in (19),

$$3c \leq \text{tr } \mathcal{B}_h \leq R^2 h^2 \sum_a (k_a^+ + k_a^-).$$

This gives (217). The final assertion follows by substitution. $\square$

B.4 Finite Krein orientation from the protected sign

Let $\epsilon_X \in \{+1, -1\}$ be the actual protected orientation Read of (22) on the finite support-minimal relational carrier $\mathcal{H}_J$. Functional calculus of this classical two-point Read gives orthogonal projections $P_\pm$ with $P_+ + P_- = I$; define $J = P_+ - P_-$. Thus

$$\mathcal{H}_J = \mathcal{H}_J^+ \oplus \mathcal{H}_J^-, \quad \mathcal{H}_J^\pm = \text{Ran } P_\pm, \quad J|_{\mathcal{H}_J^\pm} = \pm I. \quad (218)$$

Assume both summands are nonzero. Define

$$[u, v]_J = \langle u, Jv \rangle. \quad (219)$$

Then $[\cdot, \cdot]_J$ is a nondegenerate Hermitian form with one positive and one negative sector in the orientation factor, hence a finite Krein structure. This statement uses only the protected sign involution and the support-minimal Hilbert realization; no continuum metric is involved.

Lemma B.12 (Positive geometry and Krein signature are distinct layers). *The positive word-Gram hierarchy of Theorem A.17 remains positive after adjoining the protected orientation record, while the indefinite sign is carried entirely by J . Consequently no positive combination of the spatial root dyadics can replace the Krein involution, and the Lorentzian signature of the principal symbol requires the signed layer in addition to the A_3 bracket.*

Proof. The word-Gram hierarchy is a Gram matrix for the faithful positive regular trace and is therefore positive semidefinite. Likewise every spatial bracket is a positive sum $\sum_a c_a r_a r_a^\top$ with $c_a > 0$. Such a form has no negative direction. By contrast, J has both $+1$ and -1 spectral subspaces by (218); the form (219) is therefore indefinite. The two structures are independent in exactly the required sense: the spatial geometry remains positive while J supplies the sign subsequently represented by the temporal Clifford generator. $\square$

Theorem B.13 (Irreducible sign and Lorentzian inertia before Clifford realization). *Let W_4 carry the positive spatial form reconstructed by Theorem B.10. The complete positive Grand-Tensor packet cannot supply a negative direction, while the protected orientation row is independent of that packet by Theorem A.18. If the geometric extension retains the positive restriction on W_4 and introduces exactly one independent nondegenerate signed line $\mathbb{R}t$, then, up to overall sign,*

$$q(\tau t + w) = -a^2 \tau^2 + g(w, w), \quad a > 0, \quad \text{inertia}(q) = (1, 3). \quad (220)$$

The conclusion depends only on positivity of the spatial form, independence of the signed row, and one-line minimality; it does not depend on a choice of gamma matrices.

Proof. The positive word-Gram and root-race forms have no negative eigenvalues by Lemma B.12. The separating pair in the proof of Theorem A.18 shows that reversing the protected orientation leaves all positive operational data unchanged, so its sign is not a derived eigenvalue of the positive packet. A minimal extension introduces one real line and no additional signed directions.

Nondegeneracy makes its coefficient nonzero; choosing the relative sign opposite to the positive spatial restriction gives one negative eigenvalue and the three positive eigenvalues of g . Sylvester's law of inertia makes this count invariant under every real change of basis. Multiplying the entire form by -1 reverses the sign convention but not the Lorentzian type. No Clifford representation has been used. $\square$

B.5 Clifford realization of the signed quadratic form

The inertia has already been fixed by [Theorem B.13](#). The following matrices provide a minimal complex Clifford realization of that form; they are not used to choose its signature. Let X, Y, Z denote the Pauli matrices and define the four matrices in (26).

Lemma B.14 (Depth-two Lorentzian Clifford cell). *The matrices γ^μ satisfy*

$$(\gamma^0)^2 = -I, \quad (\gamma^a)^2 = I, \quad \{\gamma^\mu, \gamma^\nu\} = 0 \quad (\mu \neq \nu), \quad (221)$$

and generate $M_4(\mathbb{C}) \cong \text{Cl}_{1,3}(\mathbb{C})$.

Proof. The square relations follow from $X^2 = Y^2 = Z^2 = I$. Every mixed product contains exactly one local Pauli anticommutation, so the mixed anticommutators vanish. The generators contain $X_A \otimes I$ and $Y_A \otimes I$, hence also $Z_A \otimes I$. Multiplication by γ^1 and γ^2 then gives $I \otimes Z_B$ and $I \otimes Y_B$, hence $I \otimes X_B$. Both local matrix factors are generated, so the total algebra is $M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) = M_4(\mathbb{C})$. $\square$

Corollary B.15 (Lorentzian principal square). *Let $E_h E_h^\top = g_h^{-1}$ and define σ_h by (27). Then (28) holds. If $g_h \succ 0$ and $N_h > 0$, the characteristic quadratic form realizes the already determined inertia $(1, 3)$ in the convention $(-, +, +, +)$.*

Proof. The temporal-spatial cross terms vanish by the Clifford anticommutation relations. The temporal square contributes $-N_h^{-2}(\xi_0 - \beta_h \cdot \vec{\xi})^2$, while the spatial square is $\delta^{ab}(E_h)_a{}^i (E_h)_b{}^j \xi_i \xi_j = g_h^{ij} \xi_i \xi_j$. $\square$

B.6 Cofinal transport and proof of [Theorem 3.6](#)

For adjacent cutoffs, let the complete old/detail positive form have block matrix

$$M_Y = \begin{pmatrix} A_{Y/X} & B_{Y/X} \\ B_{Y/X}^* & D_{Y/X} \end{pmatrix}, \quad (222)$$

and Schur complement $S_{Y/X} = A_{Y/X} - B_{Y/X} D_{Y/X}^\dagger B_{Y/X}^*$. Define

$$\sigma_{Y/X} = \|S_{Y/X} - M_X\|_{\text{op}}, \quad \theta_{Y/X} = \|B_{Y/X} D_{Y/X}^\dagger\|_{\text{op}}, \quad \mu_{Y/X} = \lambda_{\min}^+(D_{Y/X}). \quad (223)$$

The harmonic lift

$$\mathcal{H}_{Y/X} u = J_{Y/X} u - K_{Y/X} D_{Y/X}^\dagger B_{Y/X}^* u \quad (224)$$

minimizes the fine energy over detail completions and has energy $\langle u, S_{Y/X} u \rangle$. This is ordinary completion of the square.

If

$$\sum_X (\sigma_X + \theta_X + \mu_X^{-1}) < \infty, \quad (225)$$

then the finite forms and their regular isolated spectral projections are Cauchy after harmonic transport. In particular, a separated threshold rank, a positive pair gap, and a finite-dimensional Clifford source persist. This is the finite-dimensional core of the common-screen limit used here.

Proof of Theorem 3.6. The predictive carrier is supplied by Theorem A.9. For the regular-graph route, the local dimension follows directly from Theorem B.7. For the pair-homogeneous simplex route, the local graph is complete by Lemma B.2 and the same K_4/A_3 module follows from Theorems B.3 and B.6. Thus the alternating form is needed only on the second route. The root sums and differences, faithful rate coordinates, ADM inversion, and clock lower bound are Lemmas B.8, B.9 and B.11 and Theorem B.10. The positive/signed separation is Lemma B.12; Theorem B.13 fixes Lorentzian inertia before the Clifford matrices are introduced, and Lemma B.14 and Corollary B.15 realizes that form. Finally, (225) makes every corrected finite matrix packet Cauchy in operator norm on each protected finite screen; spectral separation therefore preserves the ranks and positive gaps, while the Schur mismatch and leakage tails vanish. The canonical limit quotient consequently retains the K_4/A_3 module, the physically occurring temporal calibration, the ADM packet, and the Lorentzian Clifford cell. $\square$

Remark B.16 (Exact dimension scope). The result applies to the derived future-minimal, one-source/one-target, reciprocal, pair-homogeneous, faithful, and minimal faithful branch. A deliberately higher-arity boundary, a relation-coloured theory, a nonfaithful cycle quotient, or an enriched cross-product packet changes the primitive query and is not a counterexample. In particular, a seven-dimensional vector cross product $\Lambda^2 W \rightarrow W$ has a fourteen-dimensional kernel and requires an additional stable G_2 three-form; it is an enriched theory rather than an equal-cost realization of the faithful K_4 branch.

C Operational action reconstruction and the Einstein handoff

This appendix proves the reconstruction and variational statements of Section 4. The order is important: the determining field is first obtained from the complete accepted cylinder; the common action is then reconstructed from a protected reward-duration packet and finite exactness tests; the actual accepted transition is compared with that action; spatial compactness is derived from a cut margin; and only then is the gravitational first variation passed to the continuum.

C.1 Accepted determining-field quotient and kernel

Let $\tilde{\Theta}_X$ be the finite accepted ledger of Equation (29). Two states x, y are future equivalent when every declared determining coordinate and every admitted future word agrees from x and y , including all branch-domain indicators. Let $\Theta_{\text{fld}, X}^{\min}$ be the quotient.

Theorem C.1 (Terminating determining-field and accepted-kernel compiler). *The quotient $\Theta_{\text{fld}, X}^{\min}$ is the unique coarsest finite record through which every selected determining-field future and every primitive update factors. Partition refinement beginning with the immediate determining coordinates and branch domains terminates after finitely many strict rounds. The kernel K_X^{acc} of (30) is well defined, table native, and is the unique accepted transition reproducing all selected post-accepted field futures.*

Proof. Future equality is a typed right congruence. If $x \sim_{\text{fld}, X} y$, then for every primitive branch a and continuation w , every determining read satisfies

$$d(u_w u_a(x)) = d(u_{wa}(x)) = d(u_{wa}(y)) = d(u_w u_a(y)),$$

so every primitive update descends to the quotient. At each refinement round, split two states whenever an immediate coordinate differs or one primitive update reaches different blocks. The k th partition identifies precisely the states not separated by a future of length at most k . Every strict round increases the number of finite blocks, hence the refinement terminates. Its stable partition has the universal property by the usual fibre argument.

Accepted branch probabilities and target quotient classes are constant on the stable fibres. Summing the resolved accepted branches whose target lies in a specified quotient class gives (30), and these sums are exactly the probabilities of the corresponding post-accepted Reads. Uniqueness follows because those Reads separate the quotient classes. $\square$

Remark C.2 (Occurrence boundary). The theorem reconstructs the dynamical state and kernel from the *complete* accepted cylinder. It does not reconstruct that cylinder from an accepted binary clock alone. Indeed, an identity accepted kernel and a full-support reset kernel can be coupled to the same state-independent acceptance bit while producing different terminal field laws. Thus occurrence of the complete accepted field record is a genuine physical input row.

C.2 Renewal pressure and the common scalar action

Let (Ω_X, p_X) be a faithful marked first-return law, let $\tau_X > 0$ be the protected physical duration, and let $\mathcal{G}_X : E_X \rightarrow L^2(\Omega_X, p_X)$ be the reward synthesis of (31). Put

$$\mathcal{L}_X(q, r) = \mathbb{E}_X[e^{-g_{q,X} - r\tau_X}], \quad \bar{\tau}_X = \mathbb{E}_X\tau_X, \quad T_X c = c\tau_X.$$

Theorem C.3 (Table-derived renewal pressure and action-slope reconstruction). *The complete Laplace transform $\mathcal{L}_X$ and all of its finite derivatives are reconstructed from the marked-cycle table. There is a unique analytic pressure $\mathcal{P}_X$ near the origin satisfying*

$$\mathcal{P}_X(0) = 0, \quad \mathcal{L}_X(q, \mathcal{P}_X(q)) = 1.$$

Let $S_X v = D \log p_{X,q}|_{q=0}[v]$ and $\pi_X = D\mathcal{P}_X(0)$. Then

$$\pi_X(v) = -\frac{\mathbb{E}_X g_{v,X}}{\bar{\tau}_X}, \quad S_X v = -\mathcal{G}_X v - \pi_X(v)\tau_X, \quad \mathcal{G}_X = -S_X - T_X \pi_X. \quad (226)$$

Thus the physical duration restores the affine action coordinate omitted by the centered probability score.

Proof. The transform is a finite sum of known probabilities times known protected writer values. At the origin, $\mathcal{L}_X(0, 0) = 1$ and $\partial_r \mathcal{L}_X(0, 0) = -\bar{\tau}_X < 0$, so the analytic implicit-function theorem gives the pressure root. Differentiating $\mathcal{L}_X(q, \mathcal{P}_X(q)) = 1$ in direction v gives

$$0 = -\mathbb{E}_X g_{v,X} - \pi_X(v)\bar{\tau}_X,$$

which proves the first identity. Differentiating the normalized tilted law gives the score formula and hence the synthesis reconstruction. $\square$

Corollary C.4 (Pressure-jet reconstruction of mixed action blocks). *With $\bar{\tau}_X(q) = \mathbb{E}_{X,q}\tau_X$, $\delta_X = D\bar{\tau}_X(0)$, and $m_{2,X} = \mathbb{E}_X\tau_X^2$,*

$$D^2\mathcal{P}_X(0) = \bar{\tau}_X^{-1} S_X^* S_X, \quad \delta_X = T_X^* S_X, \quad (227)$$

$$\mathcal{G}_X^* \mathcal{G}_X = \bar{\tau}_X D^2 \mathcal{P}_X(0) + \delta_X^* \pi_X + \pi_X^* \delta_X + m_{2,X} \pi_X^* \pi_X. \quad (228)$$

Consequently one pressure surface reconstructs every marginal and mixed action block represented on the same marked cycle record.

Proof. Differentiate the pressure normalization twice. The first factors are the centered scores $-S_X u$ and $-S_X v$, which gives the first identity. Differentiating the tilted mean duration gives $\delta_X = T_X^* S_X$; expanding $\mathcal{G}_X = -S_X - T_X \pi_X$ gives the second formula. $\square$

Let $\mathcal{V}_X$ be the finite protected variation complex, with coboundaries $d_0 : C^0 \rightarrow C^1$ and $d_1 : C^1 \rightarrow C^2$, $d_1 d_0 = 0$, and let $\alpha_X \in C^1(\mathcal{V}_X; \mathbb{R})$ be the reconstructed physical action increment.

Theorem C.5 (Finite common-action exactness). *There is a scalar action $\mathcal{S}_X$ with $\alpha_X = d_0 \mathcal{S}_X$ if and only if $d_1 \alpha_X = 0$ on every declared 2-cell and all periods of α_X vanish on one integral basis of $H_1(\mathcal{V}_X; \mathbb{Z})$. After fixing one action anchor, $\mathcal{S}_X$ is unique and reconstructed by path integration. If P_{har} and P_{coex} are the Hodge projections and $\sigma_{\text{coex}} > 0$ is the least singular value of d_1 on the coexact range, then*

$$\text{dist}(\alpha_X, \text{Ran } d_0)^2 \leq \|P_{\text{har}} \alpha_X\|^2 + \sigma_{\text{coex}}^{-2} \|d_1 \alpha_X\|^2. \quad (229)$$

Proof. If $\alpha_X = d_0 \mathcal{S}_X$, telescoping gives zero face sums and zero periods. Conversely, fix a base configuration and define $\mathcal{S}_X(q)$ by summing α_X along a path from the base to q . Face closure makes the sum invariant under elementary 2-cell moves and the period conditions remove the remaining homology class. Anchoring removes the additive constant. For the estimate use the finite Hodge decomposition $C^1 = \text{Ran } d_0 \oplus \mathcal{H}^1 \oplus \text{Ran } d_1^*$ and the singular-value bound $\|d_1 \beta\| \geq \sigma_{\text{coex}} \|\beta\|$ on the coexact range. $\square$

Remark C.6 (Provenance is not stationarity). Reconstructing a common action does not prove that the observed accepted transition is generated by it. The accepted/action comparison below is an independent same-history test. Conversely, fitting a cost to the observed kernel after the fact would be tautological and has no stationarity content.

C.3 Fisher–Bregman stationarity and physical test lifts

Let Θ , Ψ , G , $\mathcal{A}$, and η satisfy the convex-chart and interiority hypotheses of [Theorem 4.5](#). Define D_Ψ , Φ_θ , y_θ^* , Δ_Ψ , and $\bar{\Delta}_\Psi$ by (39)–(42).

Theorem C.7 (Quantitative accepted-action stationarity). *Put*

$$\mu_\Phi = (\eta^{-1} - \lambda)m > 0, \quad \Lambda_\Phi = (\eta^{-1} + L)M, \quad \ell = \max\{\lambda, L\}. \quad (230)$$

Then

$$\int D_\Psi(y, \theta) \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \leq \eta \bar{\Delta}_\Psi, \quad (231)$$

$$\int \|y - \theta\|^2 \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \leq \frac{2\eta}{m} \bar{\Delta}_\Psi, \quad (232)$$

and

$$\int \mathfrak{S}(\theta) \nu(d\theta) \leq C_{\Psi, \mathcal{A}, \eta} \bar{\Delta}_\Psi, \quad (233)$$

where one admissible constant is

$$C_{\Psi, \mathcal{A}, \eta} = \frac{4\Lambda_\Phi^2}{m\mu_\Phi} + \frac{4M^2(1 + \eta\ell)^2}{m^2\eta}. \quad (234)$$

Proof. State independence of $\vartheta > 0$ and stationarity of ν imply $\nu K^{\text{acc}} = \nu$. Since y_θ^* minimizes Φ_θ and $\Phi_\theta(\theta) = \mathcal{A}(\theta)$,

$$\mathcal{A}(y) - \mathcal{A}(\theta) + \eta^{-1} D_\Psi(y, \theta) \leq \Delta_\Psi(\theta, y).$$

The action difference cancels on averaging, proving (231). The lower Hessian bound gives $D_\Psi(y, \theta) \geq m\|y - \theta\|^2/2$, hence (232).

The Hessian of Φ_θ lies between $\mu_\Phi I$ and $\Lambda_\Phi I$ on the convex chart. Interiority of y_θ^* gives $\nabla \Phi_\theta(y_\theta^*) = 0$. Strong convexity and the upper Hessian bound therefore imply

$$\|y - y_\theta^*\|^2 \leq \frac{2\Delta_\Psi(\theta, y)}{\mu_\Phi}, \quad \|\nabla \Phi_\theta(y)\|^2 \leq \frac{2\Lambda_\Phi^2}{\mu_\Phi} \Delta_\Psi(\theta, y).$$

For

$$r_\Psi(\theta, y) = \eta \nabla \Phi_\theta(y) = \nabla \Psi(y) - \nabla \Psi(\theta) + \eta \nabla \mathcal{A}(y)$$

we have

$$\eta \nabla \mathcal{A}(\theta) = r_\Psi - [\nabla \Psi(y) - \nabla \Psi(\theta)] + \eta [\nabla \mathcal{A}(\theta) - \nabla \mathcal{A}(y)].$$

The last two terms together have norm at most $M(1 + \eta\ell)\|y - \theta\|$. Squaring with $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, using $G^{-1} \preceq m^{-1}I$, and averaging yields (233) with (234). $\square$

Remark C.8 (Boundary minima). The first two inequalities do not require interiority, but the ordinary gradient conclusion does. For example, on $\Theta = [0, 1]$ with $\Psi(x) = x^2/2$ and $\mathcal{A}(x) = x$, the source $\theta = 0$ has proximal minimum $y_\theta^* = 0$. The stationary identity transition supported at zero has zero gap although $\mathcal{A}'(0) = 1$. A boundary formulation must retain the normal-cone or projected variational residual. It cannot be substituted for unrestricted determining metric variations without checking their admissibility.

Proposition C.9 (Cutoff-aware stationarity on the determining test core). *Let k be a fixed physical test and let its represented lift satisfy (44). Then (45) holds. If*

$$B_{X,K}(k)^2 C_X \bar{\Delta}_{\Psi,X} \longrightarrow 0 \tag{235}$$

for each test in a countable determining core, the corresponding first variations converge to zero in mean square. Under any common coupling there is a deterministic subsequence of cutoffs on which all these first variations converge to zero almost surely.

Proof. At each represented field value, duality for the positive G_X pairing gives

$$|D\mathcal{A}_X[v]|^2 \leq \|D\mathcal{A}_X\|_{G_X^{-1}}^2 \|v\|_{G_X}^2.$$

Take the supported supremum of the second factor and apply Theorem C.7. This proves the mean-square estimate. Enumerate the determining core as $k_1, k_2, \dots$. Select increasing cutoffs X_j so that the right-hand side of (45) is at most 2^{-j} for each of $k_1, \dots, k_j$. For a fixed k_m , the sum over $j \geq m$ of the expected squared residuals is finite. Tonelli's theorem implies that the corresponding sum of squared residuals is finite almost surely, so the residual tends to zero. A countable intersection gives the assertion for the whole core. No independence between cutoffs is used. Uniform continuity of the limiting variational functional on the declared test topology then extends zero from the core to all determining tests. $\square$

Corollary C.10 (Finite Gibbs/KL certificate). *For a finite reference row $p_\theta(y) > 0$ and reconstructed same-history cost $c_\theta(y)$, the row q_θ^* of (46) is the unique entropy-proximal row and*

$$\Delta_{\text{Gibbs}, \theta} = \eta^{-1} D_{\text{KL}}(K^{\text{acc}}(\cdot | \theta) \| q_\theta^*) \quad (236)$$

is exactly its row variational gap.

Proof. For any probability row q ,

$$\sum_y q(y) c_\theta(y) + \eta^{-1} D_{\text{KL}}(q \| p_\theta) = \eta^{-1} D_{\text{KL}}(q \| q_\theta^*) - \eta^{-1} \log \sum_z p_\theta(z) e^{-\eta c_\theta(z)}.$$

Relative entropy is nonnegative and vanishes only at equality. This proves a statement about probability rows; it does not identify their sampled points with interior critical points of the field action. $\square$

C.4 Global cut margin and compact-screen reconstruction

Let a finite physical endpoint graph have vertex masses $m_{v,X}$, measure μ_X , symmetric conductances $c_{uv,X}$, and efficient edge capacities $u_{e,X}$. We use the two cut constants I_X^{eff} and I_X^{sp} of (49); on the positive spatial branch they are related by the finite capacity–conductance comparison (48).

Definition C.11 (Spatial Dirichlet normalization). On the finite weighted graph $G_X = (V_X, E_X)$ set

$$\mathcal{E}_X^{\text{sp}}(f) = \sum_{\{u,v\} \in E_X} c_{uv,X} |f(u) - f(v)|^2, \quad \langle f, \Delta_X^{\text{sp}} f \rangle_{L^2(\mu_X)} = \mathcal{E}_X^{\text{sp}}(f). \quad (237)$$

All scalar Sobolev estimates below are first stated for real-valued functions; the real weighted Laplacian has the same spectrum after complexification. Let $P_{0,X}$ be the orthogonal projection onto the mean-zero subspace and put $T_X(t) = e^{-t\Delta_X^{\text{sp}}} P_{0,X}$. Since the conductances are symmetric and nonnegative, $e^{-t\Delta_X^{\text{sp}}}$ is a finite Markov semigroup and is contractive on every $L^q(\mu_X)$, $1 \leq q \leq \infty$.

Lemma C.12 (Cut inequality implies the half-volume L^1 Sobolev bound). *Assume*

$$h_X c_X(\partial A) \geq I_*^{\text{sp}} \mu_X(A)^{2/3} \quad (238)$$

whenever $0 < \mu_X(A) \leq \mu_X(V_X)/2$. If $g \geq 0$ and $\mu_X(\text{supp } g) \leq \mu_X(V_X)/2$, then

$$\|g\|_{L^{3/2}(\mu_X)} \leq \frac{h_X}{I_*^{\text{sp}}} \sum_{\{u,v\} \in E_X} c_{uv,X} |g(u) - g(v)|. \quad (239)$$

Proof. For $t \geq 0$ put $A_t = \{v : g(v) > t\}$. Since $A_t \subseteq \text{supp } g$, every nonempty A_t lies on the half-volume side of (238). The layer-cake identity gives $g = \int_0^\infty \mathbf{1}_{A_t} dt$. Minkowski's integral inequality in $L^{3/2}(\mu_X)$ therefore yields

$$\|g\|_{3/2} \leq \int_0^\infty \|\mathbf{1}_{A_t}\|_{3/2} dt = \int_0^\infty \mu_X(A_t)^{2/3} dt. \quad (240)$$

Applying the cut inequality to every level set and then using the finite coarea identity gives

$$\|g\|_{3/2} \leq \frac{h_X}{I_*^{\text{sp}}} \int_0^\infty c_X(\partial A_t) dt \quad (241)$$

$$= \frac{h_X}{I_*^{\text{sp}}} \sum_{\{u,v\} \in E_X} c_{uv,X} |g(u) - g(v)|, \quad (242)$$

which is (239). $\square$

Lemma C.13 (Discrete L^6 Sobolev and Poincaré inequalities). Assume (238), $\mu_X(V_X) \leq V_*$, and

$$\sum_u c_{uv,X} \leq D_* h_X^{-2} m_{v,X} \quad (v \in V_X). \quad (243)$$

Then every mean-zero f satisfies

$$\|f\|_6^2 \leq C_S \mathcal{E}_X^{\text{sp}}(f), \quad C_S = \frac{128D_*}{(I_*^{\text{sp}})^2}, \quad (244)$$

and

$$\|f\|_2^2 \leq C_P \mathcal{E}_X^{\text{sp}}(f), \quad C_P = V_*^{2/3} C_S. \quad (245)$$

In particular $\lambda_1(\Delta_X^{\text{sp}}|_{\mathbf{1}^\perp}) \geq C_P^{-1}$.

Proof. First suppose $f \geq 0$ and $\mu_X(\text{supp } f) \leq \mu_X(V_X)/2$. Apply Lemma C.12 to $g = f^4$. Since $\|f^4\|_{3/2} = \|f\|_6^4$ and $|a^4 - b^4| \leq 4(a^3 + b^3)|a - b|$ for $a, b \geq 0$,

$$\|f\|_6^4 \leq \frac{4h_X}{I_*^{\text{sp}}} \sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3) |f(u) - f(v)| \quad (246)$$

$$\leq \frac{4h_X}{I_*^{\text{sp}}} \left[\sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3)^2 \right]^{1/2} \mathcal{E}_X^{\text{sp}}(f)^{1/2}. \quad (247)$$

Using $(a^3 + b^3)^2 \leq 2(a^6 + b^6)$ and (243),

$$\sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3)^2 \leq 2 \sum_v f(v)^6 \sum_u c_{uv,X} \quad (248)$$

$$\leq 2D_* h_X^{-2} \|f\|_6^6. \quad (249)$$

Substitution into (247) and division by $\|f\|_6^3$ gives

$$\|f\|_6^2 \leq \frac{32D_*}{(I_*^{\text{sp}})^2} \mathcal{E}_X^{\text{sp}}(f) \quad (250)$$

for every half-volume-supported nonnegative f .

For a general real f , choose a weighted median m so that $\mu_X(f > m)$ and $\mu_X(f < m)$ are each at most half the total volume. Put $f_+ = (f - m)_+$ and $f_- = (m - f)_+$. Truncation is 1-Lipschitz and, edgewise,

$$|f_+(u) - f_+(v)|^2 + |f_-(u) - f_-(v)|^2 \leq |f(u) - f(v)|^2, \quad (251)$$

so $\mathcal{E}(f_+) + \mathcal{E}(f_-) \leq \mathcal{E}(f)$. Since the two functions have half-volume support, (250) gives

$$\|f - m\|_6^2 = (\|f_+\|_6^6 + \|f_-\|_6^6)^{1/3} \quad (252)$$

$$\leq \|f_+\|_6^2 + \|f_-\|_6^2 \leq \frac{32D_*}{(I_*^{\text{sp}})^2} \mathcal{E}(f). \quad (253)$$

If f has mean zero, then

$$|m| \mu_X(V_X)^{1/6} \leq \|f - m\|_6, \quad (254)$$

because $|m| = |\mu_X(V_X)^{-1} \int (f - m) d\mu_X|$ and Hölder gives $\|f - m\|_1 \leq \mu_X(V_X)^{5/6} \|f - m\|_6$. Therefore $\|f\|_6 \leq 2\|f - m\|_6$, which yields (244) with the stated factor four. Finally,

$$\|f\|_2^2 \leq \mu_X(V_X)^{2/3} \|f\|_6^2 \leq V_*^{2/3} C_S \mathcal{E}(f), \quad (255)$$

which is (245). Rayleigh–Ritz gives the spectral-gap bound. $\square$

Lemma C.14 (Nash inequality, ultracontractivity, and spectral counting). *Under the hypotheses of Lemma C.13, every mean-zero f obeys*

$$\|f\|_2^{10/3} \leq C_S \mathcal{E}_X^{\text{sp}}(f) \|f\|_1^{4/3}. \quad (256)$$

Moreover, for every $t > 0$,

$$\|T_X(t)\|_{L^1 \rightarrow L^\infty} \leq 4 \left(\frac{3C_S}{2t} \right)^{3/2}. \quad (257)$$

Consequently, if $N_X(R) = \text{rank } \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}})$, then for $R > 0$

$$N_X(R) \leq 1 + 4e^{3/2} V_*(C_S R)^{3/2}. \quad (258)$$

Proof. Interpolation between L^1 and L^6 gives $\|f\|_2 \leq \|f\|_1^{2/5} \|f\|_6^{3/5}$. Raising this inequality to the power $10/3$ and using (244) yields (256).

Let $F_t = T_X(t)f$ and $u(t) = \|F_t\|_2^2$. Since F_t has mean zero and the finite Markov semigroup is L^1 -contractive,

$$\|F_t\|_1 \leq \|P_{0,X} f\|_1 \leq 2\|f\|_1. \quad (259)$$

Self-adjointness and finite dimensionality permit differentiation under the spectral decomposition:

$$u'(t) = -2\mathcal{E}_X^{\text{sp}}(F_t). \quad (260)$$

Combining (256), (259), and (260) gives

$$u'(t) \leq -\frac{2}{C_S 2^{4/3} \|f\|_1^{4/3}} u(t)^{5/3}. \quad (261)$$

Integrating the differential inequality for $u^{-2/3}$ yields

$$\|T_X(t)f\|_2^2 \leq \frac{3^{3/2}}{2} C_S^{3/2} t^{-3/2} \|f\|_1^2. \quad (262)$$

Because $T_X(t/2)$ is self-adjoint and $T_X(t) = T_X(t/2)T_X(t/2)$ on the mean-zero subspace, duality and the semigroup property give

$$\|T_X(t)\|_{1 \rightarrow \infty} \leq \|T_X(t/2)\|_{1 \rightarrow 2} \|T_X(t/2)\|_{2 \rightarrow \infty} \quad (263)$$

$$= \|T_X(t/2)\|_{1 \rightarrow 2}^2 \leq 4 \left(\frac{3C_S}{2t} \right)^{3/2}, \quad (264)$$

which is (257). This is the standard Sobolev–Nash ultracontractivity route in finite weighted form; compare [30, Chs. 10–11] and [31].

Let $K_t^0(v, w)$ be the integral kernel of $T_X(t)$ with respect to μ_X . The spectral expansion gives $K_t^0(v, v) = \sum_{j \geq 1} e^{-t\lambda_j} |\varphi_j(v)|^2 \geq 0$, and therefore

$$\text{Tr}(T_X(t)) = \sum_v m_{v,X} K_t^0(v, v) \quad (265)$$

$$\leq \mu_X(V_X) \|T_X(t)\|_{1 \rightarrow \infty} \quad (266)$$

$$\leq 4V_* \left(\frac{3C_S}{2t} \right)^{3/2}. \quad (267)$$

Since the positive cut margin implies connectedness, the zero eigenvalue is simple. Hence

$$(N_X(R) - 1)e^{-tR} \leq \sum_{0 < \lambda_j \leq R} e^{-t\lambda_j} \leq \text{Tr}(T_X(t)). \quad (268)$$

Choosing $t = 3/(2R)$ gives (258). $\square$

Theorem C.15 (Operational Sobolev, Poincaré, and compact-screen bounds). *Assume uniformly that $\mu_X(V_X) \leq V_*$, (243) holds, the capacity–conductance comparison (48) holds, and $I_X^{\text{eff}} \geq I_*^{\text{eff}} > 0$. Then $I_X^{\text{sp}} \geq I_*^{\text{eff}}/a_+ = I_*^{\text{sp}} > 0$ and the estimates (244), (245), and (258) hold. In addition,*

$$\|(I - P_{X,R})f\|_2^2 \leq R^{-1}\mathcal{E}_X^{\text{sp}}(f), \quad P_{X,R} = \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}}). \quad (269)$$

Thus $P_{X,R}$ is a uniformly exhausted finite-rank compact-screen family. A sequence of positive cut constants tending to zero gives the normalized cut/potential obstruction (56); a zero-capacity cut is retained directly as disconnected support.

Proof. The comparison (50) converts the efficient-capacity margin into (238). The Sobolev and Poincaré estimates are Lemma C.13; the counting law is Lemma C.14. By the spectral theorem, $\Delta_X^{\text{sp}} \succeq R(I - P_{X,R})$, so (269) follows immediately. For a positive minimizing cut A , set $\phi = \mathbf{1}_A/u_X(\partial A)$. Then $\sum_e u_{e,X}|\text{d}\phi(e)| = 1$, and (51) gives $|\sum_v b_{A,X}(v)\phi(v)| = \mu_X(A)^{2/3}/u_X(\partial A) = (I_X^{\text{eff}})^{-1}$. This is an explicit feasible dual witness. A zero-capacity cut is recorded directly as disconnected support. $\square$

Proposition C.16 (Compact positive-screen upgrade). *Let $Y_X \rightarrow Y$ in a Hilbert space and let positive operators $\mathbb{A}_X$ satisfy $mI \preceq \mathbb{A}_X \preceq MI$ and $\mathbb{A}_X^{-1/2} \rightarrow \mathbb{A}^{-1/2}$ strongly. Put $Z_X = \mathbb{A}_X^{1/2}Y_X$ and suppose $Z_X \rightarrow Z$. If for every R there are finite-rank screens $S_{X,R} \rightarrow S_R$ in operator norm, with S_R compact, and*

$$\lim_{R \rightarrow \infty} \sup_X \|(I - S_{X,R})Z_X\| = 0, \quad \lim_{R \rightarrow \infty} \|(I - S_R)Z\| = 0,$$

then $Y_X \rightarrow Y$ strongly. Consequently every bounded quadratic insertion in Y_X converges in L^1 on the corresponding compact spacetime cylinder.

Proof. For fixed R , compactness of S_R converts weak convergence into $S_R Z_X \rightarrow S_R Z$ strongly, and operator-norm convergence gives $S_{X,R} Z_X - S_R Z_X \rightarrow 0$. Decompose

$$Z_X - Z = (I - S_{X,R})Z_X + (S_{X,R}Z_X - S_R Z) + (S_R - I)Z.$$

Taking the limsup and then $R \rightarrow \infty$ gives $Z_X \rightarrow Z$ strongly. Uniform coercivity and strong convergence of the inverse square roots then give $Y_X \rightarrow Y$. The quadratic statement follows from $\|B(Y_X, Y_X) - B(Y, Y)\|_1 \leq C\|Y_X - Y\|_2(\|Y_X\|_2 + \|Y\|_2)$. $\square$

C.5 Derivative-free face linearization and complete memory shorting

The first-order gravitational score is not assumed as a derivative at a zero face. It is obtained from renewal subdivision after all resolved cut-memory coordinates have been retained.

Theorem C.17 (Derivative-free renewal face linearization). *Let E be a finite-dimensional real normed space, $U \subset E$ a balanced convex neighborhood of zero, and $A : U \rightarrow \mathbb{R}$ with $A(0) = 0$. If*

$$\boxed{A(z) = 2A(z/2)} \quad (270)$$

for every $z \in U$, A is bounded near zero, and

$$|A(x+y) - A(x) - A(y)| \leq H\|x\| \|y\|$$

near zero, then A is the restriction of a unique real linear functional. More generally, if the renormalized scores $L_m(z) = 2^m A_m(2^{-m}z)$ have summable adjacent-cutoff defects and their additivity defect tends to zero, then L_m converges locally uniformly to one continuous linear functional.

Proof. For the exact branch iterate (270):

$$A(x+y) - A(x) - A(y) = 2^m [A(2^{-m}(x+y)) - A(2^{-m}x) - A(2^{-m}y)].$$

The right-hand side is bounded by $H2^{-m}\|x\|\|y\|$ and therefore vanishes as $m \rightarrow \infty$. Thus A is additive; boundedness near zero gives continuity and hence real homogeneity. On the stable branch the adjacent-cutoff estimate makes L_m locally uniformly Cauchy; passing to the limit in the additivity defect gives a continuous linear limit. $\square$

Theorem C.18 (Complete resolved-memory subtraction). *Let the resolved cut-memory state be $\rho_{\text{cq}} = \bigoplus_{\omega} p_{\omega} \rho_{\omega}$ with faithful blocks and linear perturbation $K_{\text{cq}}(z) = \bigoplus_{\omega} K_{\omega}(z)$. Put*

$$\boxed{M_{\text{cq}}(z) = \log \text{Tr} \exp(\log \rho_{\text{cq}} + K_{\text{cq}}(z)) - \bar{m}(z)}, \quad (271)$$

where $\bar{m}$ is the average linear slope. Then $M_{\text{cq}}(0) = DM_{\text{cq}}(0) = 0$, its Hessian is the sum of a classical branch covariance and a positive Kubo–Mori form, and

$$\boxed{A_{\text{sh}} = A_{\text{coarse}} - M_{\text{cq}}} \quad (272)$$

is the canonical fully memory-shortened face score. Thus unresolved memory is removed at the complete log-partition level rather than only to second order.

Proof. The block exponential and trace split over the classical label. Duhamel’s formula gives the first derivative in each faithful block and the Kubo–Mori quadratic form as the second logarithmic derivative. Differentiating the outer finite log-sum-exp adds the covariance of the branch means. Both terms are positive. Subtracting the complete log-partition therefore removes exactly the resolved cut-memory contribution and leaves the displayed shortened score. $\square$

C.6 Connection stationarity and torsion

Let $B^{IJ} = e^I \wedge e^J$, $T^I = De^I$, and $P_{\alpha,\beta} = \alpha I + \beta \star$, with the same spacetime-constant coefficients as in (61).

Theorem C.19 (Connection stationarity and the torsion alternative). *If $(\alpha, \beta) \neq (0, 0)$, then*

$$\boxed{P_{\alpha,\beta}^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}} \quad (273)$$

For the Palatini–Holst action the spinless connection Euler equation is $D(P_{\alpha,\beta}B) = 0$. For a nondegenerate coframe, the Cartan map

$$\mathcal{K}_e(T)^{IJ} = T^I \wedge e^J - e^I \wedge T^J$$

is injective. Consequently connection stationarity implies $T = 0$; at finite cutoff, a uniform lower singular-value bound on $\mathcal{K}_{e,X}$ controls the torsion norm by the connection Euler residual.

Proof. The inverse formula follows from $\star^2 = -I$ on Lorentzian bivectors. Varying the connection and integrating by parts gives the Euler source. To prove injectivity, expand $T^I = \frac{1}{2}T^I_{KL}e^K \wedge e^L$ in a coframe basis. Choosing a fourth index in $T^I \wedge e^J - e^I \wedge T^J = 0$ first eliminates all components with $K, L \neq I$; the remaining coefficients satisfy pairwise sign relations that, in four dimensions, force them all to vanish. Inverting the two maps gives the finite estimate. $\square$

C.7 Lorentz-natural classification, curvature propagation, and Palatini handoff

Define, up to overall normalization,

$$L_{\text{Palatini}} = \frac{1}{2}\epsilon_{IJKL}e^I \wedge e^J \wedge R^{KL}, \quad (274)$$

$$L_{\text{Holst}} = e^I \wedge e^J \wedge R_{IJ}, \quad (275)$$

$$L_{\text{vol}} = \frac{1}{4!}\epsilon_{IJKL}e^I \wedge e^J \wedge e^K \wedge e^L. \quad (276)$$

Lemma C.20 (Strong interpolation for the reconstructed coframes). *Let K be a compact spacetime cylinder with finite reconstructed volume measure. Suppose the common-cylinder interpolants of the finite coframes satisfy*

$$e_X \rightarrow e \quad \text{strongly in } L^2(K), \quad \sup_X \|e_X\|_{L^6(K)} \leq M_6 < \infty. \quad (277)$$

Then $e \in L^6(K)$ and, for every $2 \leq q < 6$,

$$\boxed{e_X \rightarrow e \quad \text{strongly in } L^q(K).} \quad (278)$$

More precisely, if $\theta_q = 3/q - 1/2 \in (0, 1]$, then

$$\|e_X - e\|_{L^q} \leq \|e_X - e\|_{L^2}^{\theta_q} \|e_X - e\|_{L^6}^{1-\theta_q}. \quad (279)$$

If $p > 3/2$ and $p' = p/(p-1)$, then $2p' < 6$ and

$$\boxed{e_X \wedge e_X \rightarrow e \wedge e \quad \text{strongly in } L^{p'}(K).} \quad (280)$$

Proof. The uniform L^6 bound gives a weakly convergent subsequence in $L^6(K)$. Because K has finite measure, weak L^6 convergence implies distributional convergence, while the strong L^2 convergence has the unique distributional limit e . Hence every weak L^6 cluster point is e , so $e \in L^6(K)$ and $\|e\|_6 \leq M_6$. The difference $e_X - e$ is therefore uniformly bounded in L^6 . The Riesz–Thorin/Hölder interpolation identity $1/q = \theta_q/2 + (1-\theta_q)/6$ gives (279); its right-hand side tends to zero for $q < 6$, proving (278).

For $p > 3/2$, $p' < 3$ and hence $q = 2p' < 6$. The exterior product is a bounded bilinear map on the finite-dimensional coframe fibres, so pointwise $|a \wedge b| \leq C|a||b|$. Therefore Hölder’s inequality gives

$$\|e_X \wedge e_X - e \wedge e\|_{L^{p'}} \leq C\|(e_X - e) \wedge e_X + e \wedge (e_X - e)\|_{L^{p'}} \quad (281)$$

$$\leq C(\|e_X\|_{L^{2p'}} + \|e\|_{L^{2p'}})\|e_X - e\|_{L^{2p'}} \rightarrow 0, \quad (282)$$

which proves (280). $\square$

C.7.1 Propagation and differentiated source control

Theorem C.21 (Finite curvature-energy propagation from initial data). *At a fixed cutoff let $y_X = (E_X, B_X)$, $M_X \succ 0$, K_X , L_X , and f_X be as in (63), and set $z_X^2 = y_X^T M_X y_X$. Then*

$$\frac{1}{2} \frac{d}{dt} z_X^2 \leq a_X z_X^2 + z_X \|f_X\|_{M_X}, \quad (283)$$

and therefore (65) holds. Under the stable incidence and uniform budgets (66)–(67), (68) follows.

Proof. Differentiate $z_X^2 = y_X^T M_X y_X$ and substitute (63). The principal term cancels exactly because $M_X K_X + K_X^T M_X = 0$. By (64), the remaining quadratic term is at most $2a_X z_X^2$, while Cauchy–Schwarz in the positive M_X -Gram bounds the source term by $2z_X \|f_X\|_{M_X}$. Applying the integrating-factor argument to $(z_X^2 + \varepsilon^2)^{1/2}$ and sending $\varepsilon \downarrow 0$ gives (65). The interpolation estimate in (66), integration against the physical lapse and slice measure, and the uniform budgets give (68). $\square$

Theorem C.22 (Midpoint propagation with the physical Gram transfer). *For the actual normalized recurrence (69), assume $\|T_j\|_{\text{op}} \leq e^{d_j}$, $K_j^T = -K_j$, $(L_j + L_j^T)/2 \preceq a_j I$, and $s_j a_j/2 \leq b < 1$. Then (71) holds. With the stable curvature incidence of (66) and stable interval interpolation, it gives the fully discrete curvature certificate of Theorem 4.9.*

Proof. Write $A_j = K_j + L_j$. For any real vector v ,

$$\langle v, (I - s_j A_j/2)v \rangle \geq (1 - s_j a_j/2) \|v\|^2.$$

Cauchy–Schwarz gives $\|(I - s_j A_j/2)v\| \geq (1 - s_j a_j/2) \|v\|$. Thus the finite matrix is invertible and its inverse norm is at most $(1 - s_j a_j/2)^{-1}$.

Let $v = T_j Y_j$ and first let w solve the step with zero residual. Pair $w - v = s_j A_j(w + v)/2$ with $(w + v)/2$ to obtain

$$\|w\|^2 - \|v\|^2 \leq \frac{s_j a_j}{2} (\|w\| + \|v\|)^2.$$

Dividing by $\|w\| + \|v\|$ when this number is nonzero, and treating the zero case directly, gives

$$\|w\| \leq \frac{1 + s_j a_j/2}{1 - s_j a_j/2} \|v\|.$$

The inhomogeneous solution adds $(I - s_j A_j/2)^{-1} \varepsilon_j$. Consequently

$$\|Y_{j+1}\| \leq e^{d_j} \frac{1 + s_j a_j/2}{1 - s_j a_j/2} \|Y_j\| + \frac{\|\varepsilon_j\|}{1 - b}.$$

For $0 \leq x \leq b$, $\log((1+x)/(1-x)) \leq 2x/(1-b)$. Every product of step factors is therefore bounded by $\exp(D_X + A_X^d/(1-b))$. Iteration and weighted Cauchy–Schwarz,

$$\sum_j \|\varepsilon_j\| \leq \sqrt{T_X \sum_j \|\varepsilon_j\|^2 / s_j},$$

prove (71). The size of the skew matrix K_j has entered neither estimate.

For raw curvature records y_j with Gram M_j , the transfer is $T_j = M_{j+1}^{1/2} \mathcal{T}_j M_j^{-1/2}$, where $\mathcal{T}_j$ is the actual recorded frame/transport map. Its norm excess is retained in d_j . Stable interval interpolation bounds the reconstructed slice norm by a fixed multiple of the largest endpoint norm.

Integrating with the physical lapse and adding the bounded reconstruction error yields, for an appropriate interpolation constant C_I ,

$$\|R_X\|_{L^2(K)} \leq C_I \sqrt{N_* T_X} e^{D_X + A_X^d/(1-b)} \left(\|Y_0\| + \frac{\sqrt{T_X \mathcal{D}_X}}{1-b} \right) + C_\zeta.$$

All interface and incoming boundary contributions must already be included in the actual transfer, residual or reconstruction comparison. This proof does not assert that arbitrary nonlinear coordinate changes preserve the midpoint recurrence: any such discrepancy is part of ε_j . $\square$

Differentiated gravitational source. The source norm is not the norm of an undifferentiated Einstein residual. On the actual shared-link branch the forcing contains the differentiated, volume-normalized Palatini metric row, together with the differentiated Cartan/torsion defect and the differential Bianchi defect. Lower-order transport terms are retained in a_X . To state a quantitative criterion, use the physical incidence (72), with all untransferred contributions collected in r_X^{src} . Its verification is an assertion about the actual action, connection and observation records; it is not obtained simply by introducing an auxiliary source variable.

Proposition C.23 (Exact amplification of the differentiated action current). *Let $G_X \succ 0$ define the action dual norm and let $H_X \succ 0$ be the target source Gram. For the actual linear transfer T_X , the least constant in*

$$\|\mathsf{T}_X \mathbf{a}\|_{H_X}^2 \leq \Gamma_X \|\mathbf{a}\|_{G_X^{-1}}^2$$

is (73). If $\|\mathbf{a}_X\|_{G_X^{-1}}^2 \leq C_X \Delta_X$ pointwise, then

$$\|f_X\|_{H_X}^2 \leq 2\Gamma_X C_X \Delta_X + 2\|r_X^{\text{src}}\|_{H_X}^2. \quad (284)$$

This proves Proposition 4.10. With deterministic supported upper bounds on Γ_X , the same inequality holds in expectation when the action-gradient bound holds only in expectation.

Proof. Write $\mathbf{a} = G_X^{1/2} v$. Then $\|\mathbf{a}\|_{G_X^{-1}} = \|v\|$ and

$$\|\mathsf{T}_X \mathbf{a}\|_{H_X} = \|H_X^{1/2} \mathsf{T}_X G_X^{1/2} v\|.$$

The maximum squared ratio is the largest squared singular value. A unit maximizing right singular vector attains it, so the constant is sharp and also supplies a finite witness to large source amplification. Apply $\|u+v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$ to (72). Time Cauchy–Schwarz proves (74); summing the same pointwise inequality with weights s_j proves its discrete counterpart. For varying field values the argument is pointwise before averaging. Without a deterministic supported bound, the expected estimate must retain $\mathbb{E}[\Gamma_X \|\mathbf{a}_X\|_{G_X^{-1}}^2]$, not a product of separately averaged factors. $\square$

Proposition C.24 (Sharp grid-scale source amplification). *On a periodic grid of m sites, with m divisible by four, spacing h and inner product $h \sum_i u_i v_i$, let $(D_h u)_i = (u_{i+1} - u_{i-1})/(2h)$. With the same positive source and target norms,*

$$\|D_h\|_{2 \rightarrow 2}^2 = h^{-2}. \quad (285)$$

In particular, a vanishing unscaled action-gradient norm does not imply a bounded differentiated current.

Proof. The Fourier multiplier is $i \sin(2\pi k/m)/h$, whose maximum modulus is h^{-1} and is attained at $k = m/4$. The same singular value occurs on the real sine/cosine subspace. Choose a unit vector v_h in that subspace and set $\mathbf{a}_h = h^{1/2}v_h$. Then $\|\mathbf{a}_h\|_2^2 = h \rightarrow 0$, whereas $\|D_h \mathbf{a}_h\|_2^2 = h^{-1} \rightarrow \infty$. $\square$

Thus, for this derivative transfer and uniformly controlled remaining factors, $C_X \Delta_X = O(h_X^2)$ suffices for bounded source and $o(h_X^2)$ suffices for vanishing source. These rates are not inferred from $\Delta_X \rightarrow 0$. Cancellations in the actual Bianchi transfer may improve Γ_X , but they must be retained in that map and its physical Gram.

Corollary C.25 (Expected curvature energy and a sufficient pathwise upgrade). *Suppose the finite propagation equation holds pathwise on $[0, T_*]$, with $\int a_X \leq A_*$ deterministically. Suppose its forcing satisfies (72), with the target norm already transferred to the curvature-energy norm, and $\mathbb{E}\|\mathbf{a}_X(t)\|_{G_X^{-1}}^2 \leq C_X(t)\Delta_X(t)$. Using deterministic supported upper bounds on Γ_X , one obtains*

$$\begin{aligned} \mathbb{E} \sup_{t \leq T_*} z_X(t)^2 &\leq 2e^{2A_*} \left[\mathbb{E} z_X(0)^2 + 2T_* \int_0^{T_*} \Gamma_X C_X \Delta_X dt \right. \\ &\quad \left. + 2T_* \mathbb{E} \int_0^{T_*} \|r_X^{\text{src}}\|_{H_X}^2 dt \right]. \end{aligned} \quad (286)$$

Together with the stable interpolation and lapse bounds, uniformity of these budgets gives a uniform expected squared spacetime curvature norm. For a countable cofinal sequence, a sufficient pathwise strengthening is

$$\sup_X z_X(0) < \infty \text{ almost surely, } \quad \sum_X \mathbb{E} \int_0^{T_*} \|f_X(t)\|_{H_X}^2 dt < \infty, \quad (287)$$

with pathwise uniform growth, interpolation, lapse and boundary budgets. Under these conditions $\sup_X \|R_X\|_{L^2(K)} < \infty$ almost surely.

Proof. The pathwise estimate gives $\sup_t z_X(t) \leq e^{A_*}(z_X(0) + \int \|f_X\|)$. Square and apply Cauchy–Schwarz in time to obtain

$$\mathbb{E} \sup_t z_X(t)^2 \leq 2e^{2A_*} \left(\mathbb{E} z_X(0)^2 + T_* \mathbb{E} \int_0^{T_*} \|f_X\|_{H_X}^2 dt \right).$$

Apply Proposition C.23 and the interpolation estimate. For the last assertion, Tonelli makes the sum of the nonnegative integrated source energies finite almost surely. Their supremum is then finite, and the deterministic propagation estimate applies to the whole realized sequence. A uniform expected norm alone gives boundedness in probability by Markov’s inequality; it is not substituted for this pathwise argument. $\square$

C.7.2 Curvature identification and interface lifting

Theorem C.26 (Weak curvature compactness with an identified connection limit). *Let K be a compact cylinder and $1 < p < \infty$. A bound*

$$\sup_X \|R_X\|_{L^p(K)} \leq C_{K,p} < \infty \quad (288)$$

gives a weakly convergent subsequence

$$\boxed{R_X \rightharpoonup R \text{ in } L^p(K).} \quad (289)$$

If, in addition, $\omega_X \rightarrow \omega$ strongly in L^2_{loc} and

$$R_X - [\text{d}_{\text{dist}}\omega_X + \omega_X \wedge \omega_X] \rightarrow 0 \quad \text{in distributions,} \quad (290)$$

then $R = \text{d}\omega + \omega \wedge \omega$ distributionally. The finite-writer conditions (75)–(76), with exhaustive determining tests and their interpolation control, give (290). In particular the propagated $p = 2$ bound proves [Theorem 4.11](#).

Proof. For $1 < p < \infty$, reflexivity gives weak sequential compactness of bounded balls of $L^p(K)$. Strong L^2 convergence of the connection gives convergence of its distributional derivative, and

$$\|\omega_X \wedge \omega_X - \omega \wedge \omega\|_{L^1(K)} \leq C(\|\omega_X\|_2 + \|\omega\|_2)\|\omega_X - \omega\|_2 \rightarrow 0.$$

Thus (290) identifies the distributional limit of R_X . Weak L^p convergence has the same distributional limit, proving the assertion. For the finite-writer version, split the tested residual into $R_X - \mathcal{R}_X(\omega_X)$ and the discrepancy (76). The first tends to zero on the exhaustive core by (75), and the second by the stated consistency. The declared test-interpolation control extends the identity to the determining test topology. A diagonal subsequence works on a countable compact exhaustion.

If the curvature norms are instead unbounded, choose a subsequence on which they diverge and retain the unit fields $R_X/\|R_X\|_{L^p(K)}$ together with their diverging scales. This witnesses failure of the specified curvature-size certificate, not nonexistence of every possible weaker limit. $\square$

For a piecewise smooth connection, its full distributional derivative includes face measures. A bound on cell interiors alone therefore does not control the curvature entering the continuum variation. The following construction specifies a sufficient volume realization without discarding these measures.

Work on a periodic rectangular four-dimensional domain with a uniform cubical mesh of side h , or on a compact interior subdomain together with its neighbouring cells. Polynomial degrees are fixed, positive Grams and physical measures are uniformly comparable to the reference ones, and all face coefficients are compared in a common local trivialization. For an oriented interior face f , set $J_f = n_f^\flat \wedge [\omega_h]_f$, with the sign chosen so that

$$\text{d}_{\text{dist}}\omega_h = \text{d}_b\omega_h + \sum_f J_f \delta_f. \quad (291)$$

Let $\eta \geq 0$ be a fixed smooth function supported in $(-1/4, 1/4)$ with integral one. On the normal prism of f , define

$$(\mathcal{L}_f J_f)(y + rn_f) = h^{-1} \eta(r/h) J_f(y), \quad \mathcal{L}_h J = \sum_f \mathcal{L}_f J_f, \quad (292)$$

and extend each term by zero outside its prism.

Lemma C.27 (Interface lifting and its distributional error). *With $\mathcal{J}_h$ defined in (79),*

$$\|\mathcal{L}_h J\|_2 \leq C \mathcal{J}_h, \quad \left| \left\langle \mathcal{L}_h J - \sum_f J_f \delta_f, \Phi \right\rangle \right| \leq C_K h \mathcal{J}_h \|\nabla \Phi\|_\infty \quad (293)$$

for each fixed smooth compactly supported or periodic test Φ .

Proof. Integration in the normal coordinate gives

$$\|\mathcal{L}_f J_f\|_2^2 = h^{-1} \|\eta\|_{L^2(\mathbb{R})}^2 \|J_f\|_{L^2(f)}^2.$$

The prisms have bounded overlap depending only on dimension. Pointwise Cauchy–Schwarz over the overlapping prisms proves the first bound. For the second, subtract $\Phi(y)$ from $\Phi(y + rn_f)$ and use $|r| \leq h/4$. The total error is at most $Ch\|\nabla\Phi\|_\infty \sum_f \|J_f\|_{L^1(f)}$. Moreover

$$\sum_f h \|J_f\|_{L^1(f)} \leq \left(\sum_f h^{-1} \|J_f\|_{L^2(f)}^2 \right)^{1/2} \left(\sum_f h^3 |f| \right)^{1/2} \leq C_K h \mathcal{J}_h.$$

There are $O_K(h^{-4})$ faces of area h^3 , which proves the last inequality and the result. $\square$

Theorem C.28 (Constructive connection-and-interface curvature criterion). *Suppose*

$$\|d_b \omega_h\|_{L^2(K)} + \|\omega_h\|_{L^4(K)}^2 + \mathcal{J}_h \leq C_K. \quad (294)$$

Then the volume curvature R_h^{lift} in (80) is uniformly bounded in $L^2(K)$ and has the displayed distributional error. If also $\omega_h \rightarrow \omega$ strongly in L_{loc}^2 , every weak L^2 cluster point equals $d\omega + \omega \wedge \omega$ distributionally.

Proof. The matrix exterior product satisfies $|\omega_h \wedge \omega_h| \leq C|\omega_h|^2$. Hölder’s inequality and Lemma C.27 give (80). The discrepancy from $d_{\text{dist}}\omega_h + \omega_h \wedge \omega_h$ is exactly the lifting error in (293), which tends to zero on each fixed test. Apply Theorem C.26 with $p = 2$. $\square$

The theorem is a sufficient identification and size criterion with its own explicit connection and jump budgets. A stationary connection estimate in L^2 does not imply (294). Nor is R_h^{lift} silently substituted for the action’s curvature writer: that writer must agree with the lift on the determining tests up to a vanishing recorded discrepancy. On a conforming connection reconstruction $J_f = 0$, and the interface part is absent.

C.7.3 The weak–strong Palatini passage

Lemma C.29 (Weak–strong curvature pairing and volume convergence). *Assume (277), let $p > 3/2$, and suppose*

$$R_X \rightharpoonup R \quad \text{weakly in } L^p(K). \quad (295)$$

Then, for every bounded compactly supported coefficient tensor Φ ,

$$\boxed{\int_K \langle \Phi(e_X \wedge e_X), R_X \rangle \longrightarrow \int_K \langle \Phi(e \wedge e), R \rangle.} \quad (296)$$

The same conclusion holds with an internal Hodge star applied to either bivector factor. Moreover

$$\boxed{e_X \wedge e_X \wedge e_X \wedge e_X \longrightarrow e \wedge e \wedge e \wedge e \quad \text{strongly in } L^1(K).} \quad (297)$$

Proof. Weak convergence in L^p implies $\sup_X \|R_X\|_{L^p} < \infty$. Put $B_X = \Phi(e_X \wedge e_X)$ and $B = \Phi(e \wedge e)$. By Lemma C.20, $B_X \rightarrow B$ strongly in $L^{p'}$. Split the difference as

$$\int_K \langle B_X, R_X \rangle - \int_K \langle B, R \rangle = \int_K \langle B_X - B, R_X \rangle + \int_K \langle B, R_X - R \rangle. \quad (298)$$

The first term tends to zero by Hölder and the uniform L^p bound on R_X ; the second tends to zero by the definition of weak L^p convergence, because $B \in L^{p'}$. The internal Hodge star is a fixed bounded linear map on the bivector fibre and changes none of these estimates.

For the volume term, [Lemma C.20](#) with $q = 4$ gives $e_X \rightarrow e$ strongly in $L^4(K)$ and uniform L^4 boundedness. Expanding the fourfold difference one factor at a time and applying Hölder with four L^4 factors gives

$$\|e_X^{\wedge 4} - e^{\wedge 4}\|_{L^1} \tag{299}$$

$$\leq C\|e_X - e\|_{L^4}(\|e_X\|_{L^4} + \|e\|_{L^4})^3 \rightarrow 0, \tag{300}$$

which is [\(297\)](#). $\square$

Remark C.30 (Weak–strong passage without compensated compactness). The product passage above does not use a div–curl or compensated-compactness lemma. Such a device would be needed if both the coframe bivector and curvature were only weakly convergent. Here the compact-screen hypothesis gives strong L^2 convergence of the coframe together with a uniform L^6 bound; the explicit interpolation in [Lemma C.20](#) upgrades this to strong $L^{2p'}$, so ordinary weak–strong duality suffices. At this stage all fields are already represented by their canonical common-cylinder interpolants; equivalently, the same estimates may be read cellwise with the physical cell measures because the displayed L^q hypotheses are precisely the cutoff-uniform interpolation bounds used in the handoff.

Theorem C.31 (Localized renewal–Palatini handoff). *Assume the face-linearization and complete memory-short branches, real Lorentz naturality, scalar residual typed commutant on the bivector carrier, and represented determinant volume. Let the constant calibrated coefficients converge to (α, β, λ) with $\beta \neq 0$, and let the spinless connection Euler residual vanish with the uniform Cartan inverse of [Theorem C.19](#). On every compact K , assume [\(277\)](#), nondegeneracy of the limiting coframe, and either the continuous or the discrete initial-energy curvature certificate. Assume also the connection identification of [Theorem C.26](#) and the bounded, convergent physical test lifts of (E1) in [Theorem 4.19](#). Then the limiting score has the classified form*

$$\boxed{\alpha L_{\text{Holst}} + \beta L_{\text{Palatini}} + \lambda L_{\text{vol}}}, \tag{301}$$

limiting torsion vanishes, and the metric first variations converge to

$$\boxed{\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad \chi \neq 0.} \tag{302}$$

Here χ is the nonzero Palatini normalization and Λ is the corresponding calibrated volume-to-Palatini coefficient ratio, in the density conventions of [\(276\)](#).

Proof. Both coframe bivectors and Lorentz curvature lie in $\Lambda^2 \mathbb{R}^{1,3}$. After complexification the bivector module splits into nonisomorphic self-dual and anti-self-dual irreducibles. The real commutant is therefore

$$\text{End}_{SO^+(1,3)}(\Lambda^2 \mathbb{R}^{1,3}) = \text{span}_{\mathbb{R}}\{I, \star\}. \tag{303}$$

The residual typed commutant supplies no additional internal tensor, and the only alternating four-coframe scalar at this order is determinant volume. Thus [\(301\)](#) follows in the stated first-order class. The calibrated coefficient limits ensure that these are the coefficients of the limiting variational functional, rather than unrelated values chosen after taking the limit.

By [Theorem C.19](#), the vanishing connection Euler residual and the uniform inverse bounds give vanishing torsion. Strong local L^2 convergence of e_X and ω_X passes their product in L^1 ; the

distributional derivative of e_X passes against every fixed smooth test. The limit is therefore a metric-compatible torsion-free represented connection. Its curvature is identified by [Theorem C.26](#), not merely by matching finite curvature labels.

The continuous or discrete propagation theorem supplies $\sup_X \|R_X\|_{L^2(K)} < \infty$. After passage to the identified weak subsequence, [Lemma C.20](#) gives $e_X \wedge e_X \rightarrow e \wedge e$ strongly in L^2 . The curvature terms therefore converge by [Lemma C.29](#) with $p = p' = 2$. The same argument applies to their physical coframe variations: their coefficient tensors are uniformly bounded and converge, so multiplying the strong L^2 bivectors by these tensors preserves strong convergence. The four-coframe volume term and its test insertions converge in L^1 by the same lemma. No product of two uncontrolled weak limits is used.

For the torsion-free connection, the first Bianchi identity removes the Holst coframe variation. The Palatini and volume variations give the Einstein and cosmological insertions in [\(302\)](#). The coefficient of the first is nonzero because $\beta \neq 0$. A nonzero coefficient pair alone would only justify the torsion step, and would not justify division by this Einstein coefficient. $\square$

Corollary C.32 (Vacuum equation from certified accepted-action variations). *Under the preceding hypotheses, vanishing realized common-action variations on the determining metric-test core imply*

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0}$$

distributionally on every compact cylinder. The scaled hypothesis [\(235\)](#) is a sufficient stochastic route on the almost-sure subsequence of [Proposition C.9](#), provided the remaining compactness and identification budgets hold on the same realized sequence.

Proof. The handoff identifies the limit of each certified first variation with [\(302\)](#). Its value is zero on the determining core, and $\chi \neq 0$ permits division. Continuity in the declared test topology extends the identity to the closure of that core. In the stochastic case, first take the realized vanishing subsequence supplied by [Proposition C.9](#); the assumed pathwise compactness budgets then permit the curvature subsequence used in the handoff. A mean-square bound is not interpreted as stationarity of every finite sample. $\square$

C.8 Proof of the certification-or-obstruction alternative

Proof of Theorem 4.21. For a complete cylinder, [Theorem C.1](#) reconstructs the finite determining field and accepted kernel. The reward-pressure and common-action theorems either reconstruct the common action or expose its curl/period failure. The accepted-action theorem and the physical test-lift estimate give a quantitative stationarity certificate with every cutoff constant retained. The spatial cut theorem gives the stated spatial screen or a cut witness, while the common-cylinder comparison packet separately records the temporal and transported-screen conditions needed for strong coframe convergence.

Face linearization and memory subtraction give the classified first-order score on their positive branches. The calibrated coupling and Cartan tests supply the nonzero Einstein coefficient and torsion control. Continuous or fully discrete propagation bounds the actual curvature from the finite initial energy and growth/transfer/source records; the sharp current transfer may supply the last budget from scaled action errors. Finally the connection and interface comparisons identify that curvature. If all certificates pass, [Theorem C.31](#) and [Corollary C.32](#) prove branch (A1), using a diagonal subsequence on the compact exhaustion.

For completeness, the cofinal witnesses in branch (A2) are obtained from the quantifiers of the failed certificates. A nonnegative residual which does not tend to zero has a subsequence bounded

below by some $\epsilon > 0$. An unbounded budget has a subsequence tending to infinity. A failed strictly positive lower margin has a subsequence tending to zero; for the cut margin the finite graph supplies the corresponding potential witness. Failure of a uniform compactness tail supplies an $\epsilon > 0$ and normalized components outside arbitrarily large declared screens. Failure of a required calibrated Cauchy comparison retains the separated pairs of finite records themselves. A finite algebraic failure retains its actual curl, period or operator residual. Discarding a finite prefix is allowed when checking the cofinal tail. These records identify which sufficient certificate failed; they do not establish that every alternative compactness argument or every other subsequence must fail. $\square$

Remark C.33 (What is discharged and what remains dynamical). Neither a continuum curvature bound nor an unidentified nonlinear connection product is inserted into the conclusion. The former follows from the continuous or discrete propagation certificate; the latter is handled by strong connection convergence and the full distributional Cartan comparison, including interfaces. The differentiated source budget can follow from (74), whereas bounded symmetric growth, physical occurrence and noncollapse remain genuine conditions of the microscopic evolution. A bounded mean curvature norm does not by itself supply the pathwise version. The fixed-amplitude nonsymmetric Cauchy theorem is supplied independently by Appendix D for its declared local writer; the Gowdy sector below remains a complementary nonlinear symmetry-reduced class with pathwise full-curvature control.

D Open nonsymmetric 3 + 1 population

This section proves Theorem 4.13 directly from the declared local writer. The purpose is to supply a fixed-radius, genuinely three-dimensional nonlinear population theorem on the same reconstructed Lorentzian coefficient chart used by the general certification theorem, without invoking (E1)–(E5) as independent assumptions. The evolution law is declared explicitly and is kept logically distinct from the exact finite Euler flow studied in Appendix F. No symmetry reduction or analytic Fourier tail is used.

D.1 Execution graph, exact difference calculus, and the local writer

Let $N = 2m + 1$, $h = N^{-1}$ and $\Lambda_h = (h\mathbb{Z}/\mathbb{Z})^3$. Let $\mathcal{I}_h$ denote the real trigonometric interpolation on the signed frequency cube $\{-m, \dots, m\}^3$, and let $\mathcal{S}_h$ denote grid sampling. The tetrahedral $A_3 \cong D_3$ root execution directions may be represented as

$$\mathcal{R}_F = \{\pm(e_i + e_j), \pm(e_i - e_j) : 1 \leq i < j \leq 3\}. \quad (304)$$

All these integer vectors have even coordinate sum. This harmless parity at fixed smooth frequency becomes relevant for a cutoff-uniform Cartesian Sobolev realization.

Lemma D.1 (Root-parity obstruction and a minimal execution connector). *On the odd periodic grid, the root Dirichlet form*

$$\mathcal{D}_F(u) = \sum_{r \in \mathcal{R}_F / \{\pm 1\}} \|(S_r - I)u/h\|_h^2 \quad (305)$$

is not uniformly coercive for $\|\nabla \mathcal{I}_h u\|_{L^2}^2$. Moreover the root-graph distance from 0 to e_1 is $N - 1$. If the oriented execution edges $\pm e_1$ are adjoined, then with

$$\mathcal{D}_+(u) = \mathcal{D}_F(u) + \|D_{e_1}^+ u\|_h^2, \quad \mathcal{D}_C(u) = \sum_i \|D_i^+ u\|_h^2, \quad (306)$$

one has

$$\boxed{\frac{1}{5}\mathcal{D}_C(u) \leq \mathcal{D}_+(u) \leq 9\mathcal{D}_C(u).} \quad (307)$$

At least one added fixed direction of odd coordinate sum is necessary within this fixed-direction class. The connector is an execution incidence only; it does not enter the A_3 moment reconstruction.

Proof. Take $u_h(x) = \exp(2\pi i m(x_1 + x_2 + x_3))$. Every difference in a minus root vanishes and every plus-root multiplier remains bounded, while $\|\nabla \mathcal{I}_h u_h\|_2 \asymp m$ diverges. The same argument applies to every fixed-radius word in even-parity directions. A root path ending at e_1 has integer displacement $e_1 + Nz$ with even coordinate sum; since N is odd, $\sum z_i$ is odd and one coordinate has modulus at least $N - 1$. The explicit alternating pair $-e_1 + e_2, -e_1 - e_2$, repeated m times, attains that distance.

For the connector, the identity

$$S_{e_2} - I = S_{e_1}^{-1}[(S_{e_1+e_2} - I) - (S_{e_1} - I)]$$

and its e_3 analogue give $\mathcal{D}_C \leq 5\mathcal{D}_+$. Conversely each root difference is bounded by the sum of the two participating coordinate differences, so $\mathcal{D}_F \leq 8\mathcal{D}_C$. Adding the connector gives the upper bound. If every added direction had even coordinate sum, the high-frequency test above would remain unchanged. $\square$

Define $S_i, D_i^\pm, D_i^0$ by (81), use $\langle u, v \rangle_h = h^3 \sum_x u(x) \cdot v(x)$, and for a multi-index put $D^\alpha = \prod_i (D_i^+)^{\alpha_i}$ and $S^\alpha = \prod_i S_i^{\alpha_i}$. Set

$$\|u\|_{r,h}^2 = \sum_{|\alpha| \leq r} \|D^\alpha u\|_h^2, \quad \|X\|_{\mathcal{X}_h^r}^2 = \|q\|_{r+1,h}^2 + \|v\|_{r,h}^2. \quad (308)$$

Lemma D.2 (Exact shifted product rule and uniform Sobolev calculus). *On the periodic array space,*

$$(D_i^+)^* = -D_i^-, \quad (D_i^0)^* = -D_i^0, \quad (309)$$

and

$$\boxed{D^\alpha(fw) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (S^{\alpha-\beta} D^\beta f) D^{\alpha-\beta} w.} \quad (310)$$

For $r \geq 3$, the norms (308) are uniformly equivalent to the trigonometric H^r norms, form a uniform product algebra, and the shifted commutator

$$\mathcal{C}_\alpha(f, w) = D^\alpha(fw) - (S^\alpha f) D^\alpha w \quad (311)$$

satisfies, for a constant coefficient f_* ,

$$\|\mathcal{C}_\alpha(f, w)\|_h \leq C_r \|f - f_*\|_{r,h} \|w\|_{r-1,h}, \quad (312)$$

$$\|D_i^- \mathcal{C}_\alpha(f, w)\|_h + \|D_i^0 \mathcal{C}_\alpha(f, w)\|_h \leq C_r \|f - f_*\|_{r+1,h} \|w\|_{r,h}. \quad (313)$$

The constants are independent of h .

Proof. Translation of the finite sum gives the adjoints, and $D_i^+(fw) = (S_i f) D_i^+ w + (D_i^+ f) w$ gives (310) by iteration. The forward symbol $d_i(k) = (e^{2\pi i h k_i} - 1)/h$ satisfies $4|k_i| \leq |d_i(k)| \leq 2\pi|k_i|$ on $|k_i| \leq m$. Parseval therefore gives uniform norm equivalence. Multiplication is convolution on

the finite frequency group and the exact identity $d_i([k+p]) = d_i(k) + e^{2\pi i h k_i} d_i(p)$ gives the usual uniform weighted convolution estimate. In every commutator term at least one difference falls on f ; one additional output difference gives the second line. Since $r > 3/2$, the required Fourier ℓ^1 weights are summable uniformly in the finite cube. Analytic coefficient compositions follow from their convergent power series on a smaller ball. $\square$

For the harmonic reduced vacuum equation use the coefficient functions in (82). The skew transport is (83) and the first-jet compensator is (84). Pointwise multiplication operators are denoted M_f .

Proposition D.3 (Locality and continuum consistency of the writer). *The acceleration in (85) at a site depends only on the site and the twelve coordinate-stencil neighbours $\{\pm e_i, \pm(e_i - e_j) : i < j\}$. These lie within two hops of the execution graph obtained in Lemma D.1. On every smooth metric jet, the continuum limit of the second row is exactly (82).*

Proof. The flux at x is

$$h^{-2} \sum_{i,j} \{c^{ij}(x)[q(x + he_j) - q(x)] - c^{ij}(x - he_i)[q(x - he_i + he_j) - q(x - he_i)]\}.$$

The skew transport uses only $x \pm he_i$, and the compensator only first forward differences. In the smooth limit the divergence contributes $(\partial_i c^{ij}) \partial_j q$ and the skew transport contributes $(\partial_i b^i) v$ in addition to the desired second- and mixed-derivative terms. The two coefficient-differential terms in (84) cancel exactly these additions by the ordinary chain rule. No discrete Leibniz rule is invoked. $\square$

D.2 Shifted nonlinear energy and a cutoff-independent lifespan

On a sufficiently small fixed $\mathcal{X}_h^s$ ball, $a \geq a_* > 0$ and $c \succeq c_* I$. Put $a_\alpha = S^\alpha a$, $c_\alpha = S^\alpha c$, $q_\alpha = D^\alpha q$, and $v_\alpha = D^\alpha v$, and define

$$\mathcal{E}_{s,h}(X) = \frac{1}{2} \sum_{|\alpha| \leq s} \left[\langle v_\alpha, a_\alpha v_\alpha \rangle_h + \sum_{i,j} \langle D_i^+ q_\alpha, c_\alpha^{ij} D_j^+ q_\alpha \rangle_h + \|q_\alpha\|_h^2 \right]. \quad (314)$$

It is uniformly equivalent to $\|X\|_{\mathcal{X}_h^s}^2$.

Theorem D.4 (All-component shifted nonlinear energy). *For $s \geq 11$, every solution of (85) remaining in the fixed chart satisfies*

$$\frac{d}{dt} \mathcal{E}_{s,h} \leq C_s \mathcal{E}_{s,h} + C_s \mathcal{E}_{s,h}^{3/2}, \quad (315)$$

with C_s independent of h .

Proof. Apply D^α to the mass row. The exact shifted product rule gives

$$a_\alpha \dot{v}_\alpha = \sum_{i,j} D_i^- (c_\alpha^{ij} D_j^+ q_\alpha) - \mathbf{K}_{b_\alpha} v_\alpha + \mathbf{R}_\alpha, \quad (316)$$

where every term in R_α contains either a shifted commutator or the quadratic first-jet source. In particular the outer-divergence commutator is retained, rather than treated as an undifferentiated error. [Lemma D.2](#) and the analytic coefficient bounds imply

$$\left(\sum_{|\alpha| \leq s} \|R_\alpha\|_h^2 \right)^{1/2} \leq C_s \|X\|_{\mathcal{X}_h^s}^2. \quad (317)$$

Pair (316) with v_α . Summation by parts cancels the divergence against the derivative of the spatial energy, while $\langle v_\alpha, K_{b\alpha} v_\alpha \rangle_h = 0$. Thus

$$\begin{aligned} \dot{\mathcal{E}}_{s,h} = \sum_{|\alpha| \leq s} & \left[\frac{1}{2} \langle v_\alpha, \dot{a}_\alpha v_\alpha \rangle_h + \frac{1}{2} \sum_{i,j} \langle D_i^+ q_\alpha, \dot{c}_\alpha^{ij} D_j^+ q_\alpha \rangle_h \right. \\ & \left. + \langle q_\alpha, v_\alpha \rangle_h + \langle v_\alpha, R_\alpha \rangle_h \right]. \end{aligned} \quad (318)$$

The coefficient time derivatives are analytic multiples of v and have ℓ^∞ size $O(\|X\|_{\mathcal{X}_h^s})$. Energy equivalence, Cauchy–Schwarz and (317) prove the result. The translated coefficient in each differentiated row is essential: replacing it by an unshifted continuum Leibniz rule would leave uncanceled top-order terms. $\square$

Corollary D.5 (Common nonlinear time and forced stability). *There are $\varepsilon_s, T_s, C_s > 0$, independent of h , such that every $\|X_h(0)\|_{\mathcal{X}_h^s} \leq \varepsilon_s$ has a unique solution through T_s satisfying (88). If an acceleration defect f_h is added, then on the same smaller chart*

$$(\mathcal{E}_{s,h}^{1/2})' \leq C_s \mathcal{E}_{s,h}^{1/2} + C_s \mathcal{E}_{s,h} + C_s \|f_h\|_{s,h}. \quad (319)$$

Two top-bounded records satisfy the corresponding lower-order difference estimate

$$\sup_{t \leq T_s} \|X_h - Y_h\|_{\mathcal{X}_h^r} \leq C_s \left(\|X_h(0) - Y_h(0)\|_{\mathcal{X}_h^r} + \int_0^{T_s} \|f_h - \tilde{f}_h\|_{r,h} dt \right) \quad (320)$$

for $3 \leq r \leq s-1$.

Proof. The finite ODE is analytic on the coefficient chart. A scalar comparison for (315) keeps its energy strictly inside a smaller fixed ball on a time chosen before the cutoff; finite-dimensional continuation then gives the common lifespan. Adding the differentiated force to the remainder proves (319). For differences, put one record's coefficients in the principal row and every coefficient difference in the source. The product and commutator estimates at one lower order, followed by Gronwall, give (320) without an $e^{CT/h}$ loss. $\square$

D.3 Compatible initial data, subsidiary propagation, and the Einstein limit

For $(\gamma, K) \in \mathcal{D}_s(\varepsilon)$ define the harmonic Cauchy normalization

$$\begin{aligned} g_{00}^0 &= -1, & g_{0i}^0 &= 0, & g_{ij}^0 &= \gamma_{ij}, \\ v_{ij}^0 &= -2K_{ij}, & v_{00}^0 &= 2 \operatorname{tr}_\gamma K, & v_{0i}^0 &= \gamma^{jk} (\partial_j \gamma_{ki} - \frac{1}{2} \partial_i \gamma_{jk}). \end{aligned} \quad (321)$$

Let $\mathcal{S}_h$ be grid sampling and initialize $q_h(0) = \mathcal{S}_h(g^0 - \eta)$, $v_h(0) = \mathcal{S}_h v^0$ with the harmless finite projection needed for the canonical interpolation. Sampling a future Einstein history is not part of this preparation.

Let $c^\mu(g) = g^{\alpha\beta}\Gamma_{\alpha\beta}^\mu$ be the harmonic gauge vector. For a smooth interpolant the reduced equation may be written

$$G(g_h) - \mathfrak{H}(g_h, c(g_h)) = r_h, \quad (322)$$

where $\mathfrak{H}$ is the standard linear first-derivative gauge insertion. The consistency estimates following from [Lemma D.2](#) and [Proposition D.3](#) give

$$\|r_h\|_{C_t H^{s-2}} + \|\partial_t r_h\|_{C_t H^{s-3}} \leq C_s h\varepsilon \quad (323)$$

on the common chart.

Lemma D.6 (Subsidiary control on the common interval). *For the harmonic initialization (321),*

$$\sup_{t \leq T_s} (\|c(g_h)\|_{H^{s-2}} + \|\partial_t c(g_h)\|_{H^{s-3}} + \|G(g_h)\|_{H^{s-3}}) \leq C_s h\varepsilon. \quad (324)$$

Proof. The continuum constraints in (87) and the harmonic normalization make the unprojected initial gauge field and its first time derivative zero. Sampling and the actual finite acceleration perturb these quantities by $O(h\varepsilon)$ at the displayed orders. The contracted Bianchi identity applied to (322) gives the inhomogeneous subsidiary system

$$\square_{g_h} c_b + \text{Ric}_b{}^d(g_h) c_d = -2\nabla^a r_{h,ab} + \mathcal{O}(h\varepsilon)$$

on the smooth canonical interpolant. The coefficients are uniformly hyperbolic on the common Cauchy ball. The standard coordinate wave energy, commuted through $s-3$ spatial derivatives, and (323) give the first two terms by Gronwall. Equation (322) then gives the Einstein residual. The argument is performed after the independent nonlinear energy has supplied the common interval; no stopped-time assumption is used. $\square$

Theorem D.7 (Whole-sequence nonsymmetric vacuum limit). *For sufficiently small $(\gamma, K) \in \mathcal{D}_s(\varepsilon)$, the complete sequence of interpolated solutions of (85) converges on $[0, T_s] \times \mathbb{T}^3$ to a unique g with those initial data,*

$$\boxed{G(g) = 0}, \quad (325)$$

and the quantitative estimate

$$\sup_{t \leq T_s} (\|g_h - g\|_{H^{s-1}} + \|\partial_t g_h - \partial_t g\|_{H^{s-2}} + \|\partial_t^2 g_h - \partial_t^2 g\|_{H^{s-3}}) \leq C_s h\varepsilon \quad (326)$$

holds. In particular the corresponding Levi-Civita curvatures converge one further derivative below, and are uniformly bounded on the slab.

Proof. The uniform $H^{s+1} \times H^s$ bound and the actual time-jet bounds obtained from the equation give compactness in every slightly weaker $C_t H^{s+1-\delta} \times C_t H^{s-\delta}$ topology. Pass to a cluster point. The exact difference calculus and consistency (323) pass the local writer to the harmonic reduced equation, while [Lemma D.6](#) gives $c(g) = 0$. Hence $G(g) = 0$ and the initial data are the prescribed pair.

If two cluster points have the same data, subtract their harmonic reduced wave systems and apply the ordinary Sobolev energy at index $s-2$; the coefficient differences are lower-order sources controlled by the bounded top norms, so Gronwall gives equality. Thus every cluster point is the same and the entire sequence converges. To obtain the rate, subtract the sampled limit equation from the finite writer. First-difference and product consistency give an $O(h\varepsilon)$ source in H^{s-2} and the sampled initial tail is $O(h\varepsilon)$ in the lower Cauchy norm. Apply (320); solving the mass-row difference for the acceleration gives the third term. Curvature is an analytic function of a nondegenerate metric and its first two jets at these Sobolev orders, proving the final assertion. $\square$

Corollary D.8 (Direct constructive Einstein limit). *The Einstein conclusion of [Theorem D.7](#) follows from the cutoff-uniform nonlinear energy, consistency of the declared local writer, harmonic subsidiary estimate, and uniqueness of the limiting reduced Cauchy problem. No item (E1)–(E5) of [Theorem 4.19](#) is invoked as an independent hypothesis. Accordingly those five items function as sufficient certification conditions for the broader class of complete renewal cylinders, while [Theorem 4.13](#) supplies an independently constructed non-symmetry-reduced positive population.*

D.4 Positive rate lift and finite renewal words

The small metric chart has an explicit positive A_3 rate lift. In the D_3 realization

$$r_a \in \{(1, 1, 0), (1, -1, 0), (1, 0, 1), (1, 0, -1), (0, 1, 1), (0, 1, -1)\}, \quad (327)$$

put $B = N_{\text{ADM}}^2 \gamma^{-1}$, $\varrho = \sqrt{\det \gamma}$ and, for a coordinate pair ij with remaining index k ,

$$c_{ij,\pm} = \frac{B_{ii} + B_{jj} - B_{kk}}{4} \pm \frac{B_{ij}}{2}, \quad d_a = \frac{1}{4} r_a \cdot \beta, \quad k_a^\pm = \frac{c_a}{2h^2} \pm \frac{d_a}{2h}. \quad (328)$$

At the flat point all $c_a = 1/4$, so a fixed small metric chart makes all rates positive and comparable to h^{-2} . Direct calculation gives

$$h^2 \sum_a (k_a^+ + k_a^-) r_a r_a^T = B, \quad h \sum_a (k_a^+ - k_a^-) r_a = \beta, \quad (329)$$

so the ADM reconstruction returns the same metric. The resulting weighted periodic graph has a cutoff-uniform cut margin by perturbation of the flat D_3 regulator and [Theorem 4.6](#).

To obtain finite histories rather than a continuous-time finite ODE, fix a segment duration $\tau_h \asymp h^2$. At a source $x = (q, v)$ use finite registers $z = (\alpha, \beta)$ and the degree-seven two-endpoint Hermite curve whose endpoint acceleration and jerk are those of the local vector field. Evaluate the normalized equation residual at two fixed interior times and apply the inverse of the resulting universal two-slot Hermite matrix. Denote the resulting local residual map by $\mathcal{D}_x(z)$.

Theorem D.9 (Finite local successor words). *On the fixed small Cauchy ball there are constants independent of h such that, for $\tau_h \leq c_s h^2$,*

$$\|D_z \mathcal{D}_x - I\|_{\text{op}} + \|D_z^2 \mathcal{D}_x\|_{\text{op}} \leq C_s \tau_h / h. \quad (330)$$

With zero initial register, six parallel residual corrections and absolute local arithmetic error at most $c_s \varepsilon h^6$ suffice, for all sufficiently small h , to commit a successor with fixed execution-graph radius. Sharing endpoint acceleration and jerk caches between adjacent segments gives a finite piecewise-Hermite history satisfying

$$\sup_t \|(Q_h, \dot{Q}_h)\|_{\mathcal{X}_h^s} \leq C_s \varepsilon, \quad \|f_h^{\text{num}}\|_{C_t H_h^s} \leq C_s \varepsilon h^4, \quad \|\partial_t f_h^{\text{num}}\|_{C_t H_h^s} \leq C_s \varepsilon h^2. \quad (331)$$

If the initial finite record is $O(\varepsilon h^4)$ from the compatible preparation, every such history has the same Einstein limit as [Theorem D.7](#).

Proof. The local acceleration loses one discrete spatial order and its jerk loses one further order. Differentiating the fixed Hermite residual therefore gives (330); every residual evaluation has spatial radius three in the coordinate stencil. For $C_s h \leq 1/2$, the correction map satisfies, with local error ν_h ,

$$d_{n+1} \leq C_s h d_n + \nu_h, \quad d_0 \leq C_s \varepsilon h.$$

Thus $d_6 \leq C_s \varepsilon h^7 + 2\nu_h = O(\varepsilon h^6)$ for the stated arithmetic floor. Fixed-order Hermite remainder estimates with $\tau_h \asymp h^2$ then give the two defects in (331). The forced energy (319) prevents exit from the common chart and the lower-order difference estimate compares the finite word to the exact local trajectory. The defects are smaller than the $O(h\varepsilon)$ consistency budget in Theorem D.7, which proves the same limit. The number of communication hops is fixed because both the residual radius and the number of corrections are fixed. This proof concerns information range; it does not provide a hardware latency bound for executing those operations inside the physical duration τ_h . $\square$

The results above prove the population statement required in the main text. They do not identify the local writer with the exact finite Euler equations of the reconstructed action. That stronger question has a different finite constraint and response structure, to which we now turn.

E Open law-space robustness and finite action-prepared initial data

This section proves Theorem 4.15. The first part enlarges the central local writer to a fixed open family whose members may differ substantially at the ultraviolet edge while sharing one nonlinear energy and one Einstein limit. The second part places finite chronological probability rows in relatively open residual-profile neighborhoods. The third constructs exact initial zeroes of the original finite lapse-shift constraint map from free tensor records. These are three distinct notions of openness: openness of differential coefficients, openness of finite transition rows, and relative openness of the original finite constraint surface.

E.1 A fixed open family of local differential writers

Put

$$\Lambda_h = - \sum_{i=1}^3 D_i^- D_i^+, \quad \ell_h(k) = 4h^{-2} \sum_i \sin^2(\pi h k_i), \quad 0 \leq h^2 \ell_h(k) \leq 12. \quad (332)$$

On the metric chart of Appendix D, let $c \succeq c_* I$. Choose $0 < b_0 < c_*/24$ and set

$$\mathcal{B} = \{B = B^T \in \mathbb{R}^{10 \times 10} : \|B\|_{\text{op}} < b_0\}. \quad (333)$$

This is an open ball of dimension 55. For $B \in \mathcal{B}$ define

$$q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^-(c^{ij}(g)D_j^+q) - \mathbf{K}_b v + \mathbf{G}_h(q, v) - h^2 B \Lambda_h^2 q. \quad (334)$$

The coefficient B is constant during one physical history. It may vary with the cutoff or be sampled once as a finite initial mark.

Lemma E.1 (Uniform fourth-order multiplier bounds). *For every array w ,*

$$h^2 \|\Lambda_h w\|_h^2 \leq 12 \sum_i \|D_i^+ w\|_h^2. \quad (335)$$

At every fixed index used below,

$$h^2 \|\Lambda_h^2 q\|_{s-1,h} \leq C_s \|q\|_{s+1,h}, \quad h^2 \|\Lambda_h^2 q\|_{s-2,h} \leq C_s h \|q\|_{s+1,h}, \quad (336)$$

$$h^2 \|\Lambda_h^2 q\|_{s-3,h} \leq C_s h^2 \|q\|_{s+1,h}, \quad h^2 \|\Lambda_h^2 v\|_{s-3,h} \leq C_s h \|v\|_{s,h}. \quad (337)$$

Proof. Parseval reduces the first claim to $h^2\ell_h(k)^2 \leq 12\ell_h(k)$. Uniform equivalence of the forward-difference Sobolev norm with the Fourier weight $(1 + \ell_h)^{r/2}$ reduces the four remaining claims to the bounded ratios

$$\frac{h^2\ell_h^2}{1 + \ell_h}, \quad \frac{h^2\ell_h^2}{(1 + \ell_h)^{3/2}}, \quad \frac{h^2\ell_h^2}{(1 + \ell_h)^2},$$

which are bounded by 12, $\sqrt{12}h$, and h^2 , respectively. $\square$

The stencil Λ_h^2 has fixed coordinate radius two. Its contribution is the negative q -gradient of $h^2\langle \Lambda_h q, B\Lambda_h q \rangle_h/2$; this identity concerns the added term only and does not identify the complete local writer with the raw finite Euler flow of the reconstructed signed action.

For $q_\alpha = D^\alpha q$ define

$$\mathcal{E}_{s,h}^B = \mathcal{E}_{s,h} + \frac{h^2}{2} \sum_{|\alpha| \leq s} \langle \Lambda_h q_\alpha, B\Lambda_h q_\alpha \rangle_h, \quad (338)$$

where $\mathcal{E}_{s,h}$ is (314).

Theorem E.2 (Fixed-radius propagation on an open writer ball). *For $s \geq 11$, uniformly for $B \in \bar{\mathcal{B}}$ on a sufficiently small common Cauchy chart,*

$$c_s \|X\|_{\mathcal{X}_h^s}^2 \leq \mathcal{E}_{s,h}^B \leq C_s \|X\|_{\mathcal{X}_h^s}^2, \quad (\mathcal{E}_{s,h}^B)' \leq C_s \mathcal{E}_{s,h}^B + C_s (\mathcal{E}_{s,h}^B)^{3/2}. \quad (339)$$

Consequently there are $\varepsilon_s, T_s > 0$, independent of h and B , such that every initial record with $\|X_h(0)\|_{\mathcal{X}_h^s} \leq \varepsilon_s$ has a unique solution through T_s and

$$\sup_{t \leq T_s} \|X_h(t)\|_{\mathcal{X}_h^s} \leq C_s \|X_h(0)\|_{\mathcal{X}_h^s}. \quad (340)$$

With an additional acceleration defect f ,

$$(\sqrt{\mathcal{E}_{s,h}^B})' \leq C_s \sqrt{\mathcal{E}_{s,h}^B} + C_s \mathcal{E}_{s,h}^B + C_s \|f\|_{s,h}. \quad (341)$$

Proof. By Lemma E.1, the absolute value of the added quadratic energy at each commuted level is at most $6b_0 \sum_i \|D_i^+ q_\alpha\|_h^2$. The pre-existing spatial energy is at least $c_* \sum_i \|D_i^+ q_\alpha\|_h^2/2$, so $b_0 < c_*/24$ leaves a cutoff-independent positive margin.

Commute (334) with D^α . Since B and Λ_h commute with every lattice shift, the new principal term is $-h^2 B \Lambda_h^2 q_\alpha$. Pairing with v_α gives exactly

$$-h^2 \langle v_\alpha, B \Lambda_h^2 q_\alpha \rangle_h = -\frac{d}{dt} \frac{h^2}{2} \langle \Lambda_h q_\alpha, B \Lambda_h q_\alpha \rangle_h.$$

Thus the additional high-frequency term cancels at principal order rather than being estimated as a small top-order force. Every remaining commutator is one of those controlled in Theorem D.4; the only change is that the actual acceleration appears in analytic mass-coefficient terms, and Lemma E.1 bounds it uniformly in H_h^{s-1} . This gives (339). The finite-dimensional vector field is analytic on the common chart, and the same first-exit argument used for (319) proves the lifespan and the forced estimate. $\square$

Theorem E.3 (Uniform Einstein limit for the whole coefficient family). *Let $X_{B,h}$ and $X_{0,h}$ have the same sufficiently small compatible initial data. On their common interval,*

$$\sum_{j=0}^3 \|q_{B,h}^{(j)} - q_{0,h}^{(j)}\|_{C_t H_h^{s-1-j}} \leq C_s h \|B\|_{\text{op}} \varepsilon. \quad (342)$$

For any sequence $B_h \in \overline{\mathcal{B}}$, with no convergence of B_h required, the whole interpolated sequence converges to the same harmonic vacuum development as the central writer. In particular the metric and its first two time derivatives satisfy the $O(h\varepsilon)$ bounds of [Theorem D.7](#), and the subsidiary field, rate positivity, cut margin and noncollapse estimates remain uniform.

Proof. For comparison only, regard the added term on the B -trajectory as an external force

$$f_{B,h} = -h^2 a(g_{B,h})^{-1} B \Lambda_h^2 q_{B,h}.$$

Both trajectories already exist on the common top chart by [Theorem E.2](#). The lower-order bounds in [Lemma E.1](#) give

$$\|f_{B,h}\|_{C_t H_h^{s-2}} + \|\partial_t f_{B,h}\|_{C_t H_h^{s-3}} \leq C_s h \|B\| \varepsilon,$$

while the top H_h^{s-1} norm need not be small. Apply the forced difference estimate (320) at the reserved lower orders to get the first two jet slots. Subtracting the acceleration rows gives the second time derivative; differentiating once and using the velocity differential of the local vector field gives the third. This proves (342). Add the central-writer convergence of [Theorem D.7](#). All constants are uniform on $\overline{\mathcal{B}}$, so no convergence of B_h is needed. $\square$

Proposition E.4 (Rapid switching is not included). *For two distinct scalar marks $B_i = \sigma_i I$ in the admissible interval, one resolved corner mode has frequency*

$$\Omega_{\sigma,h}(k)^2 = \ell_h(k)(1 + \sigma h^2 \ell_h(k)). \quad (343)$$

Holding each law for a quarter period gives transfer matrices whose two-step product has eigenvalues $-\Omega_{\sigma_1,h}/\Omega_{\sigma_2,h}$ and its reciprocal. Their modulus ratio stays separated from one at the ultraviolet corner when $\sigma_1 \neq \sigma_2$. Repeated $O(h)$ -time switching can therefore amplify that mode like $\exp(cT/h)$ on a fixed slab.

Proof. The flat scalar mode is an oscillator. A quarter-period transfer is $\begin{pmatrix} 0 & \Omega^{-1} \\ -\Omega & 0 \end{pmatrix}$. Multiplying the two transfers gives the stated diagonal matrix. At the corner $h^2 \ell_h(k) \rightarrow 12$, so the frequency ratio approaches $\sqrt{(1+12\sigma_1)/(1+12\sigma_2)} \neq 1$. $\square$

E.2 Relatively open neighborhoods of chronological transition rows

Fix $B \in \overline{\mathcal{B}}$ and $T < T_s$. Partition $[0, T]$ into cells I_j with $c_- h^2 \leq \tau_j \leq c_+ h^2$. A chronological letter is a finite physical polynomial metric segment Q_e with shared endpoint value, velocity, acceleration and jerk records. Before a possible guard exit define

$$f_e(t) = Q_e''(t) - V_{B,h}(Q_e(t), Q_e'(t)) \quad (344)$$

and the cost (94); a grammar or execution failure has cost one and terminates the word. The common-energy margin is chosen so that any word with total cost below one is evaluated entirely inside the coefficient chart.

Lemma E.5 (A stopped physical cost controls the whole word). *If $C_h \leq d_h^2 < 1$ and d_h is sufficiently small, then no failure or guard exit occurs and*

$$\|f_h\|_{L_t^2 H_h^s} + \|\partial_t f_h\|_{L_t^2 H_h^{s-3}} \leq 2\varepsilon d_h. \quad (345)$$

Relative to the exact unforced B -history with the same initial record,

$$\|Q_h - q_{B,h}\|_{C_t H_h^s} + \|Q_h' - q_{B,h}'\|_{C_t H_h^{s-1}} + \|Q_h'' - q_{B,h}''\|_{C_t H_h^{s-3}} \leq C_s \varepsilon d_h. \quad (346)$$

Proof. Until a first guard hit, Cauchy–Schwarz gives $\int_0^{t^*} \|f_h\|_{s,h} dt \leq \sqrt{T} \varepsilon d_h$. For small d_h the forced energy (341) contradicts a first exit through the strict guard margin; a failure is excluded because it would already contribute unit cost. Hence the integrals are over the whole interval and (345) follows. Subtract the exact and forced mass equations at the reserved lower order, use (320) and Gronwall, and then solve the actual acceleration difference for the last displayed slot. $\square$

Theorem E.6 (A chronological law neighborhood with the same Einstein limit). *Let $K_{h,j}(e \mid \pi)$ be an arbitrary finite transition row at each live prefix, possibly depending on the entire past. If*

$$\sum_e K_{h,j}(e \mid \pi) c_{h,j}(\pi, e) < M \tau_j h^4 \quad (347)$$

for every live prefix, then

$$\boxed{\mathbb{E} C_h \leq M T h^4, \quad \mathbb{P}\{C_h > h^2\} \leq M T h^2.} \quad (348)$$

On $\{C_h \leq h^2\}$ the history obeys Lemma E.5 with $d_h = h$ and therefore has the same Einstein limit as Theorem E.3. At fixed cutoff and finite grammar the strict inequalities (347) define nonempty relatively open convex subsets of the product of row simplexes when a uniformly accurate completion is available at every live source. They contain strictly positive rows whenever all candidate letters have finite cost. Along the odd sequence $h = (2m+1)^{-1}$, the exceptional probabilities are summable, so the conclusion holds eventually almost surely under any common coupling.

Proof. Assign zero future cost to stopped prefixes. Conditional on any live prefix, (347) bounds the expected stage cost by $M \tau_j h^4$, independently of the probability of reaching that prefix. Summation gives the expectation bound; Markov at threshold h^2 gives the probability bound. The deterministic cost lemma applies on the complement. The series $\sum_m (2m+1)^{-2}$ converges, so the first Borel–Cantelli lemma gives the eventual almost-sure assertion without independence between cutoffs. At fixed finite grammar each strict conditional mean bound is an open affine half-space in one row simplex; finite products preserve relative openness and convexity. Small positive mixing preserves a strict margin. $\square$

For an observable neighborhood description, attach to each ordinary letter the profile

$$\Phi_e = (f_e/\varepsilon, \partial_t f_e/\varepsilon, 0), \quad (349)$$

and to a failure the vector $(0, 0, 1)$ in $L^2(I_j; H_h^s) \oplus L^2(I_j; H_h^{s-3}) \oplus \mathbb{R}$. Let $d(e, e') = \|\Phi_e - \Phi_{e'}\|$ and let $W_{2,d}$ be the finite optimal transport distance induced by d .

Proposition E.7 (Physical profile transport certifies a row neighborhood). *For finite rows K, P ,*

$$\left(\sum_e K(e) c_e \right)^{1/2} \leq \left(\sum_e P(e) c_e \right)^{1/2} + W_{2,d}(K, P). \quad (350)$$

If $\sum_e P(e) c_e \leq M_0 \tau_j h^4$, then every row satisfying

$$W_{2,d}(K, P) < \gamma \sqrt{\tau_j} h^2 \quad (351)$$

obeys (347) for any $M > (\sqrt{M_0} + \gamma)^2$. These strict balls are relatively open in the finite probability simplex.

Proof. For any coupling of K and P , Minkowski in the profile Hilbert space gives $\|\Phi_e\|_{L^2} \leq \|\Phi_{e'}\|_{L^2} + \|\Phi_e - \Phi_{e'}\|_{L^2}$. Average under the coupling and minimize. The finite coupling polytope is compact, so a minimizer exists. Continuity in the row probabilities follows from finite-dimensional optimal transport. $\square$

The scale in (351) is intentional. A fixed unscaled total-variation ball is not stable under refinement: if there are $J_h \asymp h^{-2}$ stages and an absorbing failure of probability p_h is added independently at every live stage, then the failed-word probability is $1 - (1 - p_h)^{J_h}$, which tends to zero exactly when $J_h p_h \rightarrow 0$. Thus even $p_h \rightarrow 0$ can be too large. The theorem controls the accumulated physical defect rather than hiding this refined-word effect.

E.3 Exact finite initial constraints from free records

We now remove the need to start the constructive population from a continuum constraint solution. This subsection uses the original reconstructed finite action only at the initial cut. Let the odd phase derivative be

$$\widehat{\delta_i u}(k) = i\kappa_i(k)\widehat{u}(k), \quad \kappa_i(k) = 2h^{-1} \sin(\pi h k_i), \quad \Omega_h^2 = -\sum_i \delta_i^2. \quad (352)$$

For $r \geq 3$ put

$$\mathcal{P}_h^r = H_h^{r+2}(\text{Sym}_3) \times H_h^{r+1}(\text{Sym}_3), \quad \mathcal{Y}_h^r = H_h^r(\mathbb{R} \oplus \mathbb{R}^3), \quad (353)$$

and write $X = (u, \pi)$, $\gamma = I + u$. Let $\mathcal{C}_h : \mathcal{P}_h^r \rightarrow \mathcal{Y}_h^r$ be (95).

Lemma E.8 (Uniform original-action constraint calculus). *On a common ball $\mathcal{C}_h$ is analytic, $\mathcal{C}_h(0) = 0$, and*

$$D\mathcal{C}_h(0)(u, \pi) = \left(\delta_i \delta_j u_{ij} - \sum_i \delta_i^2 \text{tr } u, -2\delta_j \pi^{ij} \right) =: L_h(u, \pi). \quad (354)$$

Writing $\mathcal{C}_h = L_h + \mathcal{N}_h$,

$$\|\mathcal{N}_h(X)\|_{\mathcal{Y}_h^r} \leq C_r \|X\|^2, \quad \|D\mathcal{N}_h(X)\| \leq C_r \|X\|, \quad (355)$$

with cutoff-independent constants. No lapse or shift time derivative and no metric acceleration appears in $\mathcal{C}_h$.

Proof. Stationary connection and Legendre elimination are analytic on the retained small chart. In a lapse or shift derivative, every spatial load has one phase derivative; exact skew-adjointness transfers it to the stationary connection, so the output loses at most the declared Sobolev orders. At flat data the quadratic phase-compatible action has the Fierz–Pauli rows displayed in (354); the full link remainder starts at cubic order in the flat connection and contributes no linear initial row. Taylor’s theorem gives (355). $\square$

Let P_0 be spatial averaging in $\mathcal{Y}_h^r$ and $P_\perp = I - P_0$. On mean-zero targets define

$$R_h(f, m) = \left(\frac{1}{2}(\Omega_h^2)^{-1} f I, \frac{1}{2}\mathcal{L}_h^{\text{vec}} \mathcal{D}_h^{-1} m \right), \quad (356)$$

where $(\mathcal{L}_h^{\text{vec}} Y)_{ij} = \delta_i Y_j + \delta_j Y_i - \frac{2}{3} \delta_{ij} \delta_k Y_k$ and $\mathcal{D}_h Y = -\delta_j (\mathcal{L}_h^{\text{vec}} Y)_{ij}$.

Lemma E.9 (Range inverse and four missing means). *One has $L_h R_h = I$ on the mean-zero target and $\|R_h\| \leq C_r$ uniformly. For $d = N^3$,*

$$\text{rank } L_h = 4d - 4, \quad \dim \ker L_h = 8d + 4. \quad (357)$$

For each sufficiently small $z \in \ker L_h$ there is a unique mean-zero $y_h(z) = O(\|z\|^2)$ such that

$$P_\perp \mathcal{C}_h(z + R_h y_h(z)) = 0. \quad (358)$$

The remaining mean map $\Theta_h(z) = P_0 \mathcal{C}_h(z + R_h y_h(z))$ is analytic and has zero constant and linear jets.

Proof. For $u = \phi I$ the scalar row equals $2\Omega_h^2 \phi$. For $\pi = \mathcal{L}_h^{\text{vec}} Y$ the vector row equals $2\mathcal{D}_h Y$, whose nonzero-mode symbol is $|\kappa|^2 I + \kappa \kappa^T / 3$ and is uniformly invertible on the unit odd torus. This proves the right inverse and rank. The nonlinear range equation is $y = -P_\perp \mathcal{N}_h(z + R_h y)$; the bounds of Lemma E.8 make it a uniform contraction on a ball of radius $C\|z\|^2$. The expansion of Θ_h follows because $P_0 L_h = 0$. $\square$

To resolve the four means, let $T_i \in \text{Sym}_3$ have its two transverse off-diagonal entries equal to $1/\sqrt{2}$ and all other entries zero, and define

$$u_* = \sum_{i=1}^3 T_i \cos(2\pi x_i), \quad p(\vartheta) = -\frac{2}{3}\kappa I + \sum_{i=1}^3 b_i T_i \sin(2\pi x_i), \quad \vartheta = (\kappa, b_1, b_2, b_3). \quad (359)$$

These are balancing directions only; arbitrary complementary modes remain free.

Lemma E.10 (Exact four-mode mean calibration). *With $\omega_h = 2h^{-1} \sin(\pi h)$, the quadratic mean jet of the full retained action on the seed (359) is*

$$Q_h(\vartheta) = \begin{pmatrix} \frac{2}{3}\kappa^2 - \frac{1}{2} \sum_i b_i^2 - \frac{3}{8}\omega_h^2 \\ -\frac{1}{2}\omega_h b_1 \\ -\frac{1}{2}\omega_h b_2 \\ -\frac{1}{2}\omega_h b_3 \end{pmatrix}. \quad (360)$$

It has the exact root

$$\vartheta_h^* = (3\omega_h/4, 0, 0, 0), \quad DQ_h(\vartheta_h^*) = \text{diag}(\omega_h, -\omega_h/2, -\omega_h/2, -\omega_h/2), \quad (361)$$

whose inverse is uniformly bounded. The opposite sign of the first component gives the second branch.

Proof. Keep spatially constant lapse and shift independent and expand the canonical Hamiltonian to quadratic order about flat data. For the transverse trace-free configuration seed and momentum equal to a homogeneous trace plus transverse trace-free modes, the coefficient is

$$\mathcal{H}_h^{[2]}(u, \pi; N, \beta) = N \left(\|\pi\|_h^2 - \frac{1}{2} \|\text{tr } \pi\|_h^2 + \frac{1}{4} \sum_i \|\delta_i u\|_h^2 \right) + \sum_i \beta^i \langle \pi, \delta_i u \rangle_h.$$

The complete link remainder starts at cubic order in the stationary connection at flat data, so it does not modify this quadratic mean coefficient. Taking minus the lapse derivative and the three shift derivatives, then using orthogonality of the constant and unit Fourier modes on every odd grid, gives (360). Direct differentiation yields (361); ω_h is uniformly separated from zero on the declared grids. $\square$

Let $\mathcal{Z}_h = \ker L_h$ and remove the four balancing seed directions from an arbitrary small free record, leaving W in a fixed complementary subspace. Write $Z_h(\vartheta)$ for the seed and, for $t \neq 0$, define

$$\mathcal{Q}_h(t, \vartheta, W) = t^{-2} \Theta_h(t[Z_h(\vartheta) + W]). \quad (362)$$

Since Θ_h has zero constant and linear jets, the integral Taylor formula extends this map analytically through $t = 0$.

Theorem E.11 (Exact finite preparation from free initial records). *There are $t_0, \rho, d_0, h_0 > 0$, independent of h , such that for $0 < |t| < t_0$, $h < h_0$, and every complementary free record W with $\|W\| < \rho$, there is a unique $\vartheta_h(t, W)$ near ϑ_h^* for which (362) vanishes. The resulting record*

$$\boxed{\mathfrak{J}_h(t, W) = t[Z_h(\vartheta_h(t, W)) + W] + R_h y_h(t[Z_h(\vartheta_h(t, W)) + W])} \quad (363)$$

satisfies

$$\mathcal{C}_h(\mathfrak{J}_h(t, W)) = 0, \quad \|\mathfrak{J}_h(t, W)\|_{\mathcal{P}_h^r} \leq C_r |t|. \quad (364)$$

At every such record $D\mathcal{C}_h$ is onto and admits a right inverse with

$$\boxed{\|\mathcal{R}_{h,t} f\|_{\mathcal{P}_h^r} \leq C_r (\|P_\perp f\|_{\mathcal{Y}_h^r} + |t|^{-1} |P_0 f|)}. \quad (365)$$

For fixed $t \neq 0$, the image contains a relatively open neighborhood in the complete finite zero set; that zero set has codimension $4N^3$ in the $12N^3$ -dimensional canonical phase. No symmetry, CMC, conformal-flatness, frequency-support restriction, or continuum constraint solution is imposed on W .

Proof. The analytic extension of (362) converges in C^1 uniformly to the quadratic map of Lemma E.10 as $|t| + \|W\| \rightarrow 0$. Its root has a uniform inverse margin. Therefore $\vartheta \mapsto \vartheta - [DQ_h(\vartheta_h^*)]^{-1} \mathcal{Q}_h(t, \vartheta, W)$ is a contraction on one cutoff-independent ball after choosing its radius first and then t_0, ρ . The range equation was already solved in Lemma E.9, giving the complete zero.

For the derivative, eliminate the mean-zero block with the uniformly bounded range inverse. Along a balancing direction the remaining mean derivative is $t^2 D_\vartheta \mathcal{Q}_h$, while a range direction contributes only $O(t)$ to the mean. Solving for the balancing coefficient and multiplying by the physical factor t gives exactly the $|t|^{-1}$ term in (365). The inverse function theorem on the zero set then gives relative openness and the codimension count. The same estimate shows why uniformity through the symmetric tip $t = 0$ is neither claimed nor true in general. $\square$

The harmonic completion of a prepared record sets $g_{00} = -1$, $g_{0i} = 0$, $g_{ij} = \gamma_{ij}$ and uses its actual canonical metric velocity w for the spatial slot, with

$$v_{ij} = w_{ij}, \quad v_{00} = -\gamma^{ij} w_{ij}, \quad v_{0i} = \gamma^{jk} (\delta_j \gamma_{ki} - \tfrac{1}{2} \delta_i \gamma_{jk}). \quad (366)$$

These lapse-shift velocities do not enter the four initial rows (95).

Corollary E.12 (Action-prepared inputs populate the open law family). *Fix sufficiently small nonzero t and take a convergent sequence of free records in Theorem E.11. For any sequence $B_h \in \bar{\mathcal{B}}$, with no convergence of B_h required, the harmonically completed histories exist on the common interval and converge to the vacuum Einstein development of the limiting prepared data. At the exact initial cut all four original lapse-shift rows vanish identically, independently of B_h . The chronological transition-row theorem Theorem E.6 applies to these inputs unchanged.*

Proof. The preparation has size $O(|t|)$ in Sobolev spaces strictly above the evolution Cauchy norm, so (366) lies in the fixed ball of Theorem E.2. Consistency of the phase derivatives and the stationary/Legendre maps gives convergence of the prepared finite records to their continuum vacuum-constraint limit; the harmonic initial gauge error tends to zero. Apply Theorem E.3. The initial action rows vanish before the acceleration law is chosen and therefore remain independent of the coefficient mark. The probabilistic statement depends only on the physical residual costs and the common Cauchy estimates, so no new probability hypothesis is needed. $\square$

The three results of this section give an open population in a precise sense, but not universal attraction. The coefficient ball is fixed and genuinely open; the transition-row neighborhoods are open in the scaled complete-profile topology appropriate to $O(h^{-2})$ stages; and the initial records are relatively open in the exact finite constraint surface. The raw exact-action evolution itself remains the separate question addressed next.

F Exact-action response geometry, preparation, and dynamical tangency

This section studies the original phase-compatible finite action without identifying its Euler flow with the stable harmonic writer. The organization is mathematical rather than chronological. We first determine the exact finite constraint and response geometry, then exhibit nontrivial prepared branches and quantitative response estimates, and finally isolate the higher tangency condition that obstructs a direct cutoff-uniform flow theorem.

F.1 Canonical constraints and the finite characteristic quotient

Let $X = (u, \pi) \in \mathbb{R}^{12V}$ and let $\lambda = (N, \beta) \in \mathbb{R}^{4V}$. For the canonical Hamiltonian $H_h(X, \lambda)$ put

$$P_h = \nabla_\lambda H_h, \quad F_h^{\text{can}} = \mathbb{J} \nabla_X H_h, \quad B_h = D_X P_h F_h^{\text{can}}, \quad M_h = D_\lambda P_h. \quad (367)$$

The exact Euler system is

$$X_t = F_h^{\text{can}}(X, \lambda), \quad P_h(X, \lambda) = 0, \quad (368)$$

and differentiating the constraint gives

$$B_h + M_h \dot{\lambda} = 0. \quad (369)$$

Phase-compatible multiplier homogeneity gives

$$H_h(X, c\lambda) = cH_h(X, \lambda), \quad M_h \lambda = 0, \quad \langle \lambda, B_h \rangle_h = 0. \quad (370)$$

These are global scaling identities, not local discrete diffeomorphism identities.

Introduce multiplier momenta and the extended constraint bracket

$$\mathcal{A}_h = D_X P_h \mathbb{J} (D_X P_h)^*, \quad \mathfrak{D}_h = \begin{pmatrix} 0 & -M_h \\ M_h & \mathcal{A}_h \end{pmatrix}. \quad (371)$$

Theorem F.1 (Finite characteristic quotient on a regular one-null branch). *Suppose $D_X P_h$ is onto, $\ker M_h = \text{span}\{\lambda\}$ and (369) is solvable. Then*

$$\ker \mathfrak{D}_h = \text{span} \left\{ \begin{pmatrix} \lambda \\ 0 \end{pmatrix}, \begin{pmatrix} -M_h^\dagger B_h \\ \lambda \end{pmatrix} \right\}, \quad \text{rank } \mathfrak{D}_h = 8V - 2, \quad (372)$$

and the local characteristic quotient has dimension $12V - 2$.

Proof. If $(r, s) \in \ker \mathfrak{D}_h$, the first row gives $s = c\lambda$. Differentiating the homogeneity identity in X and pairing with the Hamiltonian vector field gives $\mathcal{A}_h \lambda = B_h$. The second row therefore gives $r = -cM_h^\dagger B_h + d\lambda$. The extended phase has dimension $20V$ and the $8V$ independent primary and secondary constraints leave dimension $12V$; quotienting the two characteristic directions gives $12V - 2$. $\square$

F.2 An exact regular transverse block and lossless source elimination

Use the odd-grid phase derivatives

$$\widehat{\delta_i u}(k) = i\kappa_i(k)\widehat{u}(k), \quad \kappa_i(k) = 2h^{-1} \sin(\pi h k_i), \quad \Omega_h^2 = -\sum_i \delta_i^2. \quad (373)$$

Let

$$\mathcal{T}_h = \{\beta : \widehat{\beta}(0) = 0, \kappa(k)^T \widehat{\beta}(k) = 0 \ (k \neq 0)\} \quad (374)$$

be the transverse-shift space, of dimension $2(V-1)$. On the scalar-momentum family the first source restricted to $\mathcal{T}_h$ has the form

$$A_h \beta = -\frac{h\kappa_h}{2} \mathcal{L}_h \left(-\frac{1}{2} \text{curl}_\delta \beta\right), \quad (375)$$

where κ_h is uniformly bounded above and below and $\mathcal{L}_h w$ is the symmetric matrix with zero diagonal and $(\mathcal{L}_h w)_{ij} = S_i w_j + S_j w_i$.

Theorem F.2 (Exact all-cutoff transverse rank and natural-order bounds). *At every odd cutoff,*

$$\text{rank } A_h = 2(V-1), \quad \text{P}_{\text{tr}} A_h = 0, \quad (376)$$

and

$$\boxed{ch^2 \|\Omega_h \beta\|_h^2 \leq \|A_h \beta\|_h^2 \leq Ch^2 \|\Omega_h \beta\|_h^2}, \quad \beta \in \mathcal{T}_h. \quad (377)$$

The same estimates hold after multiplication by a scalar Fourier Sobolev weight, and

$$A_h^* K_0^{-1} A_h = \frac{1}{2} A_h^* A_h \succ 0. \quad (378)$$

Proof. At one Fourier mode the off-diagonal map is unitarily equivalent to $w \mapsto (w_1 + w_2, w_1 + w_3, w_2 + w_3)$, whose squared singular values are 4, 1, 1. For transverse β , $\|\text{curl}_\delta \beta\|_h = \|\Omega_h \beta\|_h$. Substitution in (375) proves the two-sided estimate and injectivity. The output diagonal vanishes, so the trace projection vanishes and K_0^{-1} acts by $1/2$ on this range. $\square$

Write the complete leading graded source as $F_h = (A_h, G_h)$ with target $(0, d_h)$, and put $\overline{G}_h = (I - \Pi_{A,h})G_h$.

Corollary F.3 (Lossless elimination of the transverse source columns). *One has*

$$\text{rank } F_h = (2V-2) + \text{rank } \overline{G}_h, \quad (379)$$

and

$$F_h^* K_0^{-1} F_h \sim \text{diag} \left(\frac{1}{2} A_h^* A_h, \overline{G}_h^* K_0^{-1} \overline{G}_h \right). \quad (380)$$

The triangular column transformation producing this congruence leaves the actual target unchanged and preserves every remaining source equation.

Proof. Use the coefficient $C_h = (A_h^* A_h)^{-1} A_h^* G_h$ and right-multiply by $\begin{pmatrix} I & -C_h \\ 0 & I \end{pmatrix}$. The transformed source is $(A_h, \overline{G}_h)$ with orthogonal column ranges. Since $K_0^{-1} A_h = A_h/2$, the signed cross term vanishes as well. $\square$

F.3 Exact scalar–trace reconstruction of the signed response

Work after any regular positive block elimination and let F denote the remaining full-rank source with actual target b . Define

$$\Pi = F(F^*F)^{-1}F^*, \quad w = F(F^*F)^{-1}b, \quad E\tau = \tau I/\sqrt{3}, \quad (381)$$

then

$$C = E^*\Pi E, \quad H = 2I - 3C, \quad \zeta = E^*w. \quad (382)$$

Theorem F.4 (Exact scalar–trace response reconstruction). *The signed Gram $F^*K_0^{-1}F$ is invertible if and only if H is invertible. If $\chi = H^{-1}\zeta$, then the original physical response is*

$$\boxed{v = w - 3(I - \Pi)E\chi, \quad E^*v = -\chi,} \quad (383)$$

and

$$\boxed{\|v\|^2 = \|w\|^2 + 9\langle \chi, (I - C)\chi \rangle.} \quad (384)$$

Consequently

$$v_h \rightarrow 0 \iff w_h \rightarrow 0 \text{ and } H_h^{-1}E^*w_h \rightarrow 0 \quad (385)$$

on any regular cofinal family.

Proof. Write $F = QR$ with $Q^*Q = I$. The signed Gram is congruent to $(2I - 3Q^*EE^*Q)/4$. The nonzero eigenvalues of Q^*EE^*Q and $E^*QQ^*E = C$ agree, proving the invertibility equivalence. The response is characterized by $F^*v = b$ and $K_0v \in \text{Ran } F$. The first identity gives $\Pi v = w$ and the second gives $(I - \Pi)v = 3(I - \Pi)EE^*v$. Applying E^* yields $H(E^*v) = -\zeta$, hence $E^*v = -\chi$ and (383). Orthogonality of the two displayed terms gives (384). $\square$

Diagonalizing the compression C gives the trace-angle form. Equivalently, with the positive polar decomposition $F = QR$ and $T = Q^*P_{\text{tr}}Q$, $\eta = R^{-1}b$, one obtains

$$v = (2I - 3P_{\text{tr}})Q(2I - 3T)^{-1}\eta. \quad (386)$$

If $Tu_j = \mu_j u_j$ and $\eta_j = \langle u_j, \eta \rangle$, then

$$\boxed{\|\mathcal{E}^0\|^2 = \sum_j \frac{4 - 3\mu_j}{(2 - 3\mu_j)^2} |\eta_j|^2,} \quad (387)$$

so $\mu = 2/3$ is the unique signed trace resonance. This is the spectral form of [Theorem F.4](#), not a different response law.

For a cofinal regular family define $\nu_h = \sum_j |\eta_{h,j}|^2 \delta_{\mu_{h,j}}$. Then $\|\mathcal{E}_h^0\| \rightarrow 0$ if and only if $\nu_h([0, 1]) \rightarrow 0$ and

$$\lim_{\delta \downarrow 0} \limsup_{h \rightarrow 0} \int_{|2 - 3\mu| < \delta} \frac{d\nu_h(\mu)}{|2 - 3\mu|^2} = 0. \quad (388)$$

The proof is the direct split of (387) into a region separated from $2/3$ and a shrinking neighborhood of the resonance.

F.4 Time-dependent response and the necessary clock test

Static response control is insufficient for a spacetime comparison. Let $F(t), b(t)$ be C^r with full column rank and regular signed Gram in a fixed physical tangent space, and let $\Pi, w, C, H, \zeta, \chi, v$ be the preceding objects.

Theorem F.5 (Derivative hierarchy in scalar–trace coordinates). *Put $\chi_0 = \chi$ and, for $1 \leq j \leq r$,*

$$\chi_j = H^{-1} \left[\zeta^{(j)} + 3 \sum_{\ell=1}^j \binom{j}{\ell} C^{(\ell)} \chi_{j-\ell} \right]. \quad (389)$$

Then $\chi_j = \chi^{(j)} = -E^* v^{(j)}$. In particular

$$\chi' = H^{-1}(\zeta' + 3C'\chi), \quad v' = w' + 3\Pi'E\chi - 3(I - \Pi)E\chi'. \quad (390)$$

Thus a small static response does not control the differentiated response without bounds on the transported source geometry.

Proof. Differentiate $H\chi = \zeta$ repeatedly and use $H^{(\ell)} = -3C^{(\ell)}$. The identity $E^*v = -\chi$ holds at every time; differentiating (383) gives the second formula. $\square$

The normalized exact continuation has an active/weak amplitude grading $S_a = \text{diag}(aI_A, a^2I_W)$. If u_a is the normalized source coefficient entering the multiplier solve, then

$$\|\lambda_t\|_h^2 = a^{-2} \|(u_a)_A\|_h^2 + a^{-4} \|(u_a)_W\|_h^2. \quad (391)$$

Proposition F.6 (Necessary lapse–shift clock condition for harmonic shadowing). *Suppose $a(h) \rightarrow 0$ and an exact-action metric is to shadow a small harmonic metric in a topology controlling the initial first time derivative, with the same lapse, shift and spatial metric at the initial cut. Then necessarily*

$$a(h)^{-1} \|(u_{a(h)})_A\|_h \rightarrow 0, \quad a(h)^{-2} \|(u_{a(h)})_W\|_h \rightarrow 0. \quad (392)$$

This requirement is independent of smallness of the instantaneous electric coefficient.

Proof. At $N = 1$, $\beta = 0$ in ADM coordinates, $\partial_t g_{00} = -2\partial_t N$ and $\partial_t g_{0i} = \gamma_{ij}\partial_t \beta^j$. First-jet shadowing therefore forces $\|\lambda_t\| \rightarrow 0$. Equation (391) gives the two conditions. $\square$

F.5 Prepared exact-action families

The generic probe-source calculation remains useful for comparison. For the smooth first jet used in the original response audit, the actual weak source obeys

$$\|\mathcal{I}_h d_{h,\rho} - h^2 d_\rho^{(2)}\|_{H^r} \leq C_r h^4, \quad c_r h^2 \leq \|d_{h,\rho}\|_{H_h^r} \leq C_r h^2. \quad (393)$$

Thus the generic weak Ward defect is genuinely second order, but Theorem F.4 shows why its norm alone does not control the physical response.

We next record two exact preparations with different purposes.

A rationally detuned all-cutoff active family. Let $N = 2m + 1$, $h = N^{-1}$ and define the one-dimensional product-defect operator

$$D_h u = \delta(fu) - f\delta u - (\delta f)u, \quad f(x) = \cos(2\pi x). \quad (394)$$

It is self-adjoint, has kernel the constants, and its nonconstant part is an irreducible zero-diagonal Jacobi matrix. For positive amplitudes $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ put

$$U_\alpha = \sum_{i=1}^3 \alpha_i T_i \cos(2\pi x_i), \quad p_{h,\alpha} = -s_{h,\alpha} I, \quad s_{h,\alpha} = \omega_h \sqrt{\frac{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}{6}}, \quad (395)$$

where T_i is the transverse off-diagonal tensor used in (359). The diagonal lapse part of the complete first source is

$$\mathcal{D}_{h,\alpha} n = \begin{pmatrix} (\alpha_2 D_{2,h} - \alpha_3 D_{3,h})n \\ (\alpha_3 D_{3,h} - \alpha_1 D_{1,h})n \\ (\alpha_1 D_{1,h} - \alpha_2 D_{2,h})n \end{pmatrix}. \quad (396)$$

Lemma F.7 (Finite rational detuning). *For every sufficiently fine odd cutoff there is a rational θ_h with $1 < \theta_h < 1 + h$ such that, for $\alpha_h = (1, \theta_h, 1)$,*

$$\|\mathcal{D}_{h,\alpha_h} n\|_h \geq ch^6 \|n\|_h \quad (\bar{n} = 0). \quad (397)$$

The selection can be made by a finite algebraic search at each cutoff.

Proof. The Jacobi representation of D_h has simple algebraic spectrum Σ_h . Its reciprocal-eigenvalue sum is $O(h^{-3})$. The union of ratios θ for which $|\lambda_a - \theta\lambda_b| < 4h^6$ has measure $O(h^2)$, smaller than a fixed fraction of the interval $(1, 1 + h)$ for small h . Choose a dyadic rational outside that union and sufficiently close to the selected point. In the tensor product eigenbasis, the three squared components of (396) vanish simultaneously only for the constant mode, and the spectral separation gives the stated margin. $\square$

Theorem F.8 (All-cutoff regular active source). *The exact original-action initial constraints have analytic preparations*

$$\mathcal{C}_h(X_{h,\alpha_h}(a)) = 0, \quad X_{h,\alpha_h}(a) = a(U_{\alpha_h}, p_{h,\alpha_h}) + O(a^2), \quad (398)$$

uniformly for all sufficiently fine odd cutoffs. On their first jets the complete leading active source has

$$\ker \mathcal{A}_h^{\text{act}} = \mathbb{R}(1, 0) \oplus \{(0, \beta) : \text{curl}_\delta \beta = 0\}, \quad \text{rank } \mathcal{A}_h^{\text{act}} = 3V - 3, \quad (399)$$

its range is trace-free, and

$$\boxed{(\mathcal{A}_h^{\text{act}})^* K_0^{-1} \mathcal{A}_h^{\text{act}} = \frac{1}{2} (\mathcal{A}_h^{\text{act}})^* \mathcal{A}_h^{\text{act}} \succeq ch^{16} I.} \quad (400)$$

Hence the complete remaining signed rank problem is confined to the weak columns.

Proof. The exact preparation follows from the finite initial-constraint chart and the nondegenerate four-mean calibration; the $O(h)$ detuning changes the prepared data by $O(|a|h)$ and not their continuum first jet. The diagonal lapse margin is Lemma F.7. The transverse shift block is controlled by Theorem F.2. Together they leave only constant lapse and curl-free shifts in the first-source kernel. The entire first source is trace-free on this three-amplitude family, so K_0^{-1} acts by $1/2$ on its range. The lapse estimate costs h^{-6} and the transverse reconstruction an additional two powers, giving a sufficient smallest singular value ch^8 and therefore the displayed Gram bound. $\square$

A source-neutral nontrivial family. For each axis let T_i be the transverse off-diagonal tensor and let P_i be the transverse diagonal trace-free tensor with entries $+1, -1$ on the two coordinates orthogonal to axis i . Put

$$U = \sum_{i=1}^3 r_i [T_i \cos(2\pi x_i + \phi_i) + \sigma_i P_i \sin(2\pi x_i + \phi_i)], \quad r_i > 0, \quad \sigma_i \in \{-1, 1\}. \quad (401)$$

The two polarizations on each axis are orthogonal and have equal Frobenius norm.

Theorem F.9 (Exact source-neutral finite preparation). *For the family (401), the complete weak quadratic target vanishes at every odd cutoff. On compact nonzero parameter sets the original finite constraint map admits exact preparations*

$$\boxed{\mathcal{C}_h(X_h(a)) = 0, \quad X_h(a) = a(U, p_h) + O(a^2)} \quad (402)$$

uniformly for sufficiently fine odd cutoffs. At a concrete three-site seed the complete leading signed Gram is nonsingular; hence a relatively open balanced neighborhood has an exact local original-action continuation for small nonzero a . On that branch the literal initial temporal and spatial link curvatures obey

$$\boxed{\|E_h(a, 0)\|_h + \|F_h^{\text{sp}}(a, 0)\|_h \leq C_h |a|}. \quad (403)$$

The constant in this last estimate is fixed-cutoff; no common physical time is inferred.

Proof. The weak quadratic drift on one axis is a sum of two polarization invariants: the norm difference and the Frobenius cross product. Equal norms and orthogonality make both vanish, independently of h . The linear constraints hold because the configurations and oscillatory momenta are transverse and trace-free. The nonconstant rows are solved by the uniform range inverse of the initial-constraint chart. The four means are solved by a homogeneous trace momentum and three oscillatory balancing parameters; the derivative of the quadratic mean map is uniformly invertible on compact nonzero polarization sets. This gives (402). At the explicit three-site seed, finite exact elimination gives a nonzero determinant for the complete signed Gram, and analyticity preserves it on a neighborhood. Because the normalized source target starts one amplitude order later on the balanced branch, its true finite-amplitude response is $O_h(a)$. Substitution in the literal temporal-link derivative and spatial plaquette formulas gives (403). $\square$

F.6 Direct transverse response and the positive scalar flux equation

A further weak-neutral preparation can be chosen so that the leading active range is trace-free at every sufficiently fine odd cutoff. On this family the transverse source admits an exact least-squares factorization. Let T_h be the transverse source, $B_h = T_h^* T_h$, and let the normalized positive normal matrix H_h be the finite-band operator obtained by factoring the phase curl and coefficient multiplication. It satisfies

$$\frac{9}{16} I \preceq H_h \preceq 324 I. \quad (404)$$

The actual pseudoinverse has the factorization

$$T_h^\dagger = \frac{2\sqrt{2}}{h\omega_h} \Omega_h^{-1} H_h^{-1} R_h \hat{\mathcal{L}}_h^* t_o^*, \quad (405)$$

with the operators defined by the physical Hodge and coefficient factorization of the source; no inverse-then-project substitution is made.

Theorem F.10 (Actual direct transverse geometric suppression). *For the actual prepared transverse target $b_{\beta,h}$ and direct solution $\beta_{0,h} = -2B_h^{-1}b_{\beta,h}$,*

$$\boxed{\|\beta_{0,h}\|_{H_h^{s+3}} \leq C_s h^{-1}, \quad \|\mathfrak{E}_h(\beta_{0,h})\|_{H_h^s} \leq C_s h,} \quad (406)$$

while the folded-face tail is exponentially small. These are estimates of the actual direct source response, not of a prescribed low-frequency projection.

Proof. The factorization (405) and (404) give two derivatives of Sobolev gain for B_h^{-1} at cost h^{-2} . The prepared transverse target is $O(h)$ on a fixed Fourier bank. Applying the geometric observation operator, whose symbol contributes the compensating h^2 factor at the required order, gives the second estimate. Exponential off-diagonal decay of H_h^{-1} in cyclic Fourier distance gives the folded-face tail. $\square$

After eliminating the transverse variable, the actual leading lapse solves

$$\mathcal{S}_h n_h = g_h. \quad (407)$$

Let Q_h project onto the physical off-diagonal complement and let $K_{Q,h}$ be the induced positive coefficient metric.

Theorem F.11 (Exact positive scalar energy). *At every odd cutoff,*

$$\frac{1}{324}I \preceq K_{Q,h} \preceq \frac{16}{9}I, \quad (408)$$

and the unchanged scalar operator is

$$\boxed{\mathcal{S}_h = \frac{1}{2}\mathcal{D}_h^* \mathcal{D}_h + \frac{h^2}{2}\sigma_h^* K_{Q,h}^{-1} \sigma_h.} \quad (409)$$

Hence

$$2\langle n, \mathcal{S}_h n \rangle_h = \|\mathcal{D}_h n\|_h^2 + h^2 \langle \sigma_h(n), K_{Q,h}^{-1} \sigma_h(n) \rangle_h. \quad (410)$$

On the bare diagonal soft subspace the normalized full energy diverges under refinement, so a pole predicted by the diagonal compression alone is not a pole of the full physical scalar energy. This compression result does not by itself prove a uniform inverse for (407).

Proof. The coefficient map giving the off-diagonal source is uniformly invertible, so compression of its positive inverse metric gives (408). Orthogonal projection of the physical source yields the exact residual factor $hM_h^{-*}Q_h K_{Q,h}^{-1}\sigma_h(n)$; squaring and adding the diagonal source gives (409). On the finite-dimensional soft subspace the limiting flux is nonzero on every nonzero vector, while the bare diagonal contribution is higher order; compactness therefore forces the normalized full energy to diverge. $\square$

F.7 Dynamical tangency: an exact obstruction and an upstream escape direction

Exact preparation is not the same as invariance of the prepared set under the original action vector field. Let $\mathcal{W}$ be the weak curl-free multiplier space and let $\mathbb{K}_1(Y)$ be the first Poisson matrix at a canonical first jet Y . For a leading multiplier tilt ℓ define the weak null space

$$\mathcal{N}_W(Y) = \{\nu \in \mathcal{W} : \mathbb{K}_1(Y)[\nu, \mu] = 0 \text{ for every multiplier } \mu\}, \quad (411)$$

and the first tangency functional

$$\boxed{\Delta_\nu(Y, \ell) = Dd_{2,W}(Y)[F_1Y + G\ell, \nu] + \mathbb{K}_1(F_1Y + G\ell)[\nu, \ell].} \quad (412)$$

It is determined entirely by the canonical first jet and leading multiplier tilt; higher preparation coefficients do not enter this leading null test.

Theorem F.12 (Initial-layer obstruction on an exactly prepared branch). *There is a nontrivial exactly prepared three-site original-action family X_a, λ_a with*

$$B_W(X_a, \lambda_a) = 0, \quad (\lambda_t)_W(a, 0) = 0, \quad \|\text{Riem}(g_h)(a, 0)\| = O_h(a), \quad (413)$$

but with a weak null test ν_ satisfying $\Delta_{\nu_*}(Y, \ell) \neq 0$. Along the exact continuation,*

$$\left. \frac{d}{dt} B_W[\nu_*] \right|_{t=0} = a^2 \Delta_{\nu_*}(Y, \ell) + O_h(a^3), \quad (414)$$

which forces a weak multiplier acceleration of order a^{-2} in the normalized grading. For every fixed S the exact histories exist on $0 \leq t \leq a^3 S$ for sufficiently small a , and the unscaled rates and curvature remain $O_h(a)$ there. Nevertheless no constants $T, C, a_0 > 0$ can make

$$\sup_{0 \leq t \leq T} \|R_{0i0j}(g_h)(a, t)\| \leq Ca \quad (415)$$

hold for every $0 < a < a_0$ on this family.

Proof. Taylor expansion of the exact weak drift gives $D_X B_W = a\mathcal{C}_\ell + O_h(a^2)$ and $D_\lambda B_W = a\mathcal{K}_W + O_h(a^2)$. Pairing with a Poisson-null ν_* removes the unknown active multiplier acceleration at leading order and yields (414). Rescale time by $t = a^3 s$; the finite-dimensional analytic vector field converges on every fixed s -interval to a linear displacement system, giving the shrinking layer and the nonzero limiting curvature slope. If a uniform Ca bound held on a fixed physical interval, choose a fixed s beyond the corresponding normalized bound and then take a sufficiently small so that $a^3 s < T$, contradicting that limiting slope. $\square$

The leading multiplier tilt has a finite affine freedom after the lower quadratic, normalization and cubic-mean conditions are imposed. On the preceding canonical first jet the tangency functional is constant and nonzero on that entire admissible tilt family; hence changing only the leading multiplier tilt cannot repair the obstruction.

The canonical first jet itself has additional admissible directions. Write $\mathcal{L}(Y, \ell)$ for the complete lower incidence consisting of the quadratic drift, multiplier normalizations and cubic mean rows. At the selected base point its multiplier derivative has constant rank, and one canonical direction Z_2 admits an exact multiplier tangent m_2 with

$$D\mathcal{L}(Y, \ell)[Z_2, m_2] = 0. \quad (416)$$

Theorem F.13 (The fixed-first-jet tangency obstruction is deformable). *For the exact lower-incidence tangent (Z_2, m_2) above, one weak null test $\nu_\diamond$ remains Poisson-null to first order and*

$$\boxed{D\Delta_{\nu_\diamond}(Y, \ell)[Z_2, m_2] \neq 0.} \quad (417)$$

Thus the nonzero tangency obstruction on the preceding prepared first jet is not invariant under all admissible canonical first-jet deformations of the unchanged finite action.

Proof. Differentiate every row of the finite lower-incidence map in the selected canonical direction and solve the resulting characteristic-zero linear system for m_2 . The same weak test has zero first variation of its Poisson-null condition in each coordinate direction of the local first-jet chart. The tangency functional is quadratic in the simultaneous first-jet/multiplier increment, so its derivative is obtained exactly by polarization. Direct finite algebra gives a nonzero rational coefficient; hence the derivative is nonzero over characteristic zero. The calculation establishes a tangent, not an exact nonlinear root. $\square$

F.8 Exact scope of the raw-action route

The preceding results change the status of the direct exact-action problem in three ways. First, a large part of the multiplier response is analytically regular and exactly eliminable at every cutoff. Second, source-neutral and active-regular finite preparations exist and give genuine original-action curvature information. Third, the first unresolved dynamical condition is now an explicit tangency hierarchy rather than an unspecified failure of a finite determinant.

A direct raw-action Einstein theorem would still require one common family on which all of the following hold:

- (R1) an exact nonlinear root of the complete transported tangency system in a regular canonical first-jet chart;
- (R2) persistence of the active and higher weak signed ranks and of the physical source/observation bounds along a cofinal refinement;
- (R3) a source-specific uniform estimate for the full scalar solve $\mathcal{S}_h n_h = g_h$, not merely for a diagonal compression;
- (R4) differentiated multiplier and curvature-source estimates strong enough to control the literal connection/metric observations; and
- (R5) a cutoff-independent physical interval on which the exact constrained Euler history remains in the common noncollapsed chart.

These are not assumed elsewhere in the paper. Conversely, the finite characteristic-rank difference is not a no-go theorem for singular continuum restoration of gauge symmetry. The open-law Einstein population and the exact-action hierarchy are therefore complementary: the first supplies a proved nonsymmetric $3 + 1$ continuum class, while the second identifies how much of that conclusion can already be recovered from the unchanged finite action and isolates the remaining dynamical bridge.

G Homogeneous renewal dynamics and operational action matching

This appendix proves the homogeneous dynamical statements of Section 5. Arbitrary positive homogeneous renewal variables are first varied with the full lapse constraint retained; the flat/de Sitter profiles are solved only after the stationary equations have been obtained. A second layer constructs a finite positive trial law whose measured probabilities reconstruct the corresponding finite action off shell.

G.1 Homogeneous ADM map and reduced vacuum action

On the root-symmetric branch $k_{\alpha,h} = \kappa/(8h^2)$ and $m_h = \varrho h^3$. Since $\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I$, one has $\mathcal{B} = \kappa I$ and $\beta = 0$. Substitution in (20) gives (122).

For completeness, let

$$ds^2 = -N(t)^2 dt^2 + a(t)^2 \delta_{ij} dx^i dx^j.$$

The nonzero Christoffel symbols are

$$\Gamma_{00}^0 = \dot{N}/N, \quad \Gamma_{ij}^0 = a\dot{a} \delta_{ij}/N^2, \quad \Gamma_{0j}^i = \dot{a} \delta_j^i/a.$$

Direct contraction gives

$$R_{00} = -3 \left(\ddot{a}/a - \dot{a}\dot{N}/(aN) \right), \quad (418)$$

$$R_{ij} = \left(a\ddot{a}/N^2 + 2\dot{a}^2/N^2 - a\dot{a}\dot{N}/N^3 \right) \delta_{ij}, \quad (419)$$

$$R = 6 \left(\ddot{a}/(aN^2) + \dot{a}^2/(a^2N^2) - \dot{a}\dot{N}/(aN^3) \right). \quad (420)$$

Thus

$$Na^3 R = 6 \frac{d}{dt} \left(\frac{a^2 \dot{a}}{N} \right) - \frac{6a\dot{a}^2}{N}. \quad (421)$$

With the orientation/sign convention used here, the bare torsion-free Einstein–Hilbert representative on $[t_0, t_1] \times \mathbb{T}^3$ is therefore

$$\frac{S_{\text{EH}}}{\chi V_0} = 6 \left[\frac{a^2 \dot{a}}{N} \right]_{t_0}^{t_1} + \int_{t_0}^{t_1} \left[-\frac{6a\dot{a}^2}{N} - 2\Lambda Na^3 \right] dt. \quad (422)$$

The Gibbons–Hawking–York Dirichlet boundary term [34, 35] contributes

$$\frac{S_{\text{GHY}}}{\chi V_0} = -6 \left[\frac{a^2 \dot{a}}{N} \right]_{t_0}^{t_1}, \quad (423)$$

so $S_{\text{EH}} + S_{\text{GHY}} = \chi V_0 S[a, N]$ with $S[a, N]$ exactly (123). The corresponding Dirichlet problem fixes the induced spatial metric at the two endpoint slices, $\delta a(t_0) = \delta a(t_1) = 0$, while the lapse remains independently variable in the interior. The finite action (129) is a midpoint variational discretization of this already boundary-completed first-order functional; hence fixing q_0, q_M is precisely the reduced discrete counterpart of the same induced-metric boundary condition. Substituting $a = \varrho^{1/3}$ and $N = \varrho^{1/3} \sqrt{\kappa}$ gives (124).

Detailed proof of Theorem 5.1. Varying the lapse and scale factor independently gives

$$E_N = 2a^3(3\mathcal{H}^2 - \Lambda), \quad E_a = 6Na^2(2D_\tau \mathcal{H} + 3\mathcal{H}^2 - \Lambda). \quad (424)$$

Because $a = e^{v/3}$ and $N = e^{u/2+v/3}$,

$$E_u = \frac{N}{2} E_N, \quad E_v = \frac{a}{3} E_a + \frac{N}{3} E_N,$$

which proves (125). The Jacobian of $(u, v) \mapsto (\log a, \log N)$ is $-1/6 \neq 0$, so this change of variables loses no variation. The reconstructed Einstein residual is

$$G_{00} + \Lambda g_{00} = N^2(3\mathcal{H}^2 - \Lambda), \quad G_{ij} + \Lambda g_{ij} = -a^2(2D_\tau \mathcal{H} + 3\mathcal{H}^2 - \Lambda)\delta_{ij}, \quad (425)$$

with $G_{0i} = 0$. Hence the two renewal Euler equations are equivalent to every Einstein component within this ansatz. $\square$

Proof of Corollary 5.2. For $\Lambda > 0$, the lapse constraint gives $\mathcal{H}^2 = H_0^2$ with $H_0 = \sqrt{\Lambda/3}$. Continuity on a connected interval fixes one sign σ , while $D_\tau \mathcal{H} = 0$ makes it constant. Integrating $D_\tau \log a = \sigma H_0$ gives (127); the renewal variables then follow from (122). The cases $\Lambda = 0$ and $\Lambda < 0$ follow from the same squared constraint. $\square$

Remark G.1 (Why the lapse is varied before gauge fixing). If $q = a^{3/2}$ and one sets $N = 1$ before variation, the remaining equation is $q'' = \omega^2 q$ with $\omega = \sqrt{3\Lambda/4}$. Its general solution $q = Ae^{\omega t} + Be^{-\omega t}$ satisfies the omitted lapse constraint $(q')^2 = \omega^2 q^2$ only when $AB = 0$. Thus a positive cosh-type solution is a spurious critical point of the prematurely gauge-fixed action.

G.2 Finite lapse-varied action and stationary recurrence

Put $q_j > 0$, $s_j > 0$, $\bar{q}_j = (q_j + q_{j+1})/2$, and define (129). Because this discretizes the boundary-completed first-order action, the discrete Dirichlet problem fixes q_0, q_M and varies $(q_1, \dots, q_{M-1}, s_0, \dots, s_{M-1})$ independently. A direct summation-by-parts calculation gives exactly (130); the endpoint momenta multiply δq_0 and δq_M and therefore vanish under the fixed induced endpoint geometry. No condition is imposed on $\sum_j s_j$, since that would modify the lapse variation and hence the Hamiltonian constraint. The change of variables (131), together with $q_j = \sqrt{\varrho_j}$, is a smooth bijection on the positive finite domain. Hence critical points in (q, s) and in the corresponding positive renewal coordinates are equivalent.

Proof of the Euler equations and stationary classification in Theorem 5.3. Differentiating an edge term with respect to s_j gives

$$A_0 \frac{(q_{j+1} - q_j)^2}{s_j^2} - B_0 \bar{q}_j^2,$$

which is (132). At an interior node the right derivative of the preceding edge and left derivative of the following edge give (133).

For $\Lambda > 0$, the lapse equation says $v_j = \sigma_j \omega \bar{q}_j$ with $\sigma_j \in \{\pm 1\}$. Solving for the terminal amplitude gives

$$q_{j+1} = q_j \frac{1 + \sigma_j \omega s_j / 2}{1 - \sigma_j \omega s_j / 2}.$$

Positivity requires $\omega s_j / 2 < 1$. Substitution in the interior momentum matching condition shows $2A_0 \omega q_j (\sigma_j - \sigma_{j-1}) = 0$, so every positive stationary path has one common sign. Conversely a common-sign recurrence satisfies every finite Euler equation. For $\Lambda = 0$ the lapse equation forces constant q ; for $\Lambda < 0$ it has no positive real solution.

To prove (135), set $x_j = \omega s_j / 2$. For $0 \leq x \leq b$,

$$0 \leq 2 \operatorname{arctanh} x - 2x = 2 \int_0^x \frac{t^2}{1 - t^2} dt \leq \frac{2x^3}{3(1 - b^2)}.$$

Taking logarithms of the recurrence and summing gives

$$|\log(q_j/q_0) - \sigma \omega \tau_j| \leq \frac{\omega^3}{12(1 - b^2)} \sum_i s_i^3 \leq \frac{\omega^3 T}{12(1 - b^2)} h_t^2.$$

The power relations between q, a, ϱ, κ then give the stated convergence of renewal variables. $\square$

For later residual estimates it is useful to define

$$R_j^\sigma = q_{j+1} - q_j - \sigma\omega s_j \bar{q}_j. \quad (426)$$

A direct expansion gives the exact factorization

$$S_d = -A_0 \sum_j \frac{(R_j^\sigma)^2}{s_j} - A_0 \sigma\omega (q_M^2 - q_0^2). \quad (427)$$

Thus at fixed endpoints the finite stationary branch is exactly the zero set of the nonnegative residual energy $\mathcal{E} = A_0 \sum_j (R_j^\sigma)^2 / s_j$.

Lemma G.2 (Full finite variation controlled by the residual energy). *For endpoint-vanishing nodal variations w_j and lapse variations $\delta s_j = s_j n_j$, put*

$$\mathcal{B}_s(w, n; q) = \sum_j s_j \left(\frac{w_{j+1} - w_j}{s_j} - \sigma\omega \bar{w}_j - \sigma\omega \bar{q}_j n_j \right)^2. \quad (428)$$

Then

$$|\delta S_d[w, sn]| \leq 2\sqrt{A_0 \mathcal{E}} \sqrt{\mathcal{B}_s(w, n; q)} + \mathcal{E} \|n\|_\infty. \quad (429)$$

If q_j^* is the exact common-sign stationary recurrence with the same q_0, s_j , and $\omega \max s_j / 2 \leq b < 1$, then on a fixed positive amplitude range

$$\max_j |q_j - q_j^*| \leq C\sqrt{\mathcal{E}}. \quad (430)$$

Proof. Differentiate (427); weighted Cauchy–Schwarz gives (429). For the path estimate, write the forced recurrence as $q_{j+1} = r_j q_j + R_j / d_j$ with $d_j = 1 - \sigma\omega s_j / 2$ and $r_j = (1 + \sigma\omega s_j / 2) / d_j$. Iteration, $d_j \geq 1 - b$, and $\sum_j |R_j| \leq \sqrt{T \sum_j R_j^2 / s_j}$ give the result. $\square$

G.3 Measured quadratic costs and exact off-shell action matching

We now prove that the finite action can be read from an actual normalized trial law rather than appended as an unrelated deterministic writer. Let $E_s(x, y)$ be (136).

Proposition G.3 (Three-cost identification and rank defect). *Fix $s > 0$ and distinct $u, v > 0$. Within the declared quadratic class, $E_s(u, u)$, $E_s(u, v)$, and $E_s(v, u)$ determine*

$$B = \frac{E_s(u, u)}{su^2}, \quad (431)$$

$$C = \frac{E_s(u, v) - E_s(v, u)}{2(v^2 - u^2)}, \quad (432)$$

$$A = \frac{s}{2(v - u)^2} \left[E_s(u, v) + E_s(v, u) - 2Bs \left(\frac{u + v}{2} \right)^2 \right]. \quad (433)$$

Moreover, with $\mathfrak{D} = AB - C^2$, $c = -C/A$, and $\nu = \mathfrak{D}/A$,

$$E_s(x, y) = \frac{A}{s} \left[y - x - cs \frac{x + y}{2} \right]^2 + \nu s \left(\frac{x + y}{2} \right)^2. \quad (434)$$

Thus $\mathfrak{D} = 0$ is exactly the rank-one compatibility condition.

Proof. The diagonal measurement determines B . Reversal exchanges x and y , leaving the first two terms invariant and changing the sign of the C term, which determines C ; the reversal sum then determines A . Completing the square gives (434). $\square$

Proof of Theorem 5.4. From (139), $Z_s Q_s = p_s a_s$. Taking logarithms and using $a_s = e^{-E_s/\varepsilon}$ proves (140). Summing over one path gives

$$\varepsilon \sum_j [\log(Q_j/p_j) + \log Z_j] = - \sum_j E_{s_j}(q_j, q_{j+1}).$$

The C term telescopes to $C(q_M^2 - q_0^2)$, which is exactly cancelled by the endpoint term in (141); this proves the off-shell identity (142). Differentiating that identity on the smooth positive parametric domain proves equality of all interior and lapse first variations.

For the mismatch statement, minimizing $y \mapsto E_s(x, y)$ gives, for small s , a ratio with logarithmic drift $c = -C/A$, so the limiting scale factor has Hubble rate $2c/3$. Hence its geometric cosmological parameter is $\Lambda_{\text{geom}} = 4C^2/(3A^2)$. Dividing the bulk action by A/A_0 shows that its parameter is $\Lambda_{\text{act}} = 4B/(3A)$. Therefore

$$G[g] + \Lambda_{\text{act}} g = (\Lambda_{\text{act}} - \Lambda_{\text{geom}})g = \frac{4(AB - C^2)}{3A^2}g,$$

which is (143). $\square$

Remark G.4 (Why the acceptance normalizer is physical). Retaining only p_s and Q_s would not determine the action: replacing $E_s(x, y)$ by $E_s(x, y) + c_s(x)$ leaves Q_s unchanged but multiplies $Z_s(x)$ by $e^{-c_s(x)/\varepsilon}$. Such a source-dependent term can change the lapse variation. The total acceptance frequency is therefore a load-bearing part of the complete physical trial table.

G.4 An explicit finite stochastic cylinder

We give the construction underlying Theorem 5.5. Set $s = T/M$ and, for large M ,

$$d = 1 - \sigma\omega s/2, \quad r = \frac{1 + \sigma\omega s/2}{1 - \sigma\omega s/2}, \quad \delta_M = M^{-3}, \quad \varepsilon_M = \frac{M^{-5}}{1 + \log M}. \quad (435)$$

Starting from $F_0 = \{q_0\}$, define support intervals recursively by

$$L_0 = q_0/2, \quad U_0 = 3q_0/2, \quad L_{j+1} = rL_j - \delta_M, \quad U_{j+1} = rU_j + \delta_M,$$

and let F_j be finite grids of mesh at most δ_M in these intervals. Each source proposes uniformly from F_{j+1} and accepts with

$$a_j(x, y) = \exp \left[-\frac{A_0 d^2 (y - rx)^2}{s\varepsilon_M} \right]. \quad (436)$$

Because $d(y - rx) = y - x - \sigma\omega s(x + y)/2$, this is exactly the rank-one cost (144).

Lemma G.5 (History-level entropy–energy estimate). *At stage j , let a finite reference row p_j and nonnegative cost E_j define $Q_j = p_j e^{-E_j/\varepsilon}/Z_j$. Let $\mathbb{Q}$ be the product of these accepted rows and $\mathbb{P} \ll \mathbb{Q}$ any complete path law with the same initial value, possibly with memory. If each relevant row contains a proposal of cost at most e_j and every proposal weight is at least $p_{*,j}$, then*

$$\mathbb{E}_{\mathbb{P}} \sum_j E_j \leq \sum_j e_j + \varepsilon \left[\sum_j \log(1/p_{*,j}) + D_{\text{KL}}(\mathbb{P} \parallel \mathbb{Q}) \right]. \quad (437)$$

Proof. For a fixed history prefix, direct substitution gives

$$\mathbb{E}_P E_j = \varepsilon D_{\text{KL}}(P_j \| Q_j) - \varepsilon D_{\text{KL}}(P_j \| p_j) - \varepsilon \log Z_j.$$

The candidate bound gives $Z_j \geq p_{*,j} e^{-e_j/\varepsilon}$. Integrating over prefixes, summing, and using the chain rule for path relative entropy proves (437). $\square$

Proof of Theorem 5.5. The rational stationary multiplier and its inverse are uniformly bounded on $0 \leq j \leq M$ when $\omega s/2 \leq b < 1$. Iterating the support recursion shows that all F_j lie in one fixed interval $0 < q_- \leq q \leq q_+ < \infty$ and have cardinality $O(M^3)$. For every source x , the exact target rx lies inside the next support and has a grid point within δ_M . The corresponding cost is at most $C\delta_M^2/s = CM^{-5}$. Hence the proposal floor is $p_* \geq cM^{-3}$ and the total acceptance probability has a positive explicit lower bound.

Apply Lemma G.5 to the ideal accepted product law $\mathbb{Q}_M$ itself. With the scales (435),

$$\mathbb{E}_{\mathbb{Q}_M} \mathcal{E}_M \leq CM^{-4}. \quad (438)$$

The deterministic estimates (429) and (430), Jensen's inequality, and the second-order stationary-profile estimate then give the two bounds in (145) and an $O(M^{-2})$ expected first variation against every fixed smooth scale/lapse test.

To make every cutoff history genuinely finite, permit at most

$$R_M = \lceil 7\underline{Z}_M^{-1} \log M \rceil$$

independent proposals at each stage, where $\underline{Z}_M$ is the explicit row-wise acceptance lower bound. A union bound gives total failure probability at most M^{-6} . Conditional on complete success, the path-law correction has relative entropy at most $-\log(1 - M^{-6})$, which changes (438) only by a summable term. The expected residual energies are summable in M , so Tonelli and the first Borel–Cantelli lemma give $\mathcal{E}_M \rightarrow 0$ almost surely under any common coupling. The deterministic path estimates then imply uniform and strong H^1 convergence to $q_0 e^{\sigma\omega\tau}$.

Finally, (146) is the positive ADM inverse on the proper-time nodal branch. The A_3 root identity gives exactly $\mathcal{B}_j = \kappa_j I$, and ordinary Taylor expansion of the symmetric root generator yields the spatial continuum coefficient $\frac{1}{2}q_*^{-4/3}\Delta$. Together with the homogeneous Einstein identity, this proves the vacuum limit with $\Lambda = 4\omega^2/3$. $\square$

Remark G.6 (Exact scope of the operational construction). The quadratic duration law, the rank-one condition $\mathfrak{D} = 0$, the nonzero action normalization, and the value of Λ are not deduced from the bare predictive quotient. They are finite operational conditions that can be measured and can fail. The construction proves that this matched class is nonempty and robust under vanishing measured action/likelihood errors; it does not prove generic microscopic attraction to that class.

G.5 Exact conservative blocking and the relative defect

Put $u = A/s + Bs/4$, $v = A/s - Bs/4$, $\mu = \sqrt{AB}$ and $\theta = 2 \operatorname{artanh}((s/2)\sqrt{B/A})$. Then $u^2 - v^2 = \mu^2$, $u = \mu \coth \theta$ and $v = \mu \operatorname{csch} \theta$, giving (148).

Lemma G.7 (Two-edge Schur identity). *Let $E_i(x, y) = p_i x^2 - 2q_i xy + r_i y^2$ with $p_i, q_i, r_i > 0$ and $d_i = p_i r_i - q_i^2 \geq 0$. Then the minimizer of $E_1(x, z) + E_2(z, y)$ is positive on positive endpoints and the blocked quadratic has*

$$p' = p_1 - \frac{q_1^2}{r_1 + p_2}, \quad q' = \frac{q_1 q_2}{r_1 + p_2}, \quad r' = r_2 - \frac{q_2^2}{r_1 + p_2}, \quad (439)$$

with

$$\boxed{p'r' - (q')^2 = \frac{r_2 d_1 + p_1 d_2}{r_1 + p_2}.} \quad (440)$$

Proof. Complete the square in z . The minimizer is $z_* = (q_1 x + q_2 y)/(r_1 + p_2) > 0$. The three coefficients in (439) are the remaining quadratic coefficients; direct expansion gives (440). $\square$

Proof of Theorem 5.6. For $\mathcal{E}_{\mu, \theta_i, C}$ put $u_i = \mu \coth \theta_i$ and $v_i = \mu \operatorname{csch} \theta_i$. The two intermediate Cz^2 terms cancel. Applying Lemma G.7 and $u_i^2 - v_i^2 = \mu^2$ gives

$$u' = \frac{u_1 u_2 + \mu^2}{u_1 + u_2}, \quad v' = \frac{v_1 v_2}{u_1 + u_2}, \quad C' = C. \quad (441)$$

The hyperbolic addition formulas identify these with $\mu \coth(\theta_1 + \theta_2)$ and $\mu \operatorname{csch}(\theta_1 + \theta_2)$, proving (150). Converting the two-identical-edge result back to midpoint coefficients gives (151). Hence AB , C , $\mathfrak{D}$, and $\Delta_{\text{rel}} = \mathfrak{D}/(AB)$ are invariant.

For a change of amplitude and time units, $x = r\tilde{x}$ and $s = \ell\tilde{s}$, followed by common action rescaling by $d > 0$, one has $\tilde{A} = r^2 A/(d\ell)$, $\tilde{B} = r^2 \ell B/d$, and $\tilde{C} = r^2 C/d$; therefore $C^2/(AB)$ is unchanged.

For the finite positive statement, let $F \subset (0, \infty)$ be finite and write the subprobability kernel as

$$L_C(x, y) = e^{Cx^2/\varepsilon} L_0(x, y) e^{-Cy^2/\varepsilon}, \quad L_0(x, y) = L_0(y, x) > 0. \quad (442)$$

Then $L_C^n = D_C L_0^n D_C^{-1}$ and hence, for $x \neq y$,

$$\boxed{\frac{\varepsilon}{2(x^2 - y^2)} \log \frac{L_C^n(x, y)}{L_C^n(y, x)} = C.} \quad (443)$$

Thus finite blocking preserves the measured endpoint twist exactly. If the intermediate positive amplitudes are integrated over a grid of covering error d and cardinality m , completion of the positive interior quadratic form around its exact bridge minimizer gives

$$0 \leq E_{F,n}^\varepsilon(x, y) - \mathcal{E}_{\mu, n\theta, C}(x, y) \leq 4u(n-1)d^2 + \varepsilon(n-1) \log m. \quad (444)$$

At fixed total duration one may take $d_n = n^{-2}$, $m_n = O(n^2)$, and $\varepsilon_n = n^{-3}/(1 + \log n)$, making the right side $O(n^{-2})$. The three-probe coefficient reconstruction is continuous on a compact calibrated range with $AB > 0$, so every initial $\Delta_{\text{rel}} > 0$ remains bounded away from zero for sufficiently fine finite blocking. This proves the operational statement. $\square$

G.5.1 The cross-scale optimizer test

Proof of Proposition 5.7. The coefficient of y^2 in $\mathcal{E}_{\mu, \theta, C}(x, y)$ is $\mu \coth \theta + C > 0$, since $|C| \leq \mu$ and $\theta > 0$. Its unique minimizer is therefore

$$y = \frac{\mu \operatorname{csch} \theta}{\mu \coth \theta + C} x = \frac{x}{\cosh \theta + \rho \sinh \theta} > 0.$$

Set $d_\theta = \cosh \theta + \rho \sinh \theta$. Hyperbolic addition gives

$$d_{\theta+\phi} - d_\theta d_\phi = (1 - \rho^2) \sinh \theta \sinh \phi,$$

which proves (153). All denominators and both hyperbolic sines are positive. Equality of the maps for even one positive pair and one nonzero amplitude therefore forces $1 - \rho^2 = 0$, equivalently $AB = C^2$. Conversely, $\rho = \pm 1$ gives $d_\theta = e^{\rho\theta}$ and exact composition for every pair. For $\Delta_{\text{rel}} > 0$ the formula also gives $r_{\theta+\phi} < r_\theta r_\phi$.

Solving the same identity for the defect gives

$$\Delta_{\text{rel}} = \frac{r_{\theta+\phi}^{-1} - (r_\theta r_\phi)^{-1}}{\sinh \theta \sinh \phi}. \quad (445)$$

On a compact range where the observed ratios and $\sinh \theta \sinh \phi$ are bounded away from zero, the right-hand side has bounded first derivatives in its measured arguments. It is consequently locally Lipschitz there. This is stability of a finite diagnostic, not transverse attraction under the conservative blocking operation. $\square$

H Nonlinear two-polarization vacuum regulator

This appendix proves Theorem 5.8. The purpose is not to establish unrestricted four-dimensional global regularity, but to show that the finite renewal evolution, constraint, readout, and independent action tests can be carried through a genuinely nonlinear spatially inhomogeneous vacuum sector. We use the unpolarized Gowdy T^3 class in areal coordinates [37]; the lapse and shift are nevertheless retained as independent variables until after the action variation.

H.1 First-order frame system and constraints

On $[t_0, t_1] \times \mathbb{T}^3$, $t_0 > 0$, consider

$$g = t^{-1/2} e^{\lambda/2} (-dt^2 + d\theta^2) + t [e^P (dx + Qdy)^2 + e^{-P} dy^2]. \quad (446)$$

With $a = P_t$, $b = P_\theta$, $c = e^P Q_t$, $d = e^P Q_\theta$ and

$$e = \frac{1}{2}(a^2 + b^2 + c^2 + d^2), \quad f = ab + cd, \quad (447)$$

the vacuum equations take the first-order form

$$\begin{aligned} a_t &= b_\theta - a/t + c^2 - d^2, & b_t &= a_\theta, \\ c_t &= d_\theta - c/t - ac + bd, & d_t &= c_\theta + ad - bc, \\ P_t &= a, & Q_t &= e^{-P} c, & \lambda_t &= 2te, \end{aligned} \quad (448)$$

with

$$P_\theta = b, \quad e^P Q_\theta = d, \quad \lambda_\theta = 2tf. \quad (449)$$

A direct differentiation using (448) gives

$$\boxed{e_t - f_\theta = -\frac{a^2 + c^2}{t}, \quad f_t - e_\theta = -\frac{f}{t}.} \quad (450)$$

Consequently $\partial_t(\lambda_\theta - 2tf) = 0$ whenever the displayed equations hold, and $\partial_t \int_{\mathbb{T}^1} tf \, d\theta = 0$. Thus the momentum constraint and the periodic zero-mean condition for λ propagate.

H.2 Independent lapse and shift variation before areal gauge

Let $(R, \lambda, P, Q, N, \beta)$ be independent fields with $R, N > 0$, put $v_u = \dot{u} - \beta u'$, and define

$$\begin{aligned} \mathcal{L}_G = & -\frac{v_R(v_\lambda - 4\beta')}{2N} + \frac{R}{2N}(v_P^2 + e^{2P}v_Q^2) \\ & + N \left[-2R'' + \frac{R'\lambda'}{2} - \frac{R}{2}\{(P')^2 + e^{2P}(Q')^2\} \right]. \end{aligned} \quad (451)$$

The canonical momenta are

$$p_R = -\frac{v_\lambda - 4\beta'}{2N}, \quad p_\lambda = -\frac{v_R}{2N}, \quad \pi = \frac{Rv_P}{N}, \quad \varpi = \frac{Re^{2P}v_Q}{N}. \quad (452)$$

Lapse and shift variation give respectively

$$\mathcal{C} = -2p_R p_\lambda + \frac{\pi^2 + e^{-2P}\varpi^2}{2R} + \frac{R}{2}\{(P')^2 + e^{2P}(Q')^2\} + 2R'' - \frac{1}{2}R'\lambda', \quad (453)$$

$$\mathcal{D} = p_R R' + p_\lambda \lambda' + \pi P' + \varpi Q' - 4p'_\lambda. \quad (454)$$

On a solution of (448), set

$$R = t, \quad N = 1, \quad \beta = 0, \quad p_\lambda = -\frac{1}{2}, \quad p_R = -te, \quad \pi = ta, \quad \varpi = te^P c. \quad (455)$$

Substitution of (449) shows $\mathcal{C} = 0$ and $\mathcal{D} = 0$; the remaining area equation follows from (450). Hence the areal-gauge solution is the restriction of a critical point of the unreduced symmetry-class action rather than a solution obtained after deleting the lapse and shift equations.

H.3 Local splitting and exact staggered momentum

Set

$$U = (u_1, u_2, u_3, u_4) = \sqrt{t}(a, b, c, d), \quad \mathcal{J}(U) = u_1 u_2 + u_3 u_4, \quad n(U) = |U|^2. \quad (456)$$

The local source flow, with the evolution parameter denoted by s , is

$$\begin{aligned} \frac{du_1}{ds} &= -\frac{u_1}{2t} + \frac{u_3^2 - u_4^2}{\sqrt{t}}, & \frac{du_2}{ds} &= \frac{u_2}{2t}, \\ \frac{du_3}{ds} &= -\frac{u_3}{2t} + \frac{-u_1 u_3 + u_2 u_4}{\sqrt{t}}, & \frac{du_4}{ds} &= \frac{u_4}{2t} + \frac{u_1 u_4 - u_2 u_3}{\sqrt{t}}, \\ \frac{dP}{ds} &= \frac{u_1}{\sqrt{t}}, & \frac{dQ}{ds} &= e^{-P} \frac{u_3}{\sqrt{t}}, & \frac{dt}{ds} &= 1, \end{aligned} \quad (457)$$

with $d\lambda/ds = 0$. Let S_σ be its near-identity implicit-midpoint map over source duration σ .

On a periodic grid of spacing ℓ , let $S_\pm$ be the two shifts, $D_+ = (S_+ - I)/\ell$, and $A_+ = (I + S_+)/2$.

Put

$$w^+ = \frac{1}{\sqrt{2}}(u_1 + u_2, u_3 + u_4), \quad w^- = \frac{1}{\sqrt{2}}(u_1 - u_2, u_3 - u_4), \quad g^\pm = |w^\pm|^2. \quad (458)$$

At frozen physical time define the transport map by

$$(w^+)^+ = S_+ w^+, \quad (w^-)^+ = S_- w^-, \quad \lambda^+ = \lambda + \frac{\ell}{2}[(I + S_+)g^+ + (I + S_-)g^-], \quad (459)$$

leaving P, Q fixed.

Theorem H.1 (Exact staggered momentum under symmetric splitting). *With synchronized full step $h = \ell$, the source-half/transport/source-half composition preserves exactly*

$$\boxed{\mathcal{K}_\ell = D_+\lambda - 2\mathbf{A}_+\mathcal{J}(U).} \quad (460)$$

The update is nearest neighbour, its background increment is a nonnegative sum of squared characteristic amplitudes, and it is second-order consistent on bounded smooth reference charts with $t \geq t_0 > 0$.

Proof. For the source flow, direct substitution in (457) gives $\nabla \mathcal{J}(U) \cdot \dot{U} = 0$. The difference of a quadratic function at two endpoints equals its derivative at the arithmetic midpoint paired with the increment, so implicit midpoint preserves $\mathcal{J}$ exactly while leaving λ fixed. For the transport step,

$$\Delta \mathcal{J} = \frac{1}{2}[(S_+ - I)g^+ - (S_- - I)g^-], \quad D_+\Delta\lambda = 2\mathbf{A}_+\Delta\mathcal{J}, \quad (461)$$

which proves exact preservation without invoking an assumed discrete product rule. The λ increment in (459) is a positive sum of squared norms. Characteristic translation is exact for the transport fields, while the background increment is a trapezoidal approximation to the integral of the translated squares with local error $O(\ell^3)$. Implicit midpoint has the same local order for the smooth source, and symmetric composition therefore gives second-order consistency. $\square$

H.4 Residual, constraint, metric, and action budgets

Let $X_j = (U_j, P_j, Q_j, \lambda_j)$ satisfy

$$X_{j+1} = \Phi_h(t_j, X_j) + r_j, \quad h = \ell, \quad (462)$$

and define

$$\mathcal{E}_h = \sum_j \frac{\|r_j\|_{2,\ell}^2}{2h}, \quad \mathcal{F}_h = \ell^{-(2r+1)} \mathcal{E}_h. \quad (463)$$

Here $\mathbf{S}_\ell$ denotes grid sampling.

Proposition H.2 (Residual and constraint control). *On a fixed smooth reference chart, with initial C_ℓ^r error ϵ_0 and either a smallness bootstrap or an enforced offset envelope,*

$$\boxed{\max_j \|X_j - \mathbf{S}_\ell X_*(t_j)\|_{r,\infty,\ell} \leq C(\epsilon_0 + h^2 + \sqrt{\mathcal{F}_h}).} \quad (464)$$

Writing $\widehat{U}_j$ for the deterministic predicted frame, the exact constraint increment is

$$\begin{aligned} \mathcal{K}_\ell(X_{j+1}) - \mathcal{K}_\ell(X_j) &= D_+r_j^\lambda \\ &\quad - 2\mathbf{A}_+[\nabla \mathcal{J}(\widehat{U}_j) \cdot r_j^U + \mathcal{J}(r_j^U)]. \end{aligned} \quad (465)$$

For $r \geq 1$, its accumulated norm is bounded by the initial residual plus $C\sqrt{\mathcal{F}_h}$.

Proof. The deterministic split map has smooth-chart Lipschitz constant $1 + Ch$ and consistency error Ch^3 . Discrete Gronwall therefore bounds the global error by $C(\epsilon_0 + h^2 + \sum_j \|r_j\|_{r,\infty,\ell})$. The one-dimensional inverse grid estimate and Cauchy–Schwarz bound the last sum by $C\sqrt{\mathcal{F}_h}$, proving (464). Expanding the quadratic $\mathcal{J}$ about $\widehat{U}_j$ and using exact preservation (460) gives (465); the same inverse estimate controls its accumulated norm. $\square$

For the pathwise curvature statement specialize the finite offset alphabet so that every accepted offset obeys

$$\|\epsilon_j\|_\infty \leq c_0 h^4, \quad (466)$$

and take initial C_h^1 error $O(h^2)$. The retained source implementation may add a solver error $O(h^5)$ in C_h^1 . Since $\|D_h v\|_\infty \leq 2h^{-1}\|v\|_\infty$, the offset has C_h^1 size $O(h^3)$. The one-step Lipschitz factor $1 + Lh$, second-order consistency, and discrete Gronwall then give, pathwise,

$$\boxed{\max_j \|X_j - S_h X_*(t_j)\|_{1,\infty,h} \leq Ch^2.} \quad (467)$$

For $f = P, Q, \lambda$, subtraction of the reference step and division by h also gives an $O(h^2)$ bound for the nodal time differences of the errors.

Use the unchanged derivative records

$$\begin{aligned} u &= \left(u_1/\sqrt{t}, e^{-P} u_3/\sqrt{t}, |U|^2 \right), \\ v &= \left(u_2/\sqrt{t}, e^{-P} u_4/\sqrt{t}, 2\mathcal{J}(U) \right), \quad w = D_0 u, \end{aligned} \quad (468)$$

where $D_0 = (S_+ - S_-)/(2h)$. These are respectively the time, space, and mixed jet records for (P, Q, λ) . On each spacetime square let $\mathcal{H}_h(f, u, v, w)$ be the tensor-product cubic Hermite interpolant using

$$\begin{aligned} H_0(q) &= 2q^3 - 3q^2 + 1, & H_1(q) &= -2q^3 + 3q^2, \\ G_0(q) &= q^3 - 2q^2 + q, & G_1(q) &= q^3 - q^2. \end{aligned}$$

The identities $H_0 + H_1 = 1$, $H'_0 + H'_1 = 0$, and $H''_0 + H''_1 = 0$ imply the stability estimate

$$\|F\|_\infty + \|DF\|_\infty \leq C\eta, \quad \|D^2 F\|_\infty \leq C\eta/h, \quad (469)$$

whenever the two nodal jet banks and the first nodal differences of their value records differ by at most η . Shared corner jets make the interpolant C^1 across cell interfaces, so its weak second derivative is the cellwise second derivative and has no interface measure.

Theorem H.3 (Pathwise full-curvature control on the Gowdy alphabet). *Assume the smooth reference has bounded derivatives through the order required by the Hermite remainder, (466), and (467). For the shared Hermite reconstruction of $f = (P, Q, \lambda)$,*

$$\boxed{\|f_h - f_*\|_{W^{1,\infty}} \leq Ch^2, \quad \|D^2 f_h - D^2 f_*\|_{L^\infty} \leq Ch.} \quad (470)$$

For the reconstructed metric (446),

$$\begin{aligned} \|g_h - g_*\|_{W^{1,\infty}} &\leq Ch^2, \\ \|D^2 g_h - D^2 g_*\|_{L^\infty} + \|\text{Riem}(g_h) - \text{Riem}(g_*)\|_{L^\infty} &\leq Ch. \end{aligned} \quad (471)$$

Hence on every compact positive subslab

$$\boxed{\sup_h \|\text{Riem}(g_h)\|_{L^2} < \infty, \quad \|\text{Ric}(g_h)\|_{L^\infty} \leq Ch.} \quad (472)$$

All estimates are pathwise and the curvature limit is the Levi-Civita curvature of g_* . No all-time curvature norm is assumed.

Proof. The right sides of (468) are smooth functions on the standing chart. Equation (467), the nodal time difference estimate, and centered-difference consistency therefore make every corner value/first/mixed jet error $O(h^2)$. Apply (469); the exact-reference cubic Hermite remainder is $O(h^{4-k})$ through derivative order $k = 2$. This proves (470).

On $t \geq t_0 > 0$ the Gowdy metric map is smooth on the bounded chart and has determinant $-te^\lambda$, so its inverse stays uniformly bounded. The metric estimates follow by differentiating this smooth composition twice. In coordinates, $\Gamma = g^{-1} * Dg$, hence $\|\Gamma_h - \Gamma_*\|_\infty \leq Ch^2$ and $\|D\Gamma_h - D\Gamma_*\|_\infty \leq Ch$. Because g_h is C^1 across interfaces there is no distributional surface term. Substitution in $\text{Riem} = D\Gamma + \Gamma * \Gamma$ gives (471). The reference is vacuum, so contraction with the uniformly bounded metric and inverse gives the Ricci estimate; finite volume gives the L^2 curvature certificate. $\square$

A finite positive realization is obtained by giving each predicted local state a finite offset list containing zero and contained in (466), a strictly positive reference row, and acceptance probability $\exp[-E_j/\varepsilon]$ for a nonnegative deviation cost. If $p_{*,j}$ is the reference mass of the zero offset, the exact Gibbs identity gives

$$\mathbb{E} \sum_j E_j \leq \varepsilon \left[\mathcal{K}_{\text{path}} + \sum_j \log(1/p_{*,j}) \right], \quad (473)$$

where $\mathcal{K}_{\text{path}}$ is the chain-rule relative entropy of the observed history law against the declared accepted law. Product reference masses count all local trials; solver residuals, finite retry failures, and iteration errors remain separate budgets. The positive selection cost is not identified with the signed ADM action. Since the curvature estimates above are deterministic and uniform over the permitted alphabet, every successful history inherits (472).

H.5 Measured proposal access and finite execution

The deterministic second-jet estimate is uniform over the stated offset alphabet. It therefore survives changes of the actual proposal weights, provided the acceptance rule retains that support and successful proposals really occur. At a fixed stage and source history write $\mu = p(\mathcal{G})$ for the measured mass of the admissible set, and use

$$a(u) = \mathbf{1}_{\mathcal{G}}(u) e^{-E(u)/\varepsilon}, \quad Z = \sum_u p(u) a(u), \quad Q(u) = p(u) a(u) / Z.$$

This is a declared selection rule, not an unrecorded replacement of the original microscopic transition.

Proof of Proposition 5.9. Since $0 \leq E \leq c$ on $\mathcal{G}$, $\mu e^{-c/\varepsilon} \leq Z \leq \mu$. For $\mu > 0$, let $p_{\mathcal{G}} = p/\mu$ on $\mathcal{G}$. Every $Q \ll p$ supported on $\mathcal{G}$ obeys the exact identity

$$D_{\text{KL}}(Q\|p) = D_{\text{KL}}(Q\|p_{\mathcal{G}}) - \log \mu \geq -\log \mu.$$

In particular $\mu = 0$ precludes such an accepted row altogether. No choice of temperature can correct this support failure.

On the accepted support the complete trial table still reconstructs its cost: $E(u) = -\varepsilon \log(ZQ(u)/p(u))$. Averaging also gives $\mathbb{E}_Q E = -\varepsilon \log Z - \varepsilon D_{\text{KL}}(Q\|p)$. These identities concern the recorded positive deviation cost; the independently varied signed action is unchanged.

With independent reset retries conditional on the same source, the probability that R_j attempts fail is $(1 - Z_j)^{R_j} \leq e^{-R_j Z_j}$. The same bound follows for history-dependent retries when the displayed lower success bound holds conditionally after every failed attempt. Taking the supported lower bound for each stage and using a union bound proves (166). For J stages and total failure target $0 < \delta < 1$, one sufficient finite choice is

$$R_j = \left\lceil \mu_j^{-1} e^{c_j/\varepsilon_j} \log(J/\delta) \right\rceil. \quad (474)$$

Conditioning the entire path on success may reweight its intermediate source states. No equality with the unconditioned product of accepted rows is needed: every successful path has the same deterministic admissible support, and hence the same curvature and action bounds. Summability of the complete failure probabilities and the first Borel–Cantelli lemma give the final assertion, without independence between refinements. $\square$

Lemma H.4 (A finite execution certificate for the proposal floor). *At a retained source snapshot let a launch race have finitely many bins. Conditional on no earlier launch, let a_ℓ be the probability of any launch and f_ℓ the probability of a predictor-marked launch in bin ℓ . Set $S_0 = 1$, $S_\ell = \prod_{i \leq \ell} (1 - a_i)$. Suppose*

$$f_\ell \geq \rho_* a_\ell, \quad -\log S_J \geq A_* > 0, \quad 0 < \rho_* \leq 1. \quad (475)$$

If, conditional on every predictor launch, the actually executed writer returns a certified member of $\mathcal{G}$ before the attempt deadline with probability at least $q_ > 0$, then*

$$\boxed{p(\mathcal{G}) \geq q_* \rho_* (1 - e^{-A_*}).} \quad (476)$$

Proof. The disjoint first-launch events give predictor launch probability $\sum_{\ell=1}^J S_{\ell-1} f_\ell$. Since $S_{\ell-1} a_\ell = S_{\ell-1} - S_\ell$, this is at least $\rho_* (1 - S_J) \geq \rho_* (1 - e^{-A_*})$. On each such event the conditional completed-safe-output probability is at least q_* . Multiplication of conditional probabilities and summation prove the claim. Launch, interruption, completion and output certificates belong to the actual same-history record; a possible but unexecuted numerical predictor supplies none of these probabilities. $\square$

The execution probability and deadline in this lemma are physical inputs, not consequences of a fast geometric root clock. For example, one root clock competing with D clocks of equal rate has share $1/(D + 1)$, regardless of their common rate scale. Moreover, a cap on the number of retries bounds their physical duration only when each attempt has a certified duration bound. With attempt durations at most d_j , the allowed stage duration must accommodate $R_j d_j$. Thus the proposition removes the need to assume a separate continuum proposal profile on a class with the displayed finite execution budgets, without asserting a uniform computation or sampling cost for every microscopic law.

For the general residual/action estimate below put

$$\kappa_h := \epsilon_0 + h^2 + \sqrt{\mathcal{F}_h}. \quad (477)$$

Finally discretize (451) by midpoint time sampling and centered spatial differences, keep N and β independent, and replace $-2NR''$ by $2N'R'$ under the periodic boundary convention. Evaluation at the recorded areal-gauge fields gives, for every fixed smooth compactly supported independent test $(\delta R, \delta \lambda, \delta P, \delta Q, \delta N, \delta \beta)$,

$$\boxed{|\delta S_h - \delta S_*| \leq C \kappa_h.} \quad (478)$$

The estimate follows from the same first-jet comparison used above, and the variation is taken before the areal-gauge restriction. Hence $\mathbb{E}\mathcal{F}_h = O(h^4)$ and $\epsilon_0 = O(h^2)$ yield second-order mean metric and action-test errors. Exact preservation of $\mathcal{K}_\ell$ is one constraint statement and is not substituted for the independent lapse/shift action tests.

Proof of Theorem 5.8. Item (G1) is Theorem H.1. Item (G2) is Proposition H.2. The pathwise C_h^1 estimate and Theorem H.3 give item (G3). The unreduced variational calculation (451)–(455), the action test (478), and the Gibbs identity (473) give item (G4) while retaining the lapse and shift variations before gauge restriction. $\square$

I Flat and de Sitter regulator limits

This section proves the spatial and Cartan-continuum parts of Theorem 5.10. The temporal profiles are no longer posited here: they are the stationary renewal solutions derived in Appendix G. The present appendix verifies that the associated A_3 spatial generators have the required normalization and compact screens, and that the reconstructed coframes and exact link transports converge to the corresponding flat and de Sitter Cartan geometries.

I.1 The A_3 identity and periodic regulators

Let

$$W_0 = \left\{ x \in \mathbb{R}^4 : \sum_{i=1}^4 x_i = 0 \right\}, \quad \Lambda_{A_3} = W_0 \cap \mathbb{Z}^4, \quad (479)$$

and let

$$\Phi_{A_3} = \{e_i - e_j : 1 \leq i \neq j \leq 4\} \quad (480)$$

be the twelve oriented roots. The scalar product on W_0 is the restriction of the standard Euclidean scalar product on $\mathbb{R}^4$.

Lemma I.1 (A_3 second-moment identity). *On W_0 ,*

$$\boxed{\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I_{W_0}.} \quad (481)$$

Equivalently, for every $v \in W_0$,

$$\sum_{i \neq j} (v_i - v_j)^2 = 8\|v\|^2. \quad (482)$$

Proof. For $v \in W_0$, expand

$$\sum_{i \neq j} (v_i - v_j)^2 = \sum_{i \neq j} (v_i^2 + v_j^2 - 2v_i v_j).$$

Each v_i^2 occurs six times in the first two sums, while

$$\sum_{i \neq j} v_i v_j = \left(\sum_i v_i \right)^2 - \sum_i v_i^2 = - \sum_i v_i^2.$$

Hence the displayed expression is $6 \sum_i v_i^2 + 2 \sum_i v_i^2 = 8\|v\|^2$. Polarization gives (481). $\square$

For an integer $d \geq 2$, put $h = d^{-1}$ and

$$V_h = \Lambda_{A_3}/d\Lambda_{A_3}. \quad (483)$$

We regard hx , $x \in V_h$, as a periodic mesh of the flat torus $\mathbb{T}_{A_3}^3 = W_0/\Lambda_{A_3}$. An oriented root α connects x to $x + \alpha$ in V_h . Let $m_h > 0$ be the mass of each vertex and let $k_{\alpha,h} > 0$ be the jump rate along the oriented root α . The Markov generator on functions $f : V_h \rightarrow \mathbb{C}$ is

$$(L_h f)(x) = \sum_{\alpha \in \Phi_{A_3}} k_{\alpha,h} [f(x + \alpha) - f(x)]. \quad (484)$$

Its positive Dirichlet form on $L^2(V_h, m_h)$ is

$$\mathcal{E}_h(f) = -\langle f, L_h f \rangle_{L^2(m_h)} = \frac{m_h}{2} \sum_{x \in V_h} \sum_{\alpha \in \Phi_{A_3}} k_{\alpha,h} |f(x + \alpha) - f(x)|^2, \quad (485)$$

whenever the rates are reversal symmetric.

Lemma I.2 (Fourier symbol and flat generator limit). *Suppose*

$$k_{\alpha,h} = \frac{1}{8h^2} \quad (\alpha \in \Phi_{A_3}). \quad (486)$$

For a dual-lattice mode $e_k(x) = e^{2\pi i h \langle k, x \rangle}$, $-L_h$ has eigenvalue

$$\lambda_h(k) = \frac{1}{4h^2} \sum_{a=1}^6 [1 - \cos(2\pi h \langle r_a, k \rangle)], \quad (487)$$

where $r_1, \dots, r_6$ is one orientation of the six root pairs. For every fixed k ,

$$\lambda_h(k) \longrightarrow 2\pi^2 \|k\|^2, \quad (488)$$

and, under the natural piecewise-constant embeddings,

$$L_h \longrightarrow \frac{1}{2} \Delta_{\mathbb{T}_{A_3}^3} \quad (489)$$

in strong resolvent sense and strongly at the semigroup level.

Proof. Pair the roots α and $-\alpha$ in (484). This gives (487). Taylor expansion yields

$$\lambda_h(k) = \frac{\pi^2}{2} \sum_{a=1}^6 \langle r_a, k \rangle^2 + O_k(h^2).$$

Because the sum over all twelve roots is twice the sum over the six unoriented roots, [Lemma I.1](#) gives $\sum_{a=1}^6 r_a r_a^\top = 4I$, and therefore the leading term is $2\pi^2 \|k\|^2$.

The trigonometric symbol is nonnegative and converges pointwise on the common Fourier core. On each fixed finite Fourier band the convergence is uniform. The corresponding quadratic forms converge on trigonometric polynomials and have the common lower bound zero. Their closure gives Mosco convergence to the form of $\frac{1}{2} \Delta$; the resolvent and semigroup conclusions follow from the Mosco-resolvent equivalence. Equivalently, one may apply dominated convergence to the Fourier multipliers and then use density. $\square$

Lemma I.3 (Uniform spatial screens for the periodic A_3 graph). *Let the vertex mass be $m_h = a^3 h^3$ and the symmetric root rate be $k_{\alpha,h} = a^{-2}/(8h^2)$, where a ranges in a compact interval $0 < a_- \leq a \leq a_+ < \infty$. Then there are constants $c_P, C_W, c_{\text{cut}} > 0$, depending only on a_-, a_+ and the fixed torus, such that for all sufficiently small h :*

- (i) *the first nonzero eigenvalue of $-L_h$ is at least c_P ;*
- (ii) *the low-energy counting function satisfies*

$$N_h(R) := \#\{\lambda \in \text{spec}(-L_h) : \lambda \leq R\} \leq C_W(1 + R^{3/2}); \quad (490)$$

- (iii) *with edge conductance $c_{e,h} = m_h k_{\alpha,h} = ah/8$, every vertex set A with $0 < \mu_h(A) \leq \mu_h(V_h)/2$ obeys*

$$h c_h(\partial A) \geq c_{\text{cut}} \mu_h(A)^{2/3}. \quad (491)$$

Consequently the spectral projections $P_{h,R} = \mathbf{1}_{[0,R]}(-L_h)$ have uniformly bounded rank and

$$\|(I - P_{h,R})f\|_{L^2(m_h)}^2 \leq R^{-1} \mathcal{E}_h(f). \quad (492)$$

Thus they form a uniformly exhausted compact-screen family.

Proof. The scalar a^{-2} multiplies the flat symbol (487), and the compact interval for a changes all spectral comparisons only by fixed constants. For $|2\pi h \langle r_a, k \rangle| \leq \pi$, the inequalities $2s^2/\pi^2 \leq 1 - \cos s \leq s^2/2$, together with the root-frame identity, give

$$c_1 \|k\|^2 \leq \lambda_h(k) \leq c_2 \|k\|^2 \quad (493)$$

for modes in a fundamental dual cell, with constants independent of h . The first nonzero dual vector has fixed positive length, proving (i). Counting dual-lattice points in a three-dimensional ball gives (ii).

For (iii), retain only the three basis-root directions $\pm e_1, \pm e_2, \pm e_3$ from (18). Their Cayley subgraph is a periodic three-dimensional grid and each retained edge has conductance $ah/8 \geq a_- h/8$. The discrete edge-isoperimetric inequality on the d^3 periodic grid follows by slicing A along the three coordinate directions and applying the Loomis–Whitney inequality to the three coordinate projections; in physical units it reads $h^2 \# \partial A \geq c_0 (h^3 \# A)^{2/3}$. Since $\mu_h(A) = a^3 h^3 \# A$ and $c_{e,h} = ah/8$, it follows that $h c_h(\partial A) \geq [c_0/(8a_+)] \mu_h(A)^{2/3}$. Adding the other A_3 edges only increases this boundary capacity, proving (491). Finally, $-L_h \succeq R(I - P_{h,R})$ gives (492) by the spectral theorem. $\square$

I.2 Flat renewal vacuum

Construction I.4 (Flat periodic renewal regulator). At mesh $h = d^{-1}$, take the persistent spatial state $Z_n \in V_h$, one point in every nongeometric fibre, the twelve root branches (486), vertex mass

$$m_h = h^3, \quad (494)$$

the identity root router, the identity slow row, and zero nonvacuum reward. The event alphabet consists of the root-race outcomes together with the protected orientation and one-point spectator fibres. Its conditional law is normalized directly, so the predictive carrier and scheduler records are literal parts of the operational regulator.

Theorem I.5 (Flat ADM and vacuum limit). *The regulator of [Construction I.4](#) satisfies*

$$\boxed{\mathcal{B}_h = I_3, \quad \varrho_h = 1, \quad g_h = I_3, \quad N_h = 1, \quad \beta_h = 0.} \quad (495)$$

Every undirected root edge has conductance $h/8$. The spatial generators converge to $\frac{1}{2}\Delta_{\mathbb{T}_{A_3}^3}$, and the Lorentzian continuum is

$$\boxed{(M_0, g_0) = (\mathbb{R} \times \mathbb{T}_{A_3}^3, -dt^2 + dx_1^2 + dx_2^2 + dx_3^2).} \quad (496)$$

Its Levi–Civita connection and curvature vanish, so $G(g_0) = 0$. The finite torsion, curvature, and vacuum Palatini first-variation residuals vanish identically.

Proof. The root identity and rates give

$$\mathcal{B}_h = h^2 \sum_{\alpha \in \Phi_{A_3}} \frac{1}{8h^2} \alpha \alpha^\top = I_3.$$

Reversal symmetry gives $\beta_h = 0$. The mass density per physical volume is one, so $\varrho_h = 1$. Substitution in the ADM inversion theorem [Theorem B.10](#) gives (495). Multiplication of one directed rate by one vertex mass gives $h/8$.

Generator convergence is [Lemma I.2](#). The coframe $\vartheta^0 = dt$, $\vartheta^a = dx^a$ is constant, and the exact identity router gives the zero spin connection. Hence $T^A = d\vartheta^A + \omega^A_B \wedge \vartheta^B = 0$ and $\Omega^A_B = d\omega^A_B + \omega^A_C \wedge \omega^C_B = 0$. Therefore $\text{Ric} = 0$, $R = 0$, and $G = 0$. Any face-based finite Cartan and Palatini writer constructed from these exact constant data is identically zero, as is its first variation. Normalization and predictive persistence follow directly from the operational event law and [Theorem A.9](#). $\square$

I.3 Homogeneous de Sitter renewal vacuum

Let $\Lambda > 0$, $H = \sqrt{\Lambda/3}$, and choose the expanding proper-time stationary branch of [Corollary 5.2](#) with $a_0 = 1$ and $t_0 = 0$. On a compact time slab the resulting renewal rates and masses are

$$k_{\alpha,h}(t) = \frac{e^{-2Ht}}{8h^2}, \quad m_h(t) = e^{3Ht}h^3, \quad (497)$$

with physical speed density $\varrho_h(t) = e^{3Ht}$. These profiles are stationary outputs of the homogeneous renewal action, not assumptions made in this appendix.

Theorem I.6 (Reconstructed de Sitter ADM packet). *For every h and t ,*

$$\boxed{\mathcal{B}_h(t) = e^{-2Ht}I_3, \quad \varrho_h(t) = e^{3Ht}, \quad g_h(t) = e^{2Ht}I_3, \quad N_h(t) = 1, \quad \beta_h(t) = 0.} \quad (498)$$

Each undirected edge has conductance

$$\boxed{c_{e,h}(t) = \frac{e^{Ht}}{8}h.} \quad (499)$$

On every compact time slab, the cut, Poincaré, counting, and compact-screen constants are uniform in h and t .

Proof. By [Lemma I.1](#),

$$\mathcal{B}_h(t) = h^2 \frac{e^{-2Ht}}{8h^2} \sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = e^{-2Ht} I_3.$$

The rates are reversal symmetric, so $\beta_h = 0$. Formula (20) gives

$$g_h = e^{2Ht} I_3, \quad N_h = (e^{3Ht})^{1/3} (e^{-6Ht})^{1/6} = 1.$$

The conductance is the product of vertex mass and directed rate: $e^{3Ht} h^3 e^{-2Ht} / (8h^2) = e^{Ht} h / 8$. On $[-T, T]$, the scalar $a(t) = e^{Ht}$ is bounded above and below by positive constants, so [Lemma I.3](#) applies uniformly. $\square$

Let internal indices $A, B \in \{0, 1, 2, 3\}$ be raised and lowered with $\eta = \text{diag}(-1, 1, 1, 1)$. Define

$$\vartheta^0 = dt, \quad \vartheta^a = e^{Ht} dx^a, \quad \omega^a_0 = H\vartheta^a, \quad \omega^a_b = 0, \quad (500)$$

with $\omega_{AB} = -\omega_{BA}$.

Theorem I.7 (Cartan curvature and the vacuum Einstein equation). *The connection (500) is the torsion-free Levi-Civita spin connection of*

$$g_H = -dt^2 + e^{2Ht} \sum_{a=1}^3 (dx^a)^2. \quad (501)$$

Its curvature two-form is

$$\boxed{\Omega^A_B = H^2 \vartheta^A \wedge \vartheta_B.} \quad (502)$$

Consequently

$$\text{Ric}_{\mu\nu} = 3H^2 g_{\mu\nu}, \quad R = 12H^2, \quad \boxed{G_{\mu\nu} + 3H^2 g_{\mu\nu} = 0.} \quad (503)$$

The slices $t = \text{const}$ are compact Cauchy hypersurfaces, so the spatial quotient is globally hyperbolic.

Proof. Since $d\vartheta^a = H\vartheta^0 \wedge \vartheta^a$,

$$T^a = d\vartheta^a + \omega^a_0 \wedge \vartheta^0 = H\vartheta^0 \wedge \vartheta^a + H\vartheta^a \wedge \vartheta^0 = 0,$$

and $T^0 = 0$. The coframe is nondegenerate, so this metric-compatible connection is Levi-Civita.

Cartan's second equation gives

$$\Omega^a_0 = d(H\vartheta^a) = H^2 \vartheta^0 \wedge \vartheta^a, \quad \Omega^a_b = \omega^a_0 \wedge \omega^0_b = H^2 \vartheta^a \wedge \vartheta^b,$$

which is (502) in invariant notation. Hence $R_{ABCD} = H^2(\eta_{AC}\eta_{BD} - \eta_{AD}\eta_{BC})$, so contraction gives $\text{Ric}_{AB} = 3H^2 \eta_{AB}$ and $R = 12H^2$. Therefore $G = \text{Ric} - \frac{1}{2}Rg = -3H^2g$, proving (503).

The function t has timelike gradient and every inextendible causal curve meets each compact slice $\{t\} \times \mathbb{T}_{A_3}^3$ exactly once; this is the standard global-hyperbolicity criterion for a product metric with positive scale factor. $\square$

I.4 Exact links and summable finite Cartan–Palatini defects

Let K_h be the spacetime cell complex obtained by combining the spatial root mesh with a time mesh of size h . For an oriented edge e , let

$$U_{e,h} = \mathcal{P} \exp \left(- \int_e \omega \right) \quad (504)$$

be exact parallel transport of the reconstructed continuum connection along that edge. Then $U_{e,h}^\top \eta U_{e,h} = \eta$ exactly. For an oriented face p with area $|p|$, define the normalized curvature defect

$$\mathfrak{r}_{p,h} = |p|^{-1} \log U_{\partial p} + \Omega(x_p), \quad (505)$$

where x_p is the face midpoint and the logarithm is taken on the principal chart for sufficiently small h . Define the normalized torsion defect by the covariant developed-edge closure

$$\mathfrak{t}_{p,h} = |p|^{-1} \sum_{e \subset \partial p} U_{x_p \leftarrow s(e),h} \int_e \vartheta. \quad (506)$$

Finally, let $\delta S_h(e_h, U_h)$ be the midpoint quadrature of the Palatini–cosmological first variation and let $\delta S(e, \omega)$ be its continuum value.

Theorem I.8 (First-order consistency of the exact-link regulator). *For every compact time slab $[-T, T]$ there is $C_T < \infty$, independent of h , such that*

$$\sup_p (\|\mathfrak{r}_{p,h}\| + \|\mathfrak{t}_{p,h}\|) + \sup_{\|v_h\|_{C^1} \leq 1} |\delta S_h[v_h] - \delta S[v_h]| \leq C_T h. \quad (507)$$

For the flat regulator the left-hand side is zero. For the de Sitter regulator, take

$$h_n = 2^{-n^2}, \quad T_n = n. \quad (508)$$

Then every normalized defect is summable along the diagonal exhaustion:

$$\sum_{n=1}^{\infty} C_{T_n} h_n < \infty. \quad (509)$$

Proof. On a compact slab, the coframe, connection, curvature, and the finitely many derivatives needed below are bounded. The nonabelian Stokes expansion for the exact link gives

$$\log U_{\partial p} = -|p|\Omega(x_p) + O_T(h^3),$$

because $|p| = O(h^2)$ and the first variation of the curvature across the face is $O_T(h)$. Division by $|p|$ gives the curvature part of (507). The covariant sum of developed edge coframes is the integral of the torsion over the face plus an $O_T(h^3)$ quadrature remainder. Since the continuum torsion vanishes, division by area gives the torsion bound.

The Palatini–cosmological first variation is a finite sum of products of coframe, curvature, and test-field coefficients. Midpoint quadrature on each cell has local error $O_T(h^5)$ for a four-dimensional cell of volume $O(h^4)$; summing $O(h^{-4})$ cells on a fixed slab gives $O_T(h)$. Equivalent estimates follow directly from Taylor expansion of the finite wedge and holonomy writers. In the flat case all coefficients are constant and the exact links are identities, so the defects vanish.

For de Sitter, derivatives of order bounded independently of n grow at most exponentially on $[-n, n]$: $C_{T_n} \leq C e^{cn}$ for fixed constants C, c . Hence $C_{T_n} h_n \leq C \exp(cn - (\log 2)n^2)$, whose series converges. $\square$

I.5 Proof of Theorem 5.10

Proof of Theorem 5.10. The predictive carrier at every cutoff is reconstructed by Theorem A.9. The homogeneous temporal profiles and their finite stationary origin are Theorems 5.1 and 5.3 and Corollary 5.2; the nonempty operational matching family is Theorem 5.5. The spatial ADM reconstruction is Theorems I.5 and I.6. The strong-resolvent and semigroup spatial limits follow from Lemma I.2; the scalar time-dependent factor is uniformly controlled on compact slabs. Uniform cut, Poincaré, counting, and compact-screen bounds are Lemma I.3. The flat Einstein equation follows from the zero connection in Theorem I.5. The de Sitter Cartan and Einstein calculation is Theorem I.7. Exact metric-unitary links and the $O_T(h)$, diagonally summable consistency estimates are Theorem I.8. These statements establish the six numbered clauses and the operational realization asserted in the main theorem. $\square$

Remark I.9 (Meaning of constructive vacuum emergence). The temporal rate profiles used here are not chosen in order to reproduce a target metric: on the homogeneous branch they solve the lapse-varied renewal Euler system of Appendix G, and a nonempty finite accept/reject class reconstructs the same reduced action from its measured trial probabilities. The present appendix verifies the downstream spatial and Cartan geometry of those stationary profiles. The result still does not claim that an arbitrary renewal table selects the quadratic operational class, the rank-one compatibility condition, the value of Λ , or the expanding sign, nor that de Sitter is a universal attractor under coarse graining.